

Supporting Information for

Two-Dimensional Metal–Organic Framework with High-Performance Single-Molecule Magnets as Nodes Showing Magnetic Coercivity Photomodulation

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1. Materials and Instrumentation

Materials

All reactions were carried out under a dry and oxygen-free argon atmosphere by using Schlenk techniques or in a glovebox. All solvents were dried and degassed by standard technique. Anhydrous DyCl_3 , YCl_3 and GdCl_3 salts were prepared according to the literature procedure.¹

Instrumentation

Elemental analyses (C, H and N) were performed on a Elementar vario EL cube elemental analyzer. Powder X-ray diffraction (PXRD) measurements were recorded on a Bruker D8 ADVANCE diffractometer at room temperature. The solid-state UV–vis spectra were measured using polytetrafluoroethylene as a reference on a PerkinElmer Lambda950 spectrophotometer at room temperature. Electron paramagnetic resonance (EPR) data were achieved in the X band (frequency of microwave, 9.86 GHz) at room temperature on a Bruker A300-9.5/12 EPR spectrometer with a 100 kHz magnetic field. The modulation amplitude was 2 Gauss, with a sweep time of 81.92 s and a conversion time of 80 ms. The Infrared (IR) spectra were recorded on a BRUKER ALPHA. Raman spectra were recorded on a Renishaw inVia Qontor Raman spectrometer using 785 nm laser sources at room temperature. X-ray photoelectron spectroscopy (XPS) measurements were conducted on a Thermo Scientific K-Alpha spectrometer using $\text{Al K}\alpha$ as X-ray source. To compensate for surface charging effects, all XPS spectra were referenced to the C 1s neutral carbon peak at 284.6 eV. The scanning electron microscopy (SEM) images were obtained with a GeminiSEM 500. Direct-current (dc) magnetic susceptibilities were measured on a Quantum Design MPMS-squid VSM magnetometer. Diamagnetic corrections were made with Pascal's constants for all the constituent atoms and sample holder. Alternating-current (ac) magnetic susceptibility data were collected on the same instrument employing a 1.5 Oe oscillating field at frequencies up to 1000 Hz. Solid-state samples were used for all of the above measurements, and air-sensitive samples were sealed in argon/vacuum for measurement.

Density functional theory (DFT) calculations

Two extended crystal structures, $[(\text{Dy}_6(\text{OPh})_{12}(\text{BPy})_{18}(\text{THF})_6)(\text{BPh}_4)]^{5+}$ (**1A**) and $[(\text{Dy}_6(\text{OPh})_6\text{Cl}_6(\text{BPy})_{18}(\text{THF})_6)(\text{BPh}_4)]^{5+}$ (**1B**) were selected to analyze photochromic properties in the calculations. These models were derived from the crystal structures of **1pre** and **1**, based on the coordination environment of $[\text{Dy}(\text{OPh})_2(\text{BPy})_4(\text{THF})]^+$ and $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4(\text{THF})]^+$, respectively. To facilitate the DFT calculations, and given the similar ionic radii and chemical properties of Y(III) and Dy(III), we used Y(III) in the calculations as a substitute for Dy(III).² Frontier molecular orbital analysis was performed at the Becke's three parameter hybrid method with the LYP correlation function (B3LYP)^{3,4} level in gas phase with Grimme's D3 dispersion correction.⁵ The def2-TZVP basis set for Y atom⁶ and 6-311G(d) basis set for other atoms⁷ were utilized. The calculations were carried out using the GAUSSIAN 16.⁸

Ab initio calculations

Ab initio calculations with CASSCF/RASSI-SO/SINGLE_ANISO type using the OpenMolcas program package have been performed on $[\text{Dy}(\text{PhO})_2(\text{BPy})_4(\text{THF})]^+$ and $[\text{Dy}(\text{PhO})\text{Cl}(\text{BPy})_4(\text{THF})]^+$, based on the crystal structures of **1** and **1uv** without any optimizations. The basis sets were chosen from the ANO-RCC library with the paramagnetic ion treated with VTZP quality, the coordinated O, N and Cl atoms with VDZP quality, and the other atoms with VDZ quality.⁹ The state-averaged CASSCF orbitals of the sextets, quartets and doublets were optimized with 21, 224 and 490 states, respectively. 21 sextets, 128 quartets and 130 doublets were chosen to construct and diagonalize in spin-orbit (SO) coupling Hamiltonian with the RASSI module. These computed SO states were written into the SINGLE_ANISO program to compute the g-tensors, crystal field parameters and magnetic energy levels for the doublets of the ground $J = 15/2$ multiplet of the ^6H term for Dy(III). The two electron integrals were Cholesky decomposed with a threshold of 1×10^{-8} to account for the accuracy.

2. Synthesis

Syntheses of complexes 1–4

Complex $\{[\text{Dy}_{1.5}(\text{OPh})_2\text{Cl}(\text{BPy})_3(\text{THF})_{1.5}][(\text{BPh}_4)_{1.5}] \cdot 0.5\text{THF}\}_n$ (**1**)

Phenol (0.5 mmol, 47 mg) and NaH (0.5 mmol, 12 mg) were reacted in 12 mL THF for 30 min. Next, DyCl_3 (0.5 mmol, 134 mg) and sodium tetraphenylborate (NaBPh_4) (0.5 mmol, 171 mg) were added to the above solution. After heating and reflux at 85 °C for 24 hours, the resultant suspension was filtered. 4-phenylpyridine (3 mmol, 466 mg) was then added to the filtrate. After reflux overnight, the mixture was filtered. 4,4'-bipyridine (1 mmol, 156 mg) was added to the resulting filtrate, and the reaction was carried out at reflux for 3 days. After filtration, slow diffusion of hexane into the filtrate at room temperature led to the formation of light-yellow crystals suitable for single crystal X-ray diffraction (ca. 40.2 % yield, based on Dy). Elemental analysis calcd (%) for **1** ($\text{C}_{86}\text{H}_{80}\text{B}_{1.5}\text{N}_6\text{ClDy}_{1.5}\text{O}_4$): C 66.34, H 5.18, N 5.40; found: C 66.31, H 5.39, N 5.55.

Complex $\{[\text{Y}_{1.5}(\text{OPh})_2\text{Cl}(\text{BPy})_3(\text{THF})_{1.5}][(\text{BPh}_4)_{1.5}] \cdot 0.5\text{THF}\}_n$ (**2**)

The synthesis is the same as **1** with anhydrous DyCl_3 replaced by 0.5 mmol of anhydrous YCl_3 . Yield 50.1 %. Elemental analysis calcd (%) for **2** ($\text{C}_{86}\text{H}_{80}\text{B}_{1.5}\text{N}_6\text{ClY}_{1.5}\text{O}_4$): C 71.40, H 5.57, N 5.81; found: C 71.53, H 5.23, N 5.97.

Complex $\{[\text{Gd}_{1.5}(\text{OPh})_2\text{Cl}(\text{BPy})_3(\text{THF})_{1.5}][(\text{BPh}_4)_{1.5}] \cdot 0.5\text{THF}\}_n$ (**3**)

The synthesis is the same as **1** with anhydrous DyCl_3 replaced by 0.5 mmol of anhydrous GdCl_3 . Yield 46.6 %. Elemental analysis calcd (%) for **3** ($\text{C}_{86}\text{H}_{80}\text{B}_{1.5}\text{N}_6\text{ClGd}_{1.5}\text{O}_4$): C 66.68, H 5.20, N 5.43; found: C 66.94, H 5.15, N 5.92.

Complex $[\text{Dy}(\text{OPh})_2(\text{PhPy})_5](\text{BPh}_4) \cdot 5\text{Benzene}$ (**4**)

Phenol (1 mmol, 94 mg) and NaH (1 mmol, 24 mg) were reacted in 12 mL THF for 30 min. Next, DyCl_3 (0.5 mmol, 134 mg) and sodium tetraphenylborate (NaBPh_4) (0.5 mmol, 171 mg) were added to the above solution. After heating and reflux at 85 °C for 24 hours, the resultant suspension was filtered. 4-phenylpyridine (3 mmol, 466 mg) was then added to the filtrate. After 24 hours of reflux, the solution was filtered and subsequently dried under vacuum. The resulting powder was dissolved in a small amount of benzene and allowed to stand for one week, yielding colorless block-shaped crystals.

Cl⁻ content determination

The presence of chlorine (Cl) was detected by X-ray photoelectron spectroscopy (XPS) (Figure S1). Subsequently, the concentration of chloride ions (Cl⁻) in samples **1–3** was quantitatively analyzed using a chloride ion-selective electrode (ISE) method. A series of standard solutions with different Cl⁻ concentrations were prepared according to Table S1, and their corresponding potential values were measured. Based on the Nernst equation, a standard curve of potential (E) versus $-\lg c_{\text{Cl}^-}$ was fitted. Compounds **1–3** (20.0 mg each) were individually dissolved in 0.50 mL of DMF, mixed with 5.00 mL of 0.2 mol·L⁻¹ KNO₃ solution, and then diluted to 50.00 mL to prepare the test solutions. The average potentials of compounds **1–3** were measured to be 187.0 mV, 186.0 mV, and 186.0 mV, respectively. After background subtraction, the standard curve was used to determine that the molecular formulas of compounds **1**, **2** and **3** contain 1.14, 1.15, and 1.19 Cl⁻ ions, respectively.

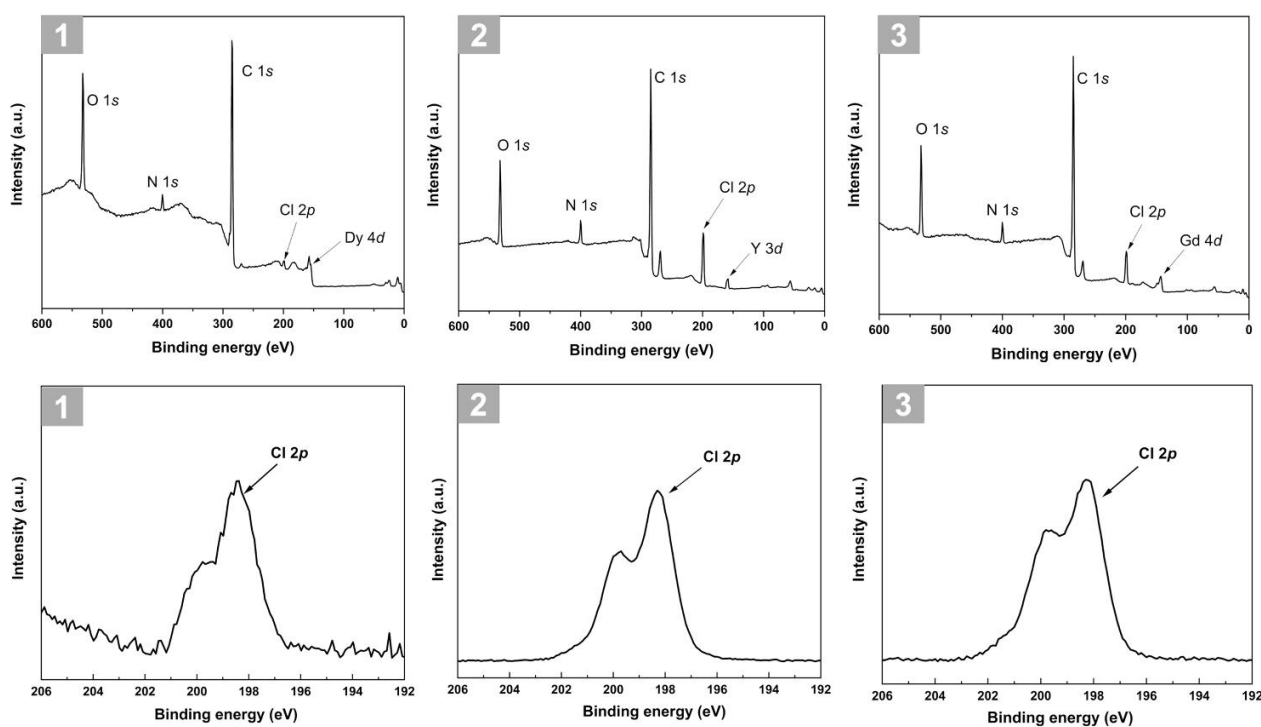


Figure S1. Survey XPS spectra (top) and Cl 2p spectra (bottom) of compounds **1**, **2** and **3**.

Table S1. Preparation of standard solutions with different concentrations.

Std. ID	A	B	C	D
0.01 mol·L ⁻¹ NaCl/mL	0.50	1.50	5.00	15.00
0.2 mol·L ⁻¹ KNO ₃ /mL	5.00	5.00	5.00	5.00
V/mL	50.00	50.00	50.00	50.00
c_{Cl^-} /mol·L ⁻¹	0.0001	0.0003	0.0010	0.0030
$-\lg c_{\text{Cl}^-}$	4.0	3.5	3.0	2.5
E/mV	213.0	191.0	163.0	140.0

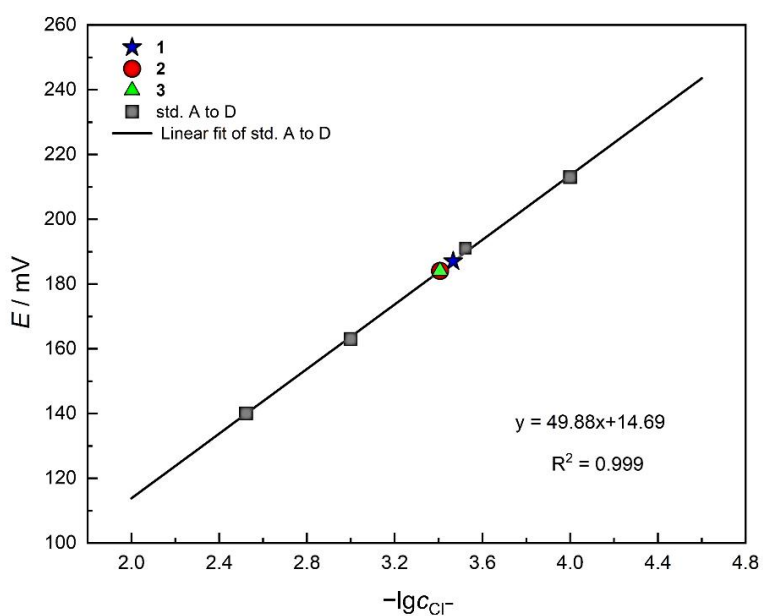


Figure S2. Plot of potential versus $-\lg c_{\text{Cl}^-}$, including the linear calibration curve fitted from stad. A to D.

3. X-ray Crystallography Data

The single-crystal X-ray diffraction data of **1** were recorded on a Bruker D8 Venture diffractometer using MoK α radiation at 150 K. The structure was solved by SHELXT and refined by full-matrix least-squares

method on F^2 with anisotropic thermal parameters for all nonhydrogen atoms.¹⁰ Hydrogen atoms were located geometrically and refined isotropically. Some $[\text{BPh}_4]^-$ anions and guest molecules in **1** were severely disordered and therefore removed by solvent mask in structural refinement. 1348 electrons were found in a volume of 8222 Å³ in 1 void per unit cell. This is consistent with the presence of 4 $[\text{BPh}_4]^-$ and 16[THF] per unit cell which account for 1316 electrons per unit cell. The crystals are naturally polycrystalline (Figure S11), and despite numerous attempts to obtain a relatively better single crystal, the crystal data still exhibit severe disorder, necessitating the application of numerous constraints during structural refinement. Nevertheless, the structure shows a high degree of consistency with a higher-quality crystal, **1pre**, formulated as $\{[\text{Dy}_{1.5}(\text{OPh})_3(\text{BPy})_3(\text{THF})_{1.5}][(\text{BPh}_4)_{1.5}]\}_n$, which further supports the definition and analysis of the structure.

The single-crystal X-ray diffraction data for **1uv** and **1O₂** were collected under identical conditions to those used for **1**. Specifically, after the diffraction data for the single crystal of **1** were collected at 150 K, the crystal was warmed to 300 K and irradiated with 365 nm UV light (6 W) under N₂ for 30 minutes. It was then cooled to 150 K under continuous irradiation, after which X-ray diffraction data for **1uv** were collected at 150 K. Subsequently, this crystal coated with perfluoropolyether was exposed to ambient air for 5~10 minutes and then cooled to 150 K, at which point the diffraction data for **1O₂** were collected. Nearly identical constraints were applied during the refinement process. Similarly, some $[\text{BPh}_4]^-$ anions and guest molecules in **1uv** and **1O₂** were severely disordered and removed by solvent mask in structural refinement. In **1uv**, 1352 electrons were found in a volume of 8056 Å³ in 1 void per unit cell. This is consistent with the presence of 4 $[\text{BPh}_4]^-$ and 16[THF] per unit cell which account for 1316 electrons per unit cell. In **1O₂**, 2208 electrons were found in a volume of 8256 Å³ in 1 void per unit cell. This is consistent with the presence of 4 $[\text{BPh}_4]^-$ and 16[THF] and 64[O₂] per unit cell which account for 2212 electrons per unit cell.

Fast scans were performed on single crystals of **2**, **2uv**, and **2O₂** utilizing CuKα radiation, while MoKα radiation was employed for **3**, **3uv** and **3O₂**. The irradiation and oxidation procedures were conducted with reference to the methods described for **1**, **1uv**, and **1O₂**.

The single-crystal X-ray diffraction data for **4** were collected utilizing MoKα radiation at 150 K.

CCDC 2427861 (**1**), 2427862 (**1uv**), 2427863 (**1O₂**), 2427864 (**4**) and 2371969 (**1pre**) contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

Table S2. Crystallographic data and structure refinement for **4**.

Compound	4
Formula	C ₁₂₁ H ₁₀₅ BDyN ₅ O ₂
<i>M</i> , g mol ⁻¹	1834.40
Temperature, K	150
Crystal system	monoclinic
Space group	<i>P</i> 2 ₁
<i>a</i> , Å	14.1759(3)
<i>b</i> , Å	46.2729(9)
<i>c</i> , Å	14.9180(3)
α , deg	90
β , deg	95.7240(10)
γ , deg	90
<i>V</i> , Å ³	9736.8(3)
<i>Z</i>	4
<i>d</i> _{cal} , g cm ⁻³	1.251
2 θ range, deg	3.808 to 52.75
Completeness	0.997
Final indices [<i>I</i> > 2 σ (<i>I</i>)]	<i>R</i> ₁ = 0.0291, <i>wR</i> ₂ = 0.0641
<i>R</i> indices (all data)	<i>R</i> ₁ = 0.0315, <i>wR</i> ₂ = 0.0650
Goodness-of-fit on <i>F</i> ²	1.020
Residual map, e Å ⁻³	0.87/-0.95

Table S3. Selected bond lengths (Å) and angles (deg) for **4**.

Dy1-O1	2.121(3)	O1-Dy1-O2	175.05(12)
Dy1-O2	2.122(3)	O1-Dy1-N1	84.27(11)
Dy1-N1	2.600(3)	O1-Dy1-N2	95.35(11)
Dy1-N2	2.512(3)	O1-Dy1-N3	88.14(11)
Dy1-N3	2.618(3)	O1-Dy1-N4	93.81(11)
Dy1-N4	2.495(4)	O1-Dy1-N5	91.66(11)
Dy1-N5	2.534(4)	O2-Dy1-N1	91.36(11)
O1-C1	1.339(5)	O2-Dy1-N2	85.29(11)
O2-C7	1.338(5)	O2-Dy1-N3	96.73(11)
N1-C13	1.344(5)	O2-Dy1-N4	88.55(11)
N1-C17	1.337(5)	O2-Dy1-N5	84.89(12)
N2-C24	1.336(6)	N1-Dy1-N2	70.08(10)
N2-C28	1.350(5)	N1-Dy1-N3	141.56(11)
N3-C35	1.342(5)	N1-Dy1-N4	146.36(11)
N3-C39	1.346(5)	N1-Dy1-N5	73.82(10)
N4-C46	1.338(6)	N2-Dy1-N3	73.22(10)
N4-C50	1.342(5)	N2-Dy1-N4	143.27(11)
N5-C57	1.339(6)	N2-Dy1-N5	142.26(10)
N5-C61	1.328(6)	N3-Dy1-N4	71.62(11)
C1-C2	1.404(6)	N3-Dy1-N5	144.19(11)
C1-C6	1.390(6)	N4-Dy1-N5	72.67(11)
C15-C16	1.393(6)	C13-N1-C17	116.3(3)
		C24-N2-C28	117.7(3)
		C35-N3-C39	114.9(3)
		C46-N4-C50	116.5(4)
		C57-N5-C61	116.6(4)
		C2-C1-C6	117.6(4)
		C19-C18-C23	118.6(4)

Table S4. Crystallographic data and structure refinement for **1**, **1uv** and **1O₂**.

Compound	1	1uv	1O₂
Formula	C ₁₈₄ H ₁₈₀ B ₃ Cl ₂ Dy ₃ N ₁₂ O ₁₁	C ₁₈₄ H ₁₇₉ B ₃ Cl ₂ Dy ₃ N ₁₂ O ₁₁	C ₁₈₄ H ₁₈₀ B ₃ Cl ₂ Dy ₃ N ₁₂ O ₃₉
<i>M</i> , g mol ⁻¹	3326.22	3325.21	3774.22
Temperature, K	150	150	150
Crystal system	monoclinic	monoclinic	monoclinic
Space group	<i>C2/c</i>	<i>C2/c</i>	<i>C2/c</i>
<i>a</i> , Å	39.447(5)	39.491(4)	39.579(4)
<i>b</i> , Å	22.811(2)	22.828(2)	22.8660(19)
<i>c</i> , Å	22.790(3)	22.730(2)	22.786(2)
α , deg	90	90	90
β , deg	105.445(4)	105.741(3)	105.914(3)
γ , deg	90	90	90
<i>V</i> , Å ³	19766(4)	19722(3)	19832(3)
<i>Z</i>	4	4	4
<i>d</i> _{cal} , g cm ⁻³	1.118	1.120	1.264
2 θ range, deg	3.676 to 49.426	3.724 to 49.426	3.562 to 49.42
Completeness	0.987	0.990	0.990
Final indices [<i>I</i> > 2 σ (<i>I</i>)]	<i>R</i> ₁ = 0.1194, <i>wR</i> ₂ = 0.2540	<i>R</i> ₁ = 0.1349, <i>wR</i> ₂ = 0.3379	<i>R</i> ₁ = 0.1324, <i>wR</i> ₂ = 0.3659
<i>R</i> indices (all data)	<i>R</i> ₁ = 0.1531, <i>wR</i> ₂ = 0.2804	<i>R</i> ₁ = 0.1478, <i>wR</i> ₂ = 0.3494	<i>R</i> ₁ = 0.1564, <i>wR</i> ₂ = 0.3855
Goodness-of-fit on <i>F</i> ²	1.042	1.036	1.170
Residual map, e Å ⁻³	1.86/-2.20	3.45/-3.41	3.73/-3.31

Table S5. Selected bond lengths (Å) and angles (deg) for **1**.

Dy1-Cl1	2.499(14)	O1-Dy1-Cl1	178.3(3)
Dy1-O1	2.034(13)	O1-Dy1-O2	172.7(11)
Dy1-O2	2.12(4)	O1-Dy1-O3	95.7(6)
Dy1-O3	2.337(14)	O1-Dy1-N1	89.7(4)
Dy1-N1	2.527(10)	O1-Dy1-N3	89.9(4)
Dy1-N3	2.553(9)	O1-Dy1-N4#1	84.6(4)
Dy1-N4#1	2.501(10)	O1-Dy1-N5	88.2(4)
Dy1-N5	2.545(10)	O2-Dy1-O3	91.5(12)
Dy2-Cl2	2.533(16)	O2-Dy1-N1	84.0(12)
Dy2-Cl2#2	2.533(16)	O2-Dy1-N3	84.7(11)
Dy2-O4	2.091(19)	O2-Dy1-N4#1	96.5(11)
Dy2-O4#2	2.091(19)	O2-Dy1-N5	94.9(12)
Dy2-O5	2.393(15)	Cl1-Dy1-O3	83.3(6)
Dy2-O5#2	2.392(16)	Cl1-Dy1-N1	90.3(5)
Dy2-N2#3	2.531(10)	Cl1-Dy1-N3	91.7(4)
Dy2-N2#4	2.531(10)	Cl1-Dy1-N5	92.8(5)
Dy2-N6	2.530(10)	O3-Dy1-N4#1	74.7(4)
Dy2-N6#2	2.530(10)	O3-Dy1-N5	71.5(4)
O1-C1	1.29(3)	N1-Dy1-N3	72.4(3)
O1-C1A	1.42(5)	N1-Dy1-N4#1	70.1(4)
O2-C7	1.41(4)	N3-Dy1-N5	72.3(4)
O4-C47	1.50(3)	O4-Dy2-Cl2#2	174.9(10)
B1-C57	1.68(3)	O4-Dy2-N6	88.8(13)
B1-C63	1.57(3)	O4#2-Dy2-N6	84.2(14)
B1-C69	1.65(3)	Cl2-Dy2-N6	93.1(6)
B1-C75	1.65(3)	Cl2#4-Dy2-O5	97.5(10)
B1-C57A	1.69(3)	Cl2-Dy2-N2#3	91.3(5)
B1-C63A	1.59(3)	O5-Dy2-N2#2	71.3(12)
N1-C13	1.26(2)	O5-Dy2-N2#3	71.4(12)
N1-C17	1.33(2)	N2#3-Dy2-N6#4	73.4(4)
N2-C20	1.36(2)	N2#2-Dy2-N6	73.4(4)
C25-C26	1.35(2)	N6-Dy2-N6#4	71.2(4)
C26-C27	1.367(18)	C13-N1-C17	116.6(12)
C28-C29	1.42(2)	C14-C15-C16	119.4(12)

^aSymmetry code: #1 1/2-X,1/2+Y,1/2-Z; #2 1-X,+Y,1/2-Z; #3 1/2+X,-1/2+Y,+Z; #4 1/2-X,-1/2+Y,1/2-Z

Table S6. Selected bond lengths (Å) and angles (deg) for **1uv**.

Dy1-Cl1	2.509(17)	O1-Dy1-Cl1	177.7(4)
Dy1-O1	1.996(11)	O1-Dy1-O2	171.2(11)
Dy1-O2	2.16(4)	O1-Dy1-O3	97.7(5)
Dy1-O3	2.354(16)	O1-Dy1-N1	89.1(4)
Dy1-N1	2.539(10)	O1-Dy1-N3	89.3(4)
Dy1-N3	2.557(11)	O1-Dy1-N4#1	83.8(4)
Dy1-N4#1	2.491(11)	O1-Dy1-N5	89.4(5)
Dy1-N5	2.595(15)	O2-Dy1-O3	90.7(12)
Dy2-Cl2	2.522(15)	O2-Dy1-N1	82.6(12)
Dy2-Cl2#2	2.522(15)	O2-Dy1-N3	85.4(11)
Dy2-O4	2.087(18)	O2-Dy1-N4#1	96.2(12)
Dy2-O4#2	2.087(18)	O2-Dy1-N5	95.6(12)
Dy2-O5	2.425(18)	Cl1-Dy1-O3	82.9(7)
Dy2-O5#2	2.424(18)	Cl1-Dy1-N1	89.0(6)
Dy2-N2#3	2.550(13)	Cl1-Dy1-N3	91.5(5)
Dy2-N2#4	2.550(13)	Cl1-Dy1-N4#1	94.2(5)
Dy2-N6	2.507(11)	Cl1-Dy1-N5	92.9(6)
Dy2-N6#2	2.507(11)	O3-Dy1-N4#1	74.1(5)
O1-C1	1.34(2)	O3-Dy1-N5	72.2(5)
O1-C1A	1.33(4)	N1-Dy1-N3	72.7(4)
O2-C7	1.33(5)	N1-Dy1-N4#1	70.4(4)
O4-C47	1.36(3)	N3-Dy1-N5	72.0(4)
B1-C57	1.69(3)	O4-Dy2-Cl2#2	172.9(6)
B1-C63	1.66(3)	O4-Dy2-N6	86.1(10)
B1-C69	1.68(3)	O4#2-Dy2-N6	83.3(9)
B1-C75	1.62(3)	Cl2-Dy2-N6	93.9(5)
B1-C57A	1.69(3)	Cl2#2-Dy2-N6	86.8(6)
B1-C63A	1.67(3)	Cl2#2-Dy2-O5	97.1(7)
N1-C13	1.24(2)	O5-Dy2-N2#3	69.5(14)
N1-C17	1.25(2)	O5-Dy2-N2#4	72.8(15)
N2-C20	1.35(2)	N2#3-Dy2-N6	74.1(4)
C25-C26	1.32(2)	N6-Dy2-N6#2	70.4(5)
C26-C27	1.38(2)	C13-N1-C17	122.1(15)
C28-C29	1.42(2)	C14-C15-C16	119.5(14)

^aSymmetry code: #1 1/2-X, 1/2+Y, 1/2-Z; #2 1-X, +Y, 1/2-Z; #3 1/2+X, -1/2+Y, +Z; #4 1/2-X, -1/2+Y, 1/2-Z

Table S7. Selected bond lengths (Å) and angles (deg) for **1O₂**.

Dy1-Cl1	2.527(14)	O1-Dy1-Cl1	177.8(4)
Dy1-O1	2.046(10)	O1-Dy1-O2	170.2(10)
Dy1-O2	2.10(4)	O1-Dy1-O3	97.0(5)
Dy1-O3	2.399(15)	O1-Dy1-N1	90.3(4)
Dy1-N1	2.542(11)	O1-Dy1-N3	89.8(4)
Dy1-N3	2.527(10)	O1-Dy1-N4#1	83.3(4)
Dy1-N4#1	2.475(10)	O1-Dy1-N5	89.3(5)
Dy1-N5	2.555(13)	O2-Dy1-O3	92.7(10)
Dy2-Cl2	2.502(14)	O2-Dy1-N1	81.2(11)
Dy2-Cl2#2	2.502(14)	O2-Dy1-N3	83.0(9)
Dy2-O4	2.078(18)	O2-Dy1-N4#1	98.4(10)
Dy2-O4#2	2.078(18)	O2-Dy1-N5	94.5(12)
Dy2-O5	2.390(18)	Cl1-Dy1-O3	83.5(5)
Dy2-O5#2	2.390(18)	Cl1-Dy1-N1	87.9(5)
Dy2-N2#3	2.520(11)	Cl1-Dy1-N3	91.0(4)
Dy2-N2#4	2.520(11)	Cl1-Dy1-N4#1	94.8(4)
Dy2-N6	2.522(11)	Cl1-Dy1-N5	92.9(5)
Dy2-N6#2	2.521(11)	O3-Dy1-N4#1	74.5(4)
O1-C1	1.34(2)	O3-Dy1-N5	73.2(4)
O1-C1A	1.31(2)	N1-Dy1-N3	73.0(3)
O2-C7	1.41(4)	N1-Dy1-N4#1	70.5(4)
O4-C47	1.55(3)	N3-Dy1-N5	70.4(4)
B1-C57	1.71(3)	O4-Dy2-Cl2#2	170.6(6)
B1-C63	1.66(3)	O4-Dy2-N6	84.9(10)
B1-C69	1.57(3)	O4#2-Dy2-N6	82.2(8)
B1-C75	1.70(3)	Cl2-Dy2-N6	94.1(5)
B1-C57A	1.72(4)	Cl2#2-Dy2-O5	98.0(7)
B1-C63A	1.67(3)	O5-Dy2-N2#3	71.7(10)
N1-C13	1.27(2)	O5-Dy2-N2#4	74.2(10)
N1-C17	1.32(2)	N2#3-Dy2-N6	72.3(5)
N2-C20	1.28(2)	N2#4-Dy2-N6#2	72.3(5)
C25-C26	1.31(3)	N6-Dy2-N6#2	70.9(6)
C26-C27	1.45(2)	C13-N1-C17	116.8(13)
C28-C29	1.41(2)	C14-C15-C16	116.2(13)

^aSymmetry code: #1 1/2-X, 1/2+Y, 1/2-Z; #2 1-X, +Y, 1/2-Z; #3 1/2+X, -1/2+Y, +Z; #4 1/2-X, -1/2+Y, 1/2-Z

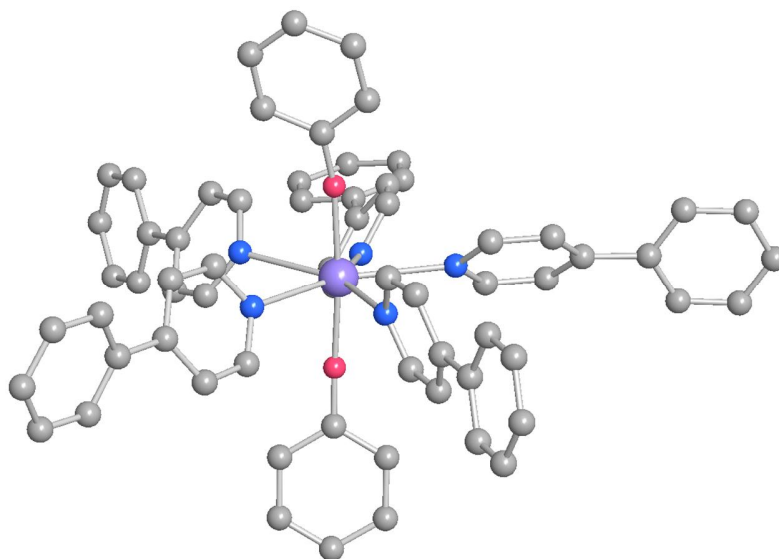


Figure S3. The crystal structure of $[\text{Dy}(\text{OPh})_2(\text{PhPy})_5]^+$ in compound **4**. Colors: O = red, C = grey, N = blue, Dy = purple. Hydrogen atoms are omitted for clarity.

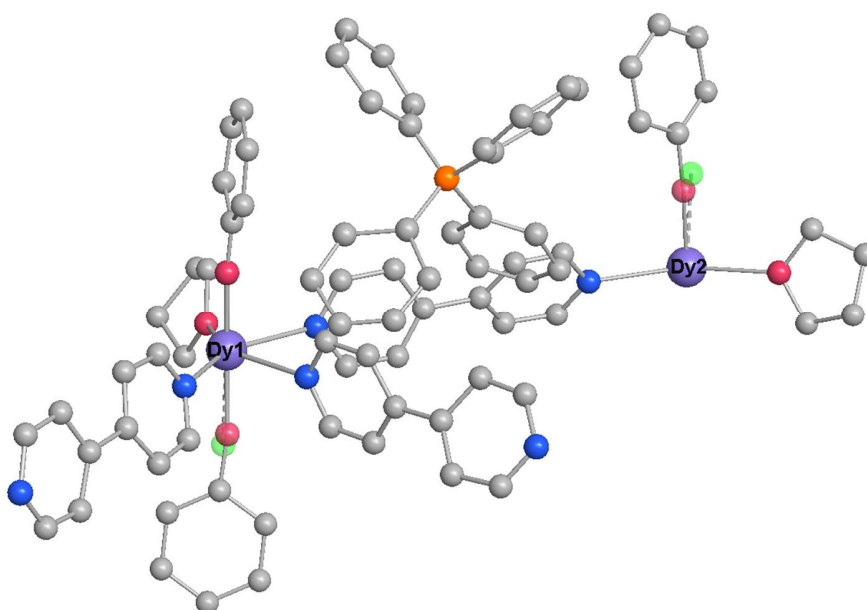


Figure S4. The asymmetric unit of compound **1**. Colors: O = red, C = grey, N = blue, Dy = purple, Cl = green, B = orange. Hydrogen atoms are omitted for clarity.

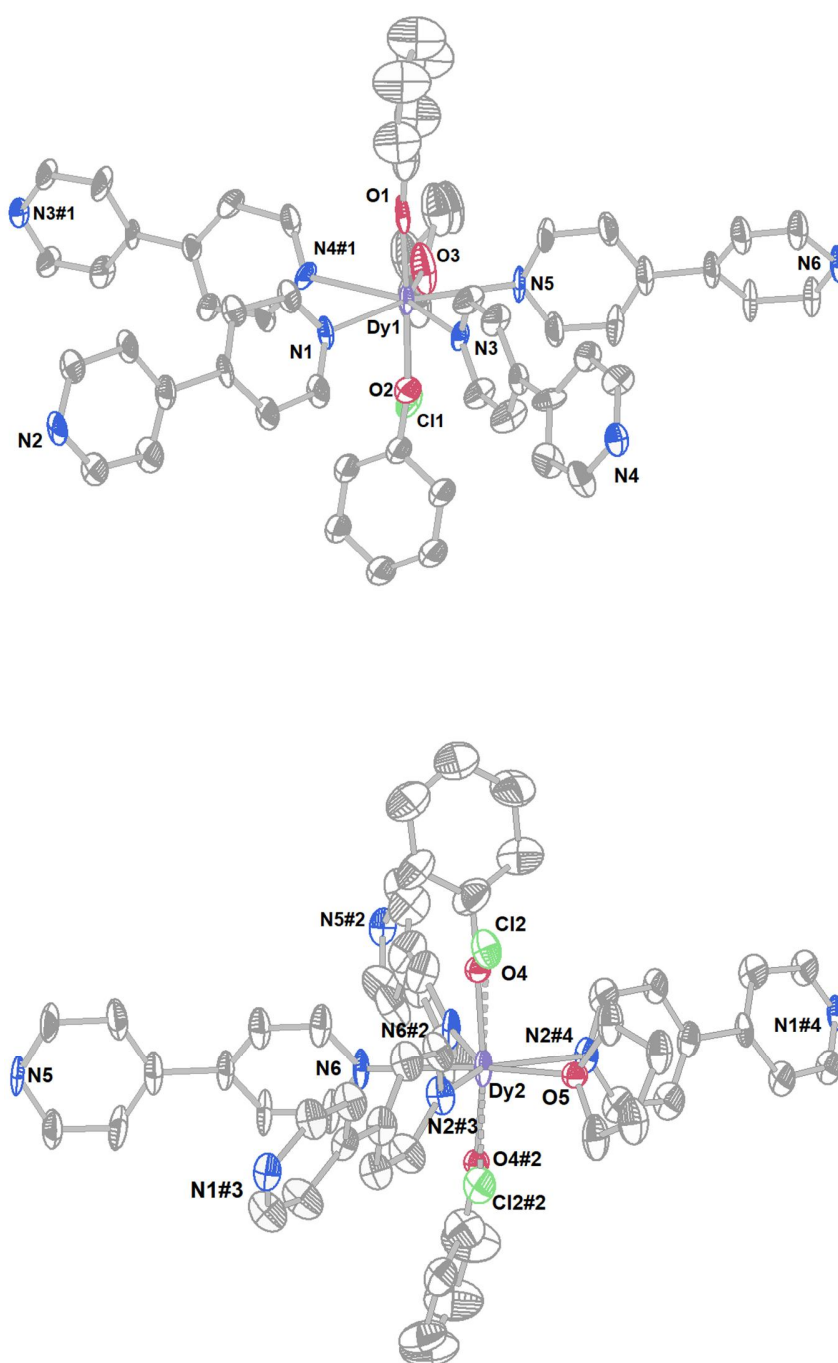


Figure S5. Crystal structure of **1**, showing the $[\text{Dy}(\text{OPh})\text{X}(\text{BPy})_4(\text{THF})]^+$ ($\text{X} = \text{Cl}/\text{PhO}^-$) cation in Dy1 and the $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4(\text{THF})]^+$ cation in Dy2 with thermal ellipsoids drawn at the 30% probability level. H atoms are omitted for clarity. Symmetry code: #1 $1/2-X, 1/2+Y, 1/2-Z$; #2 $1-X, +Y, 1/2-Z$; #3 $1/2+X, -1/2+Y, +Z$; #4 $1/2-X, -1/2+Y, 1/2-Z$.

Table S8. Continuous Shape Measures (CShM) calculation for **1**.

Structure ^a	CShM [Dy1(O) ₂ ON ₄]	CShM [Dy1(OC1)ON ₄]	CShM [Dy2(OC1)ON ₄]
HP-7	32.640	33.522	32.404
HPY-7	23.621	24.473	23.379
PBPY-7	1.010	0.733	0.605
COC-7	6.777	6.933	6.896
CTPR-7	5.341	5.472	5.103
JPBPY-7	1.616	2.777	2.888
JETPY-7	23.035	22.167	22.387

^a HP-7 = Heptagon (D7h); HPY-7 = Hexagonal pyramid (C6v); PBPY-7 = Pentagonal bipyramid (D5h); COC-7 = Capped octahedron (C3v); CTPR-7 = Capped trigonal prism (C2v); JPBPY-7 = Johnson pentagonal bipyramid J13 (D5h); JETPY-7 = Johnson elongated triangular pyramid J7 (C3v)

Table S9. Continuous Shape Measures (CShM) calculation for **1uv**.

Structure ^a	CShM [Dy1(O) ₂ ON ₄]	CShM [Dy1(OC1)ON ₄]	CShM [Dy2(OC1)ON ₄]
HP-7	32.887	33.326	32.420
HPY-7	23.631	24.387	23.372
PBPY-7	1.148	0.879	0.791
COC-7	6.486	6.637	6.190
CTPR-7	5.153	5.203	4.954
JPBPY-7	1.686	2.737	2.985
JETPY-7	22.718	22.078	22.508

^a HP-7 = Heptagon (D7h); HPY-7 = Hexagonal pyramid (C6v); PBPY-7 = Pentagonal bipyramid (D5h); COC-7 = Capped octahedron (C3v); CTPR-7 = Capped trigonal prism (C2v); JPBPY-7 = Johnson pentagonal bipyramid J13 (D5h); JETPY-7 = Johnson elongated triangular pyramid J7 (C3v)

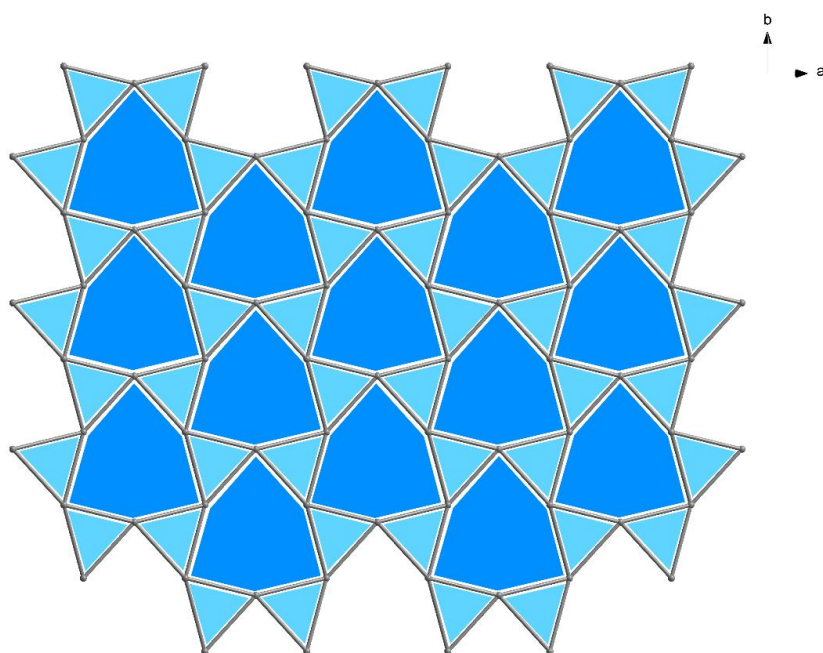


Figure S6. Archimedean tessellation of **1**.

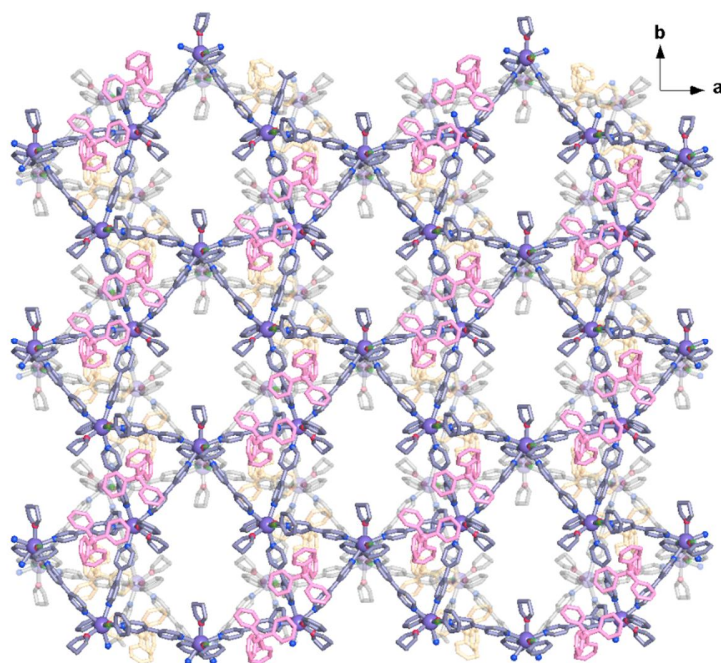


Figure S7. Crystal structure of **1** viewed along the *c*-axis, showing the inclusion of BPh_4^- in the channels.

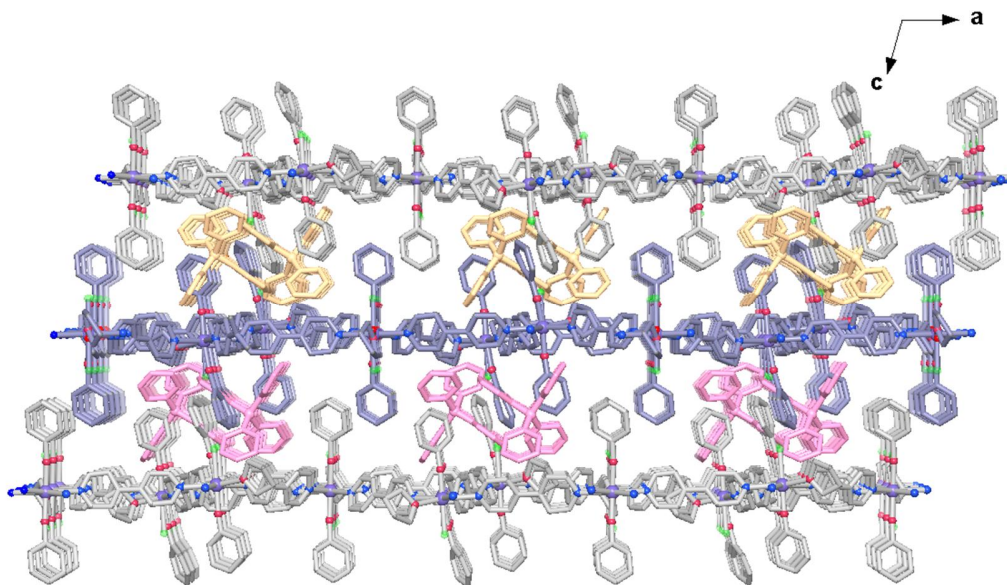


Figure S8. The 3D crystal structure of **1**, with BPh_4^- located between the layers.

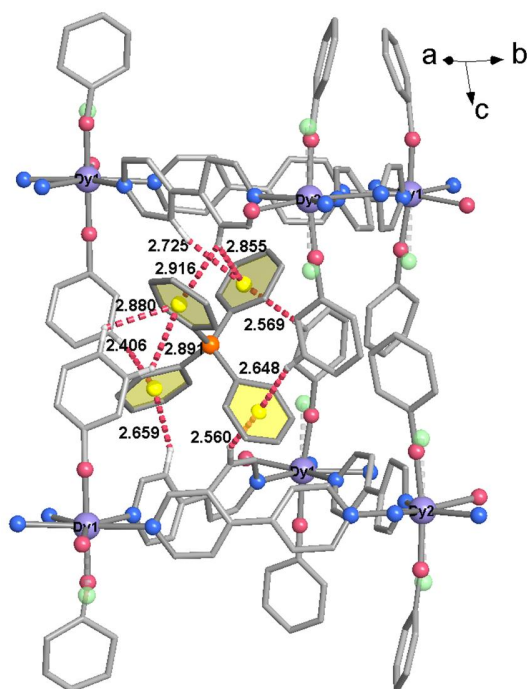


Figure S9. The interlayer C-H... π interaction in **1**. Some H atoms are omitted for clarity.

Table S10. Unit cell parameters of **2** and **3** determined at 150 K under CuK α and MoK α radiation, respectively. Note: prolonged X-ray exposure induced visible crystal coloration (see Figure S10).

Compound	2	2 (coloration)	3	3 (coloration)
Temperature, K	150	150	150	150
Crystal system	monoclinic	monoclinic	monoclinic	monoclinic
Space group	<i>C2/c</i>	<i>C2/c</i>	<i>C2/c</i>	<i>C2/c</i>
<i>a</i> , Å	39.55(4)	39.584(3)	39.94(3)	39.91(6)
<i>b</i> , Å	22.88(2)	22.893(17)	23.065(17)	23.06(3)
<i>c</i> , Å	23.05(2)	23.159(3)	23.20(2)	23.213(4)
α , deg	90	90	90	90
β , deg	105.19(4)	105.226(6)	105.36(2)	105.15(6)
γ , deg	90	90	90	90

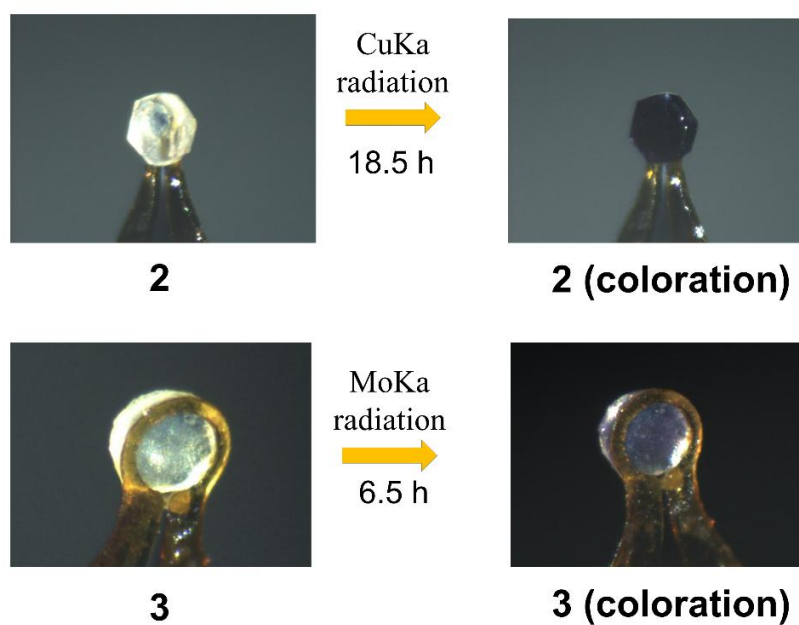


Figure S10. Photoinduced color changes of **2** after 18.5 h CuK α radiation, and **3** after 6.5 h MoK α radiation.

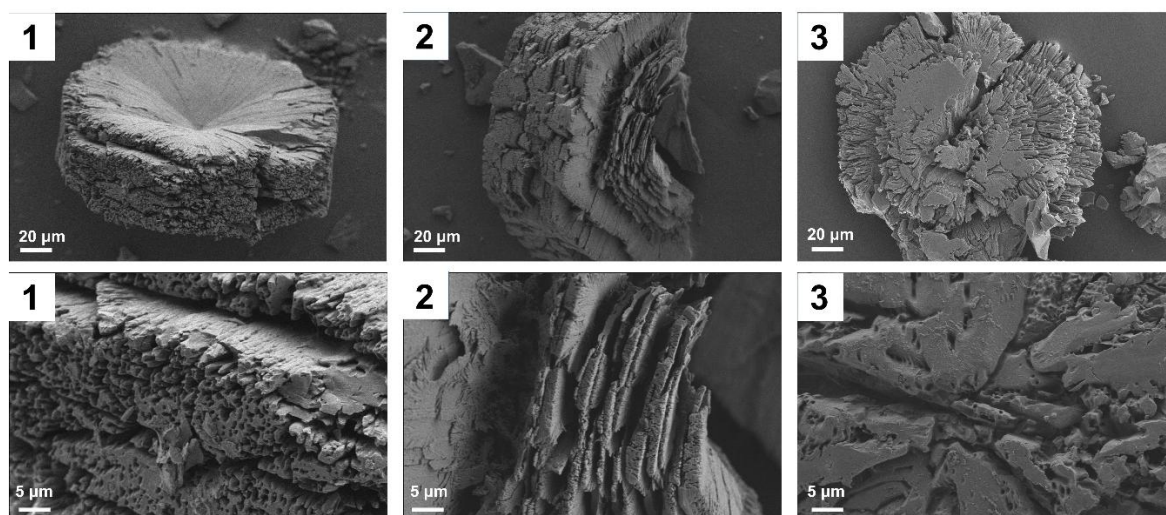


Figure S11. SEM images of dried crystals of **1**, **2** and **3**.

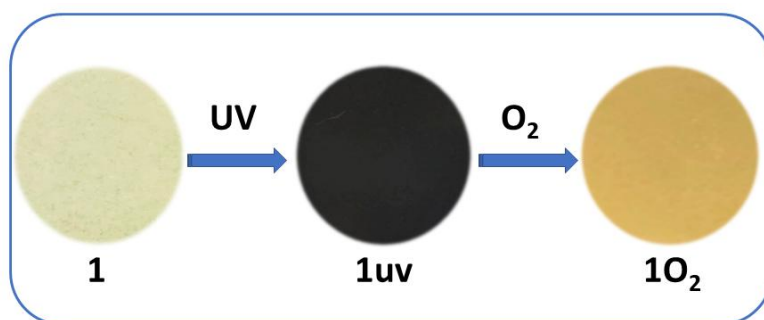


Figure S12. The powder color of **1**, **1uv** and **1O₂**.

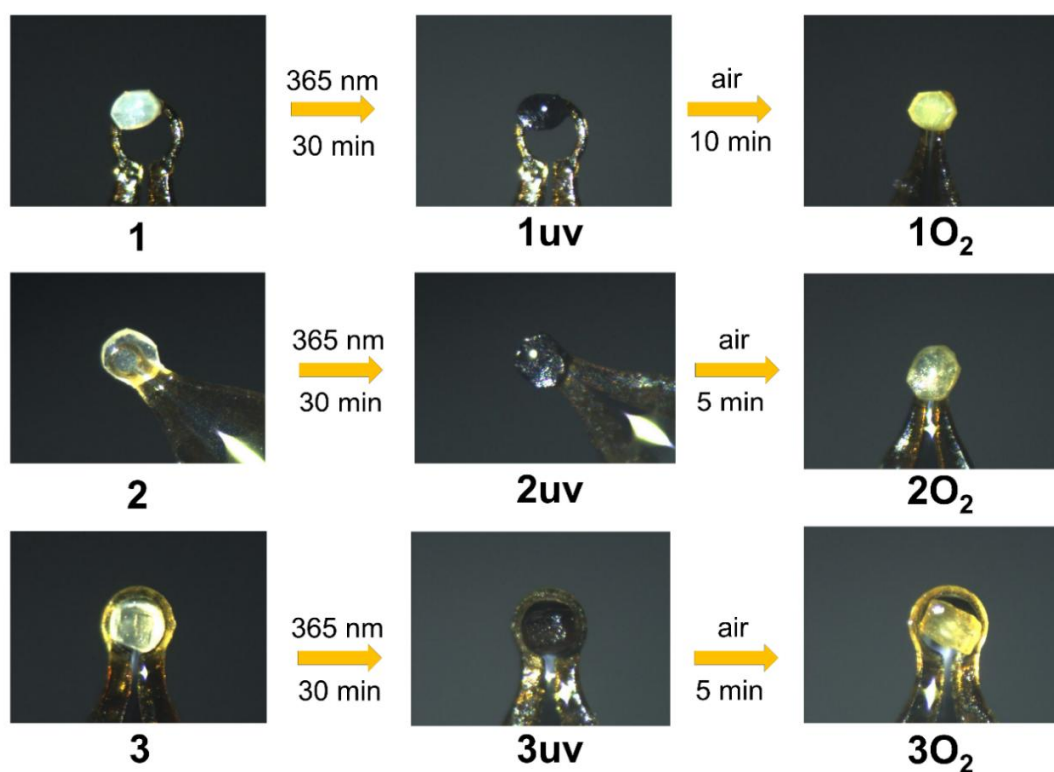


Figure S13. Color transition in a single crystal from **1**, **1uv** to **1O₂**, **2**, **2uv** to **2O₂** and **3**, **3uv** to **3O₂**.

Table S11. Unit cell parameters of **2**, **2uv**, **2O₂** and **3**, **3uv**, **3O₂** at 150 K.

Compound	2	2uv	2O₂	3	3uv	3O₂
Temperature, K	150	150	150	150	150	150
Crystal system	monoclinic	monoclinic	monoclinic	monoclinic	monoclinic	monoclinic
Space group	<i>C2/c</i>	<i>C2/c</i>	<i>C2/c</i>	<i>C2/c</i>	<i>C2/c</i>	<i>C2/c</i>
<i>a</i> , Å	39.64(4)	39.486(4)	39.583(15)	40.13(7)	39.77(9)	40.38(12)
<i>b</i> , Å	22.89(2)	22.817(2)	22.844(7)	23.16(4)	22.98(5)	23.33(7)
<i>c</i> , Å	22.92(7)	22.724(3)	22.942(9)	23.01(5)	23.22(5)	23.60(7)
α , deg	90	90	90	90	90	90
β , deg	105.28(9)	105.753(4)	105.678(15)	105.33(6)	105.47(6)	105.58(7)
γ , deg	90	90	90	90	90	90

4. Physical Measurements

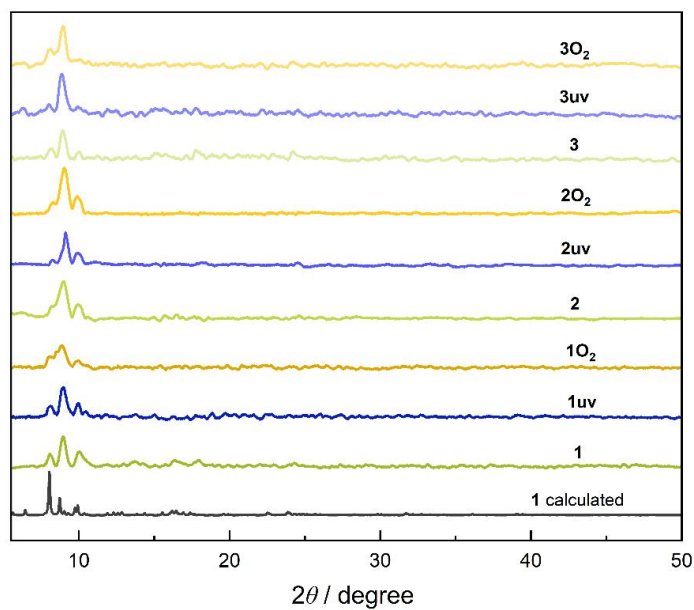


Figure S14. The PXRD curves of **1**, **1uv**, **1O₂**, **2**, **2uv**, **2O₂**, **3**, **3uv** and **3O₂**. The bottom curve is calculated from the single crystal data of **1**.

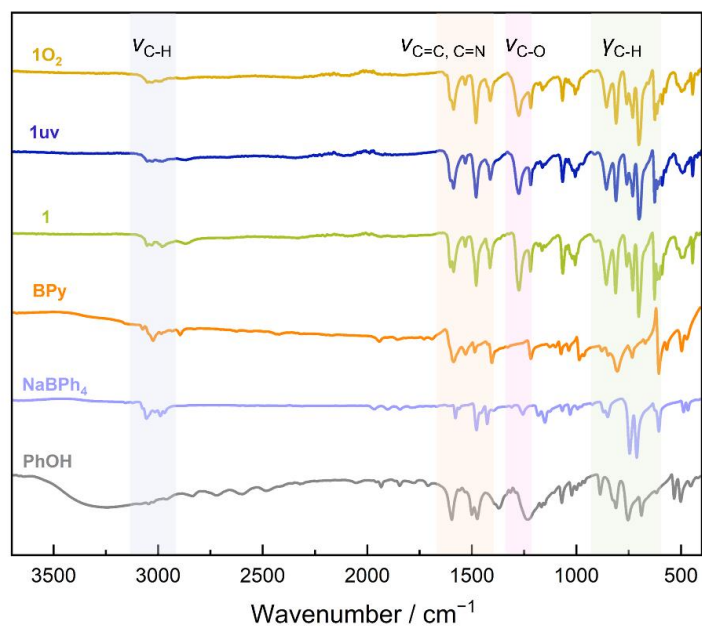


Figure S15. The IR spectra of **1**, **1uv**, **1O₂**, PhOH, NaBPh₄ and BPy.

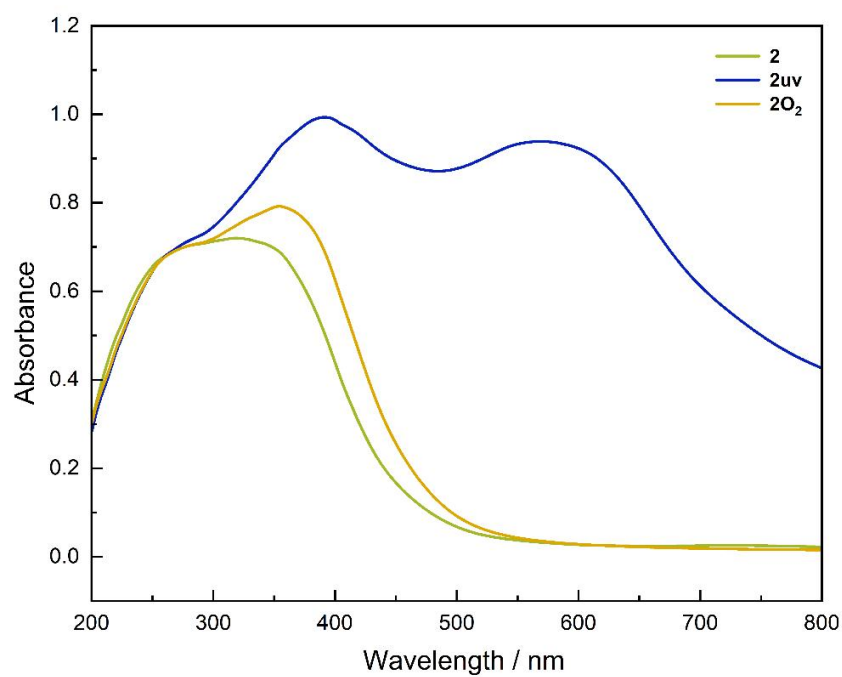


Figure S16. Solid-state UV-vis absorption spectra of **2**, **2uv** and **2O₂**.

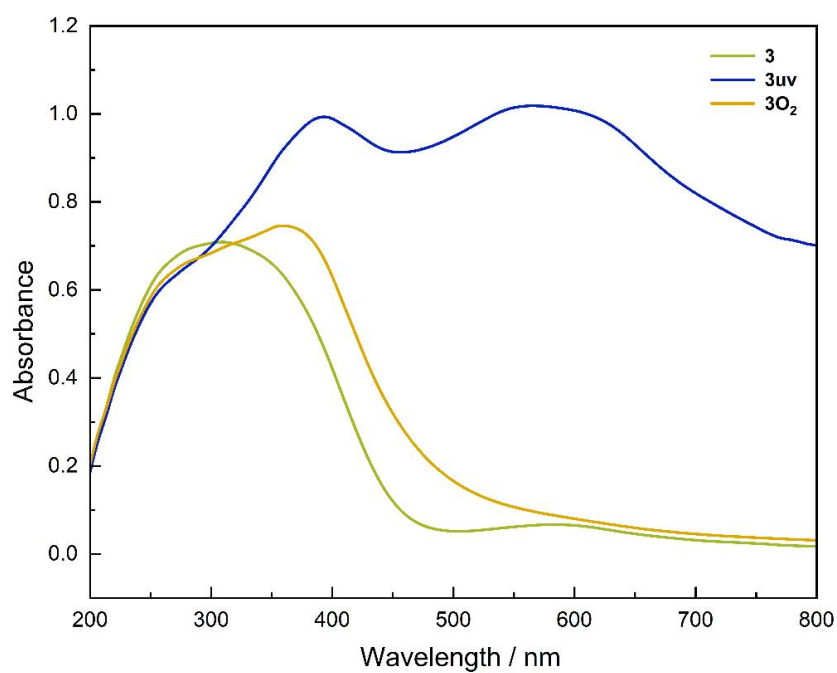


Figure S17. Solid-state UV-vis absorption spectra of **3**, **3uv** and **3O₂**.

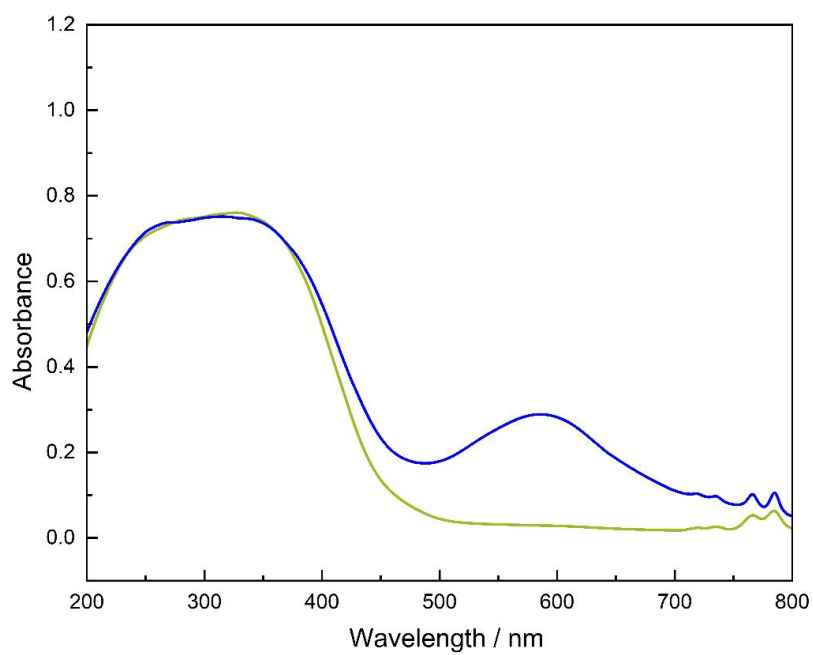


Figure S18. Solid-state UV-vis absorption spectra of **1** in its initial state (green) and under ambient light irradiation (blue).

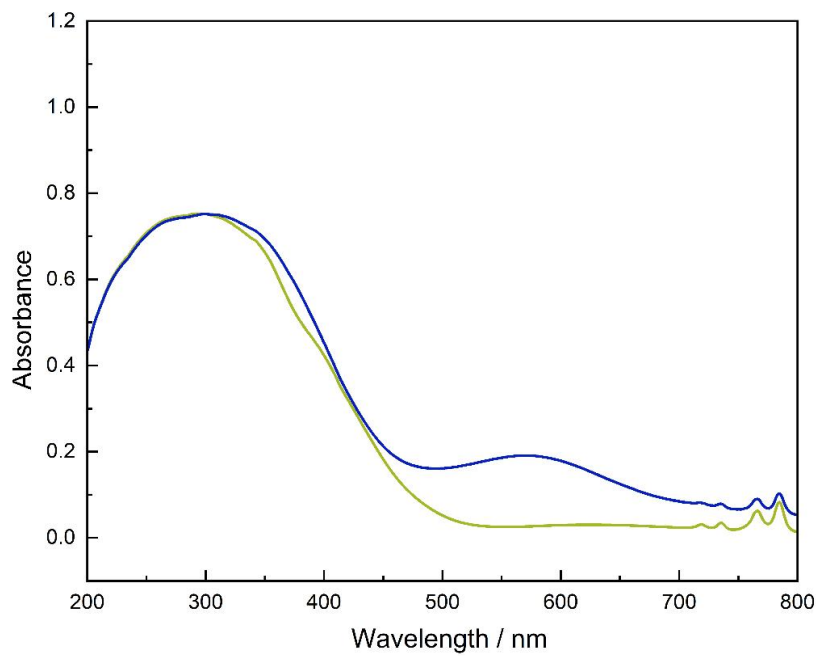


Figure S19. Solid-state UV-vis absorption spectra of **1** in its initial state (green) and after irradiation at 520 nm (blue).

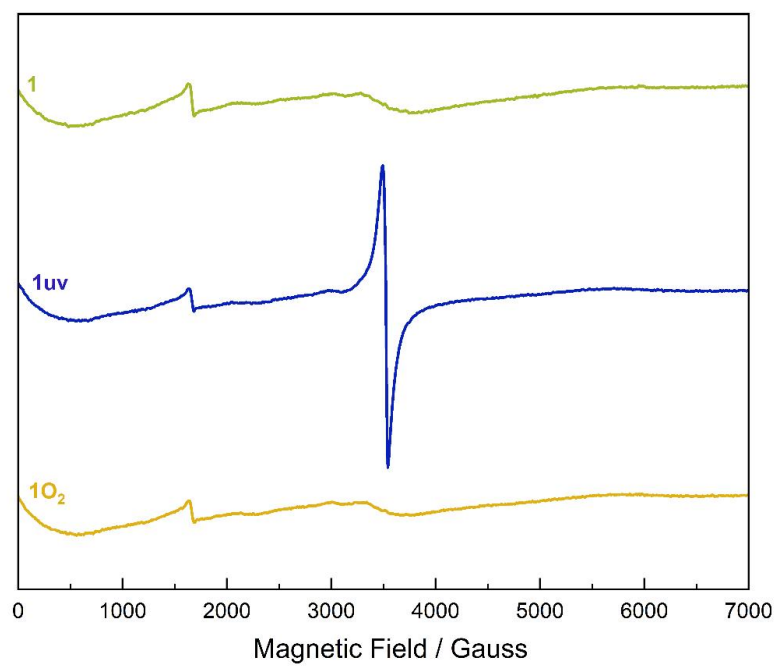


Figure S20. EPR spectra of **1**, **1uv** and **1O₂**.

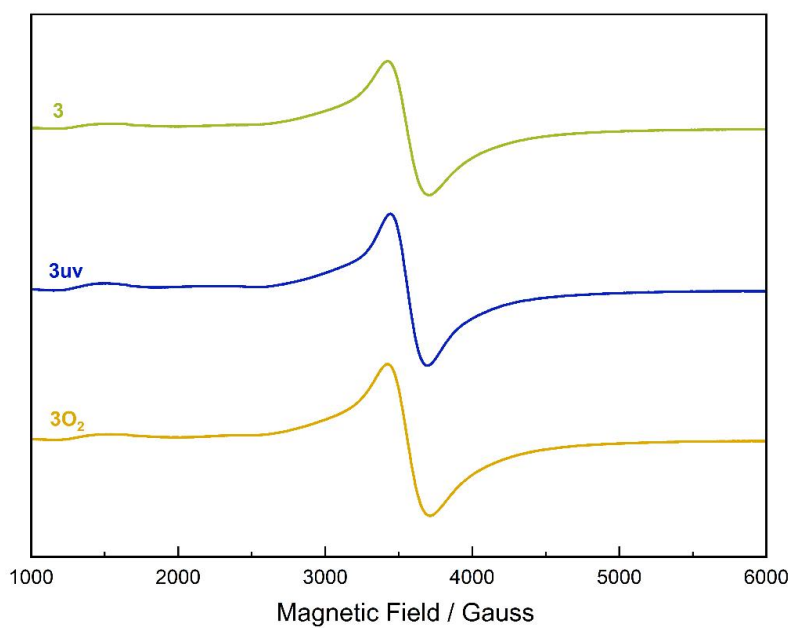


Figure S21. EPR spectra of **3**, **3uv** and **3O₂**.

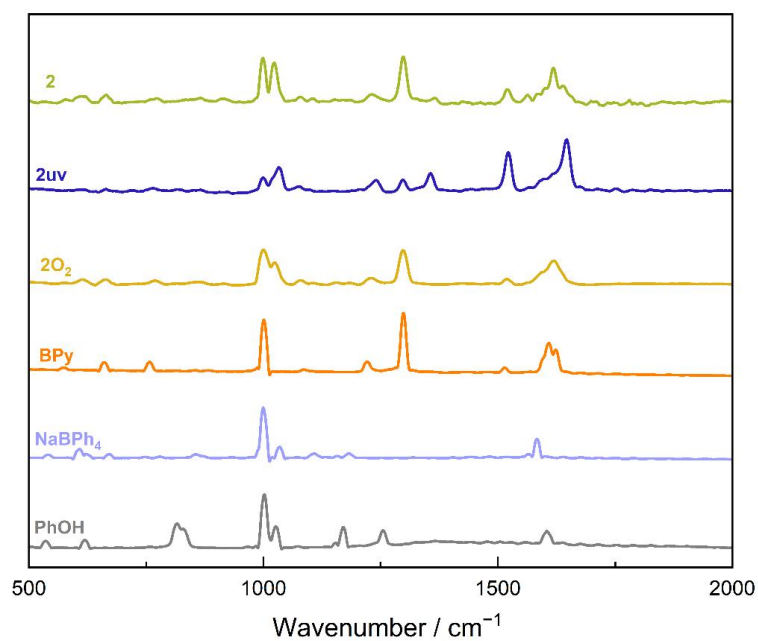


Figure S22. Raman spectra of **2**, **2uv**, **2O₂**, PhOH, NaBPh₄ and BPy. Excitation wavelength: 785 nm.

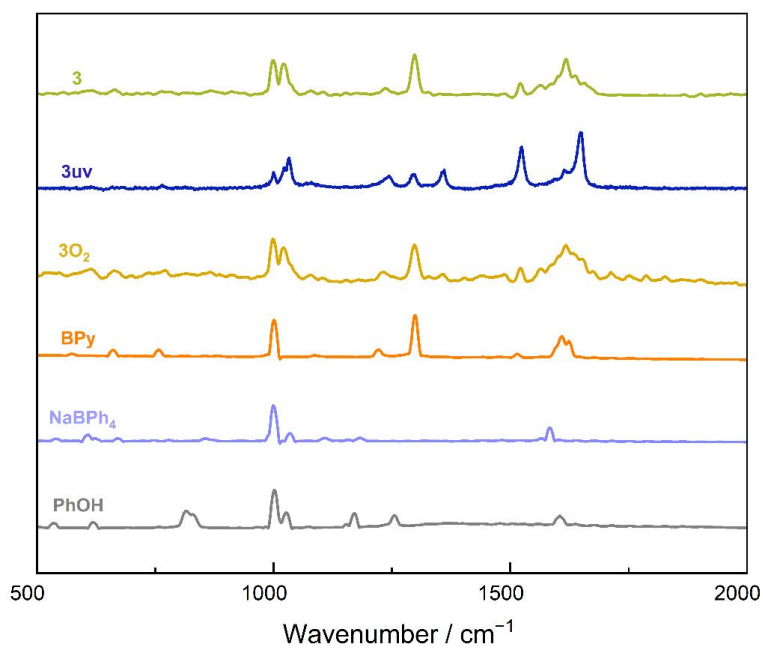


Figure S23. Raman spectra of **3**, **3uv**, **3O₂**, PhOH, NaBPh₄ and BPy. Excitation wavelength: 785 nm.

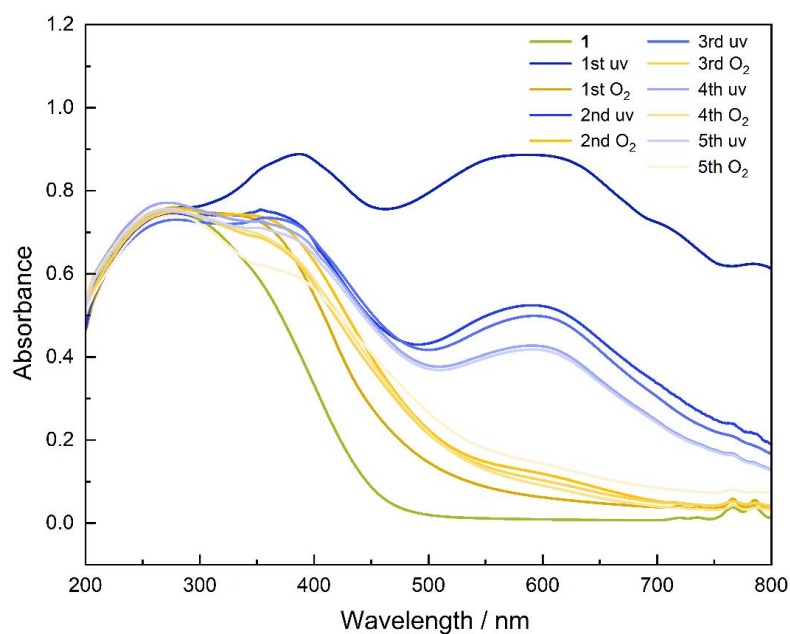


Figure S24. Solid-state UV-vis absorption spectra of **1** for five cycles of irradiation treatments and subsequent oxidation processes.

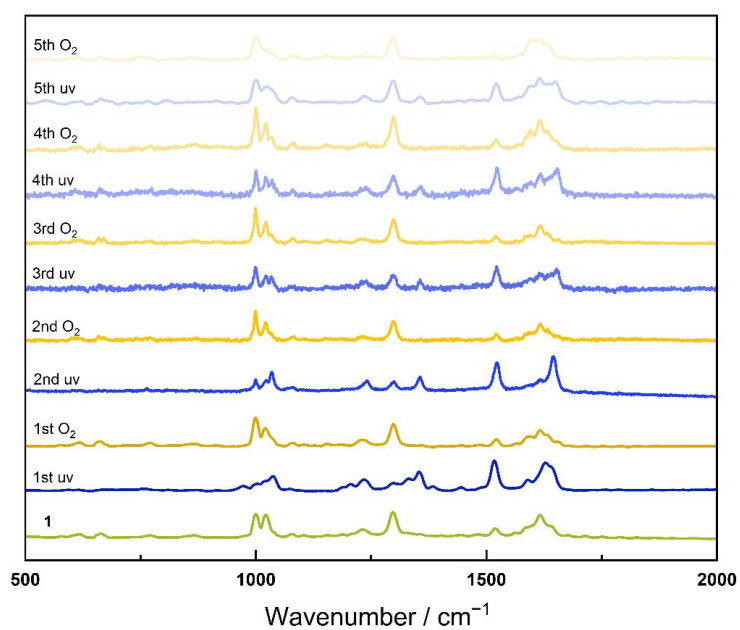


Figure S25. Raman spectra of **1** for five cycles of irradiation treatments and subsequent oxidation processes. Excitation wavelength: 785 nm.

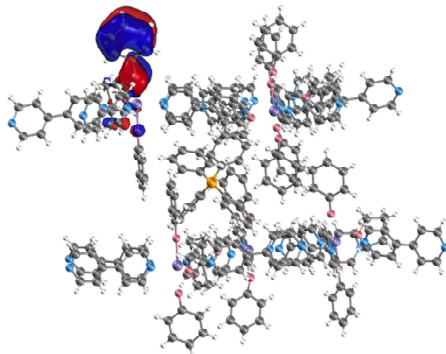
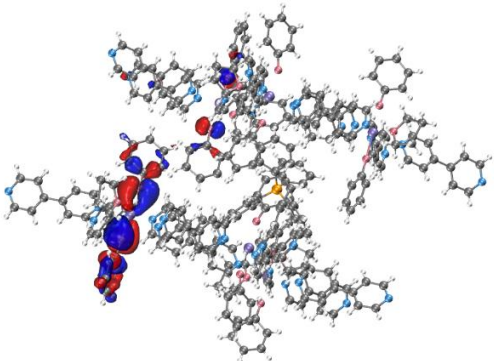
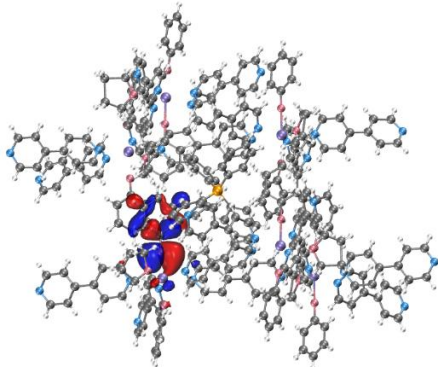
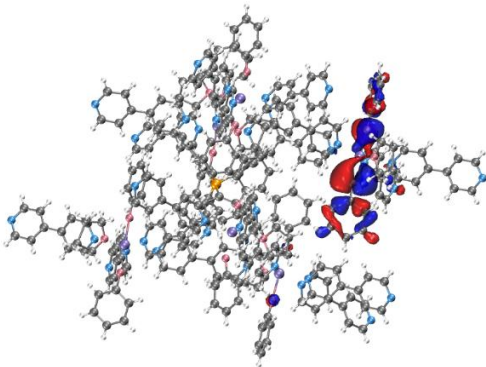
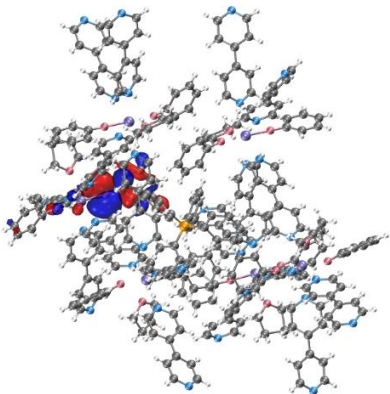
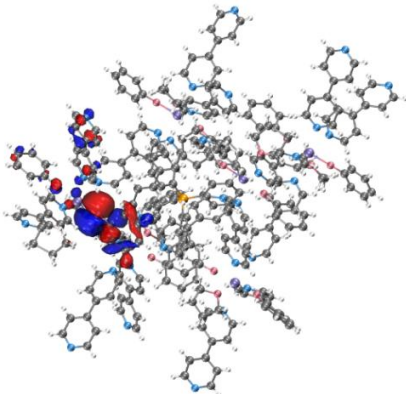
5. Density Functional Theory Calculations

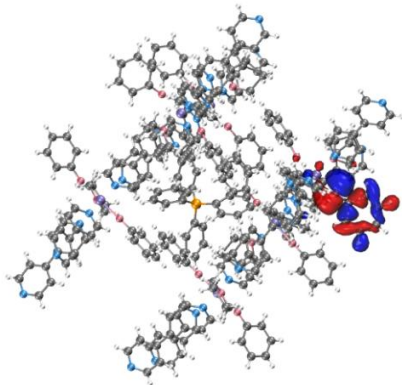
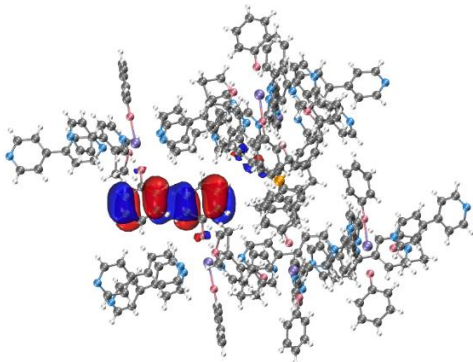
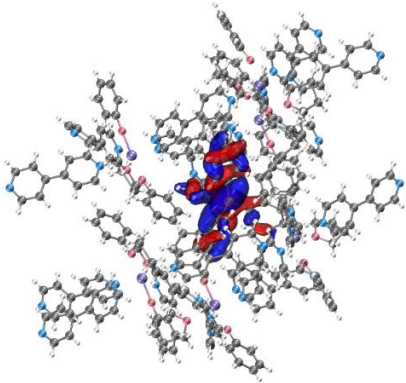
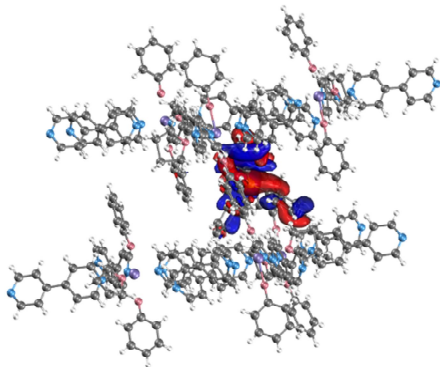
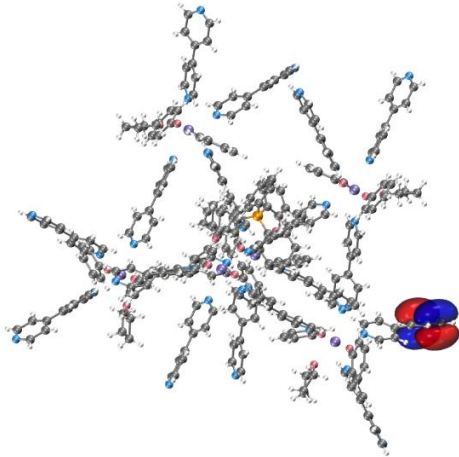
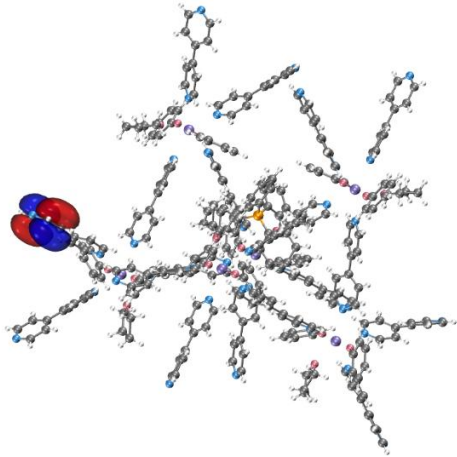
Table S12: Frontier molecular orbitals and the energy level for **1A**.

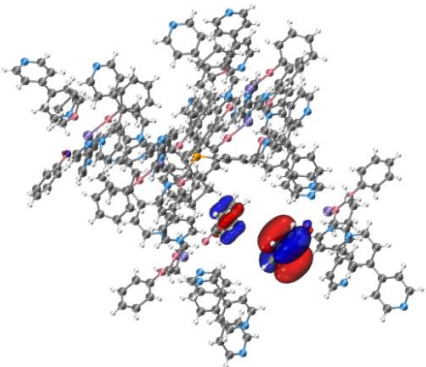
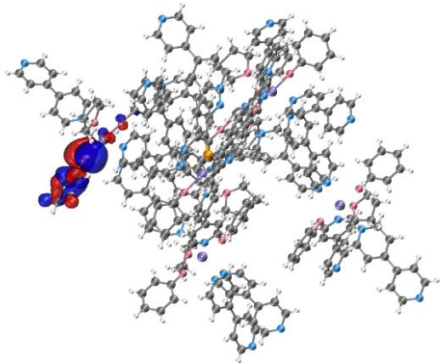
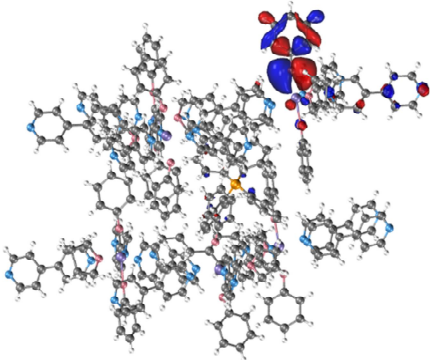
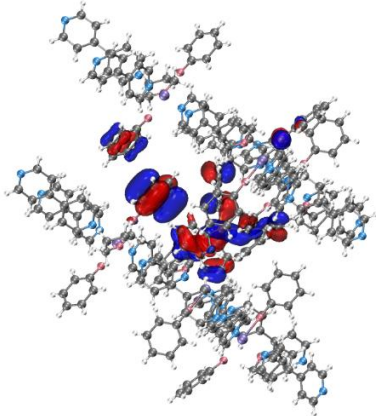
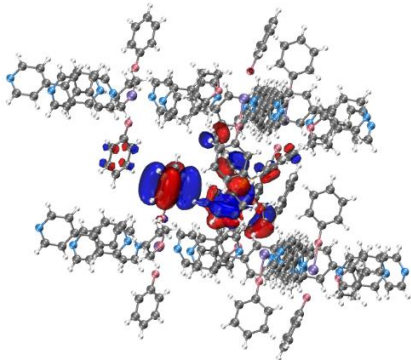
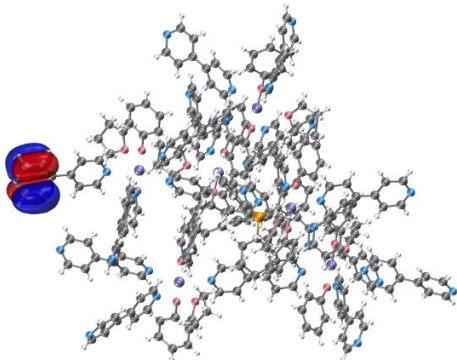
Orbital Number	Orbital Name	Energy (eV)	Orbital Number	Orbital Name	Energy (eV)
1170	HOMO-97	-13.8327	1268	LUMO	-8.8948
1171	HOMO-96	-13.8182	1269	LUMO+1	-8.846
1172	HOMO-95	-13.8131	1270	LUMO+2	-8.6101
1173	HOMO-94	-13.7889	1271	LUMO+3	-8.5488
1174	HOMO-93	-13.7486	1272	LUMO+4	-8.3447
1175	HOMO-92	-13.7113	1273	LUMO+5	-8.2744
1176	HOMO-91	-13.6982	1274	LUMO+6	-7.8525
1177	HOMO-90	-13.6917	1275	LUMO+7	-7.8433
1178	HOMO-89	-13.673	1276	LUMO+8	-7.8356
1179	HOMO-88	-13.6593	1277	LUMO+9	-7.8196
1180	HOMO-87	-13.6389	1278	LUMO+10	-7.7587
1181	HOMO-86	-13.6238	1279	LUMO+11	-7.6947
1182	HOMO-85	-13.565	1280	LUMO+12	-7.6472
1183	HOMO-84	-13.4577	1281	LUMO+13	-7.6095
1184	HOMO-83	-13.4529	1282	LUMO+14	-7.6054
1185	HOMO-82	-13.3699	1283	LUMO+15	-7.5864
1186	HOMO-81	-13.3656	1284	LUMO+16	-7.5459
1187	HOMO-80	-13.3612	1285	LUMO+17	-7.5236
1188	HOMO-79	-13.3442	1286	LUMO+18	-7.4832
1189	HOMO-78	-13.3225	1287	LUMO+19	-7.4507
1190	HOMO-77	-13.3133	1288	LUMO+20	-7.4281
1191	HOMO-76	-13.2993	1289	LUMO+21	-7.3746
1192	HOMO-75	-13.2745	1290	LUMO+22	-7.3533
1193	HOMO-74	-13.2683	1291	LUMO+23	-7.281
1194	HOMO-73	-13.2352	1292	LUMO+24	-7.2742
1195	HOMO-72	-13.2342	1293	LUMO+25	-7.2075
1196	HOMO-71	-13.2299	1294	LUMO+26	-7.207
1197	HOMO-70	-13.1848	1295	LUMO+27	-7.1418
1198	HOMO-69	-13.1663	1296	LUMO+28	-7.0746
1199	HOMO-68	-13.1453	1297	LUMO+29	-7.0627
1200	HOMO-67	-13.1183	1298	LUMO+30	-7.0291
1201	HOMO-66	-13.0658	1299	LUMO+31	-7.0259
1202	HOMO-65	-13.0532	1300	LUMO+32	-6.9461
1203	HOMO-64	-13.0067	1301	LUMO+33	-6.9263
1204	HOMO-63	-13.0034	1302	LUMO+34	-6.8966
1205	HOMO-62	-12.9898	1303	LUMO+35	-6.8911
1206	HOMO-61	-12.951	1304	LUMO+36	-6.8632
1207	HOMO-60	-12.9453	1305	LUMO+37	-6.8426
1208	HOMO-59	-12.9434	1306	LUMO+38	-6.7942
1209	HOMO-58	-12.936	1307	LUMO+39	-6.7888
1210	HOMO-57	12.9013	1308	LUMO+40	-6.7638
1211	HOMO-56	-12.8884	1309	LUMO+41	-6.6996
1212	HOMO-55	-12.8245	1310	LUMO+42	-6.6534
1213	HOMO-54	-12.8052	1311	LUMO+43	-6.6349
1214	HOMO-53	-12.7716	1312	LUMO+44	-6.5735

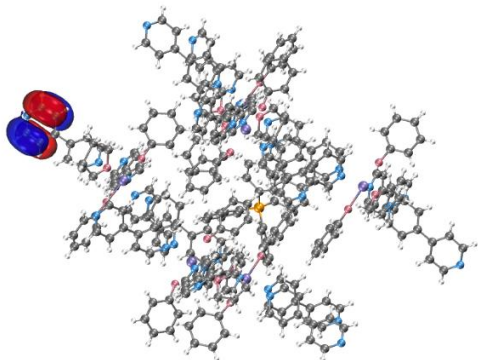
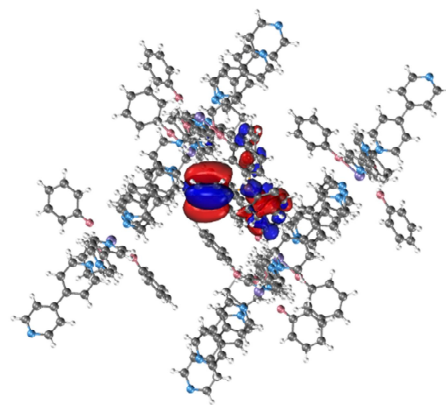
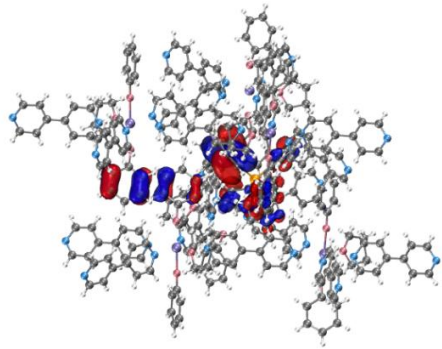
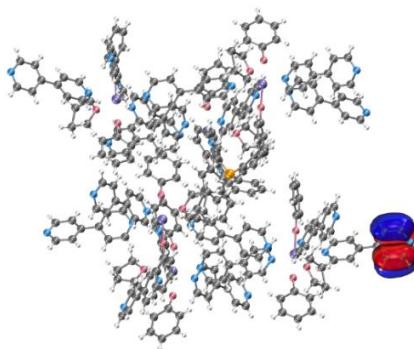
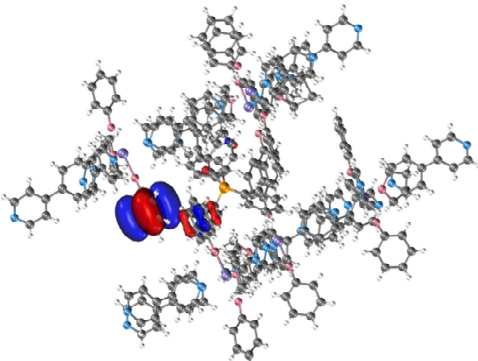
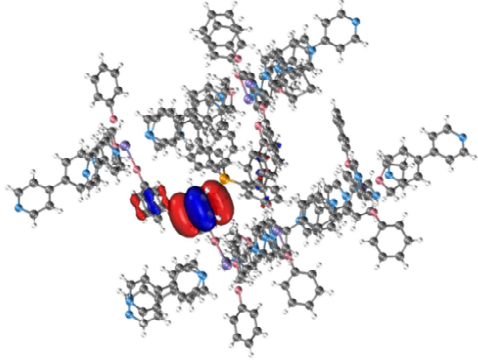
1215	HOMO-52	-12.765	1313	LUMO+45	-6.5567
1216	HOMO-51	-12.7605	1314	LUMO+46	-6.3404
1217	HOMO-50	-12.7406	1315	LUMO+47	-6.3108
1218	HOMO-49	-12.7008	1316	LUMO+48	-6.3043
1219	HOMO-48	-12.6962	1317	LUMO+49	-6.0201
1220	HOMO-47	-12.6839	1318	LUMO+50	-6.0053
1221	HOMO-46	-12.6526	1319	LUMO+51	-5.9988
1222	HOMO-45	-12.6451	1320	LUMO+52	-5.9908
1223	HOMO-44	-12.6181	1321	LUMO+53	-5.9308
1224	HOMO-43	-12.614	1322	LUMO+54	-5.9112
1225	HOMO-42	-12.6012	1323	LUMO+55	-5.8838
1226	HOMO-41	-12.562	1324	LUMO+56	-5.8771
1227	HOMO-40	-12.516	1325	LUMO+57	-5.8642
1228	HOMO-39	-12.4812	1326	LUMO+58	-5.8507
1229	HOMO-38	-12.47	1327	LUMO+59	-5.8386
1230	HOMO-37	-12.4564	1328	LUMO+60	-5.8151
1231	HOMO-36	-12.4525	1329	LUMO+61	-5.8014
1232	HOMO-35	-12.4255	1330	LUMO+62	-5.7899
1233	HOMO-34	-12.3894	1331	LUMO+63	-5.7831
1234	HOMO-33	-12.2498	1332	LUMO+64	-5.7672
1235	HOMO-32	-12.2498	1333	LUMO+65	-5.7504
1236	HOMO-31	-12.2476	1334	LUMO+66	-5.7325
1237	HOMO-30	-12.2268	1335	LUMO+67	-5.7271
1238	HOMO-29	-12.1697	1336	LUMO+68	-5.7084
1239	HOMO-28	12.1387	1337	LUMO+69	-5.7043
1240	HOMO-27	-12.1077	1338	LUMO+70	-5.6972
1241	HOMO-26	-12.0589	1339	LUMO+71	-5.6913
1242	HOMO-25	-12.0427	1340	LUMO+72	-5.6864
1243	HOMO-24	-12.0169	1341	LUMO+73	-5.6378
1244	HOMO-23	-12.0152	1342	LUMO+74	-5.62
1245	HOMO-22	-11.9806	1343	LUMO+75	-5.5905
1246	HOMO-21	-11.9562	1344	LUMO+76	-5.5765
1247	HOMO-20	-11.9469	1345	LUMO+77	-5.5632
1248	HOMO-19	-11.9376			
1249	HOMO-18	-11.9194			
1250	HOMO-17	-11.8041			
1251	HOMO-16	-11.7616			
1252	HOMO-15	-11.733			
1253	HOMO-14	-11.6996			
1254	HOMO-13	-11.655			
1255	HOMO-12	-11.6452			
1256	HOMO-11	-11.6025			
1257	HOMO-10	-11.5921			
1258	HOMO-9	-11.5478			
1259	HOMO-8	-11.4859			
1260	HOMO-7	-11.4787			
1261	HOMO-6	-11.4693			
1262	HOMO-5	-11.4562			
1263	HOMO-4	-11.4117			
1264	HOMO-3	-11.1377			
1265	HOMO-2	-10.9694			
1266	HOMO-1	-10.9602			
1267	HOMO	-10.902			

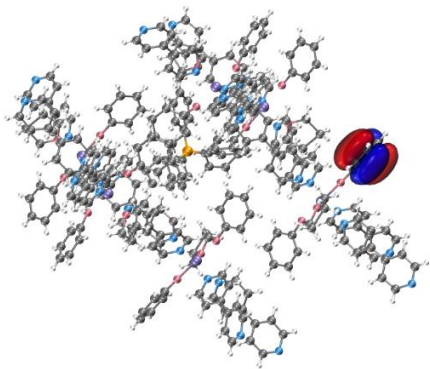
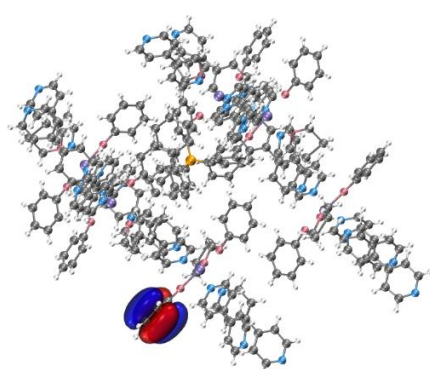
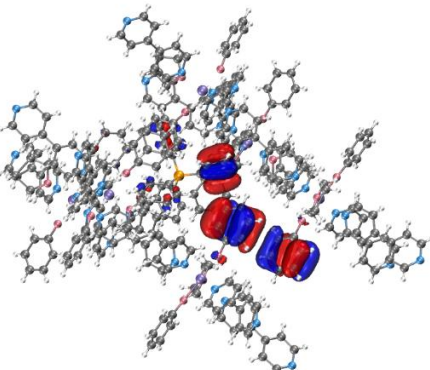
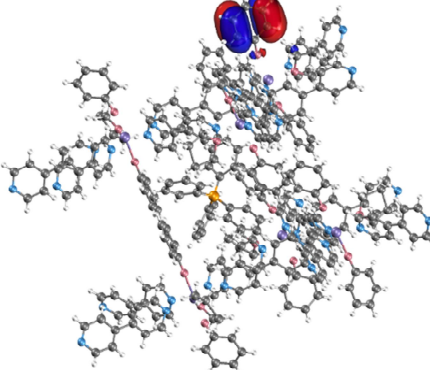
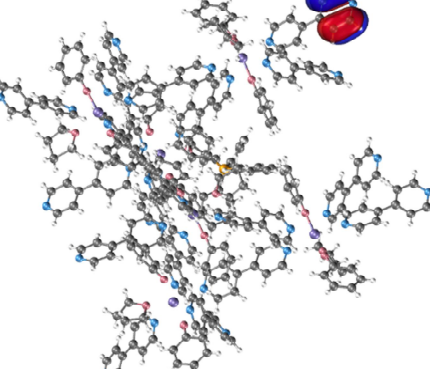
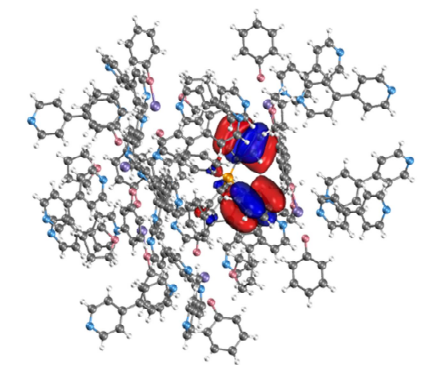
Table S13. Partial HOMOs and LUMOs orbital plots with **1A**. The densities for HOMOs are predominantly localized on BPh₄⁻ and OPh⁻, while the densities for LUMOs are mainly distributed over BPy.

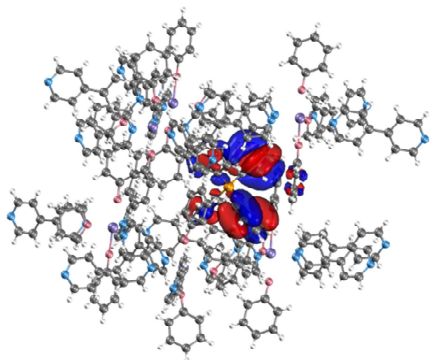
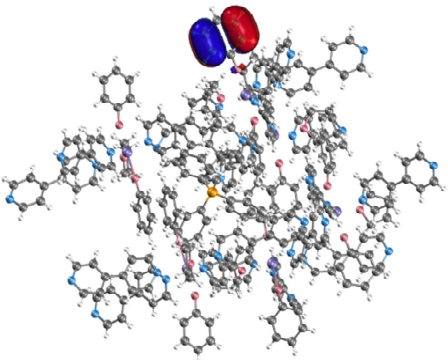
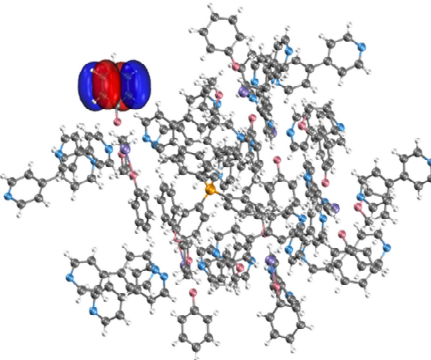
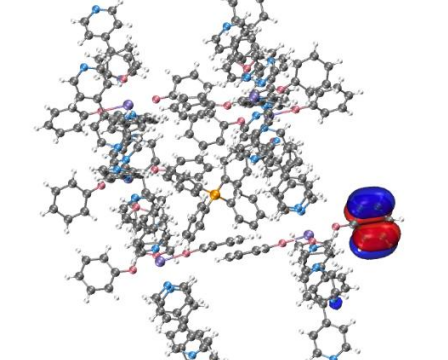
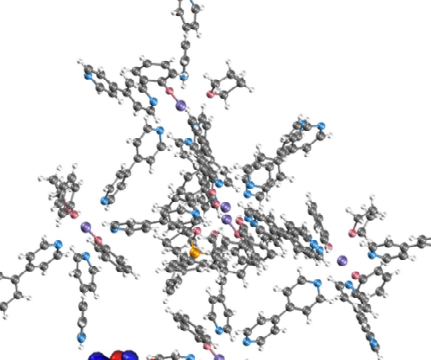
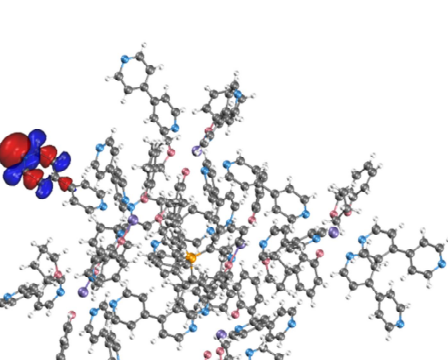
	
1170 (HOMO-97) -13.8327 eV	1187 (HOMO-80) -13.3612 eV
	
1190 (HOMO-77) -13.3133 eV	1192 (HOMO-75) -13.2745 eV
	

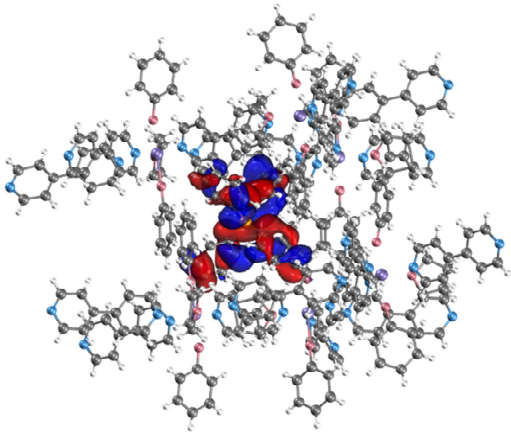
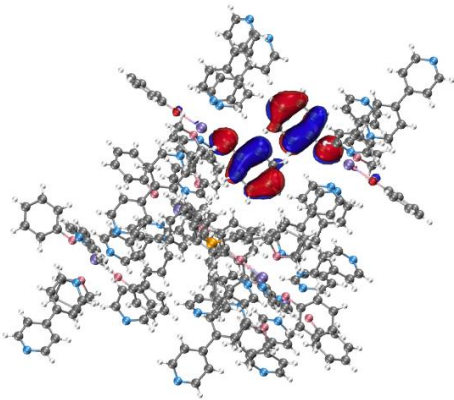
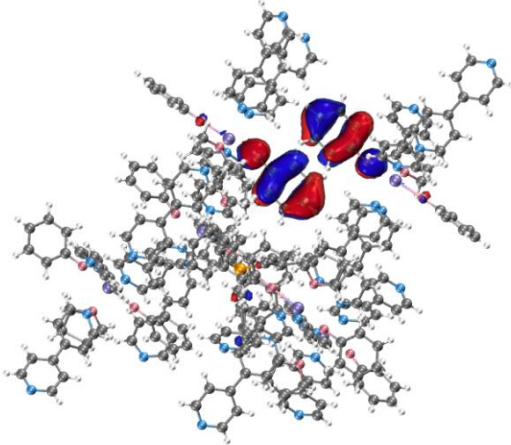
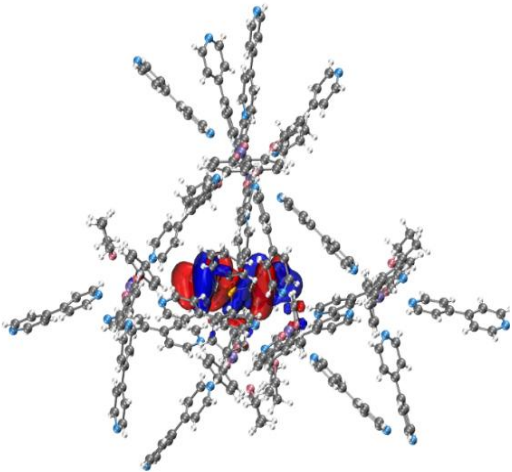
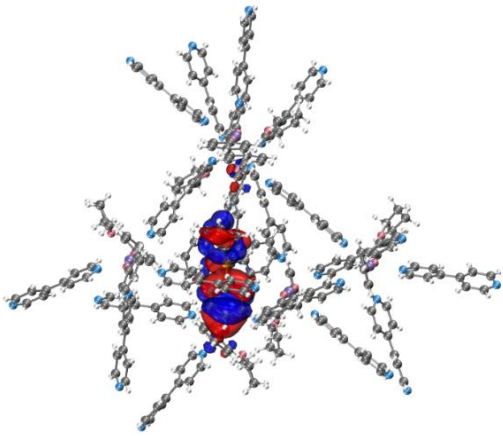
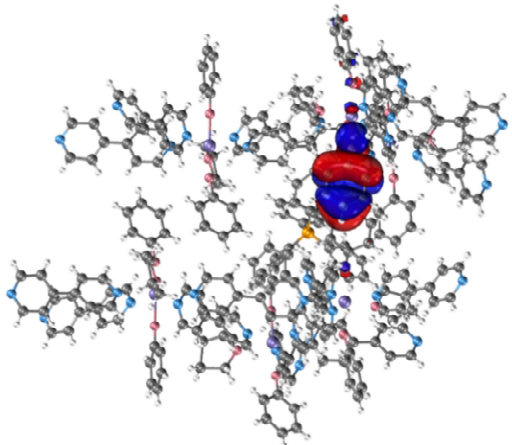
1197 (HOMO-70) -13.1848 eV	1198 (HOMO-69) -13.1663 eV
	
1200 (HOMO-67) -13.1183 eV	1202 (HOMO-65) -13.0532 eV
	
1206 (HOMO-61) -12.951 eV	1212 (HOMO-55) -12.8245 eV
	
1215 (HOMO-52) -12.765 eV	1216 (HOMO-51) -12.7605 eV

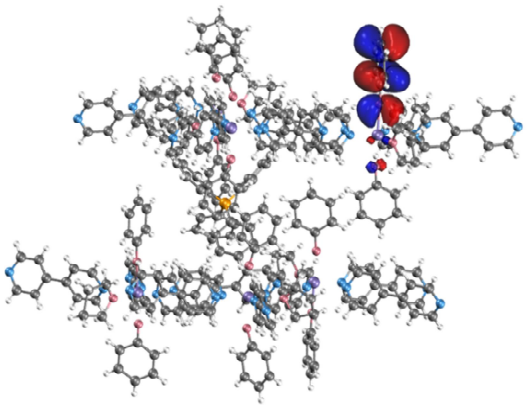
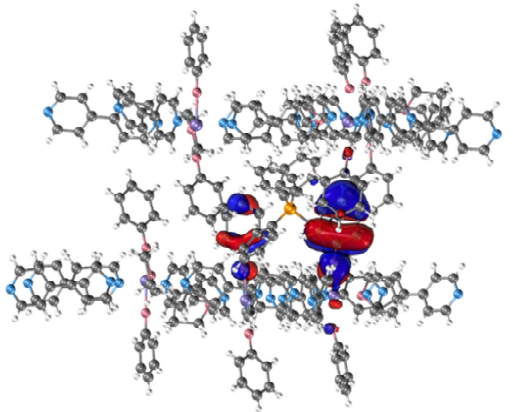
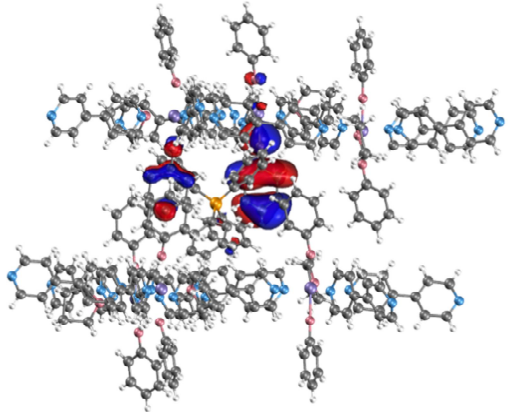
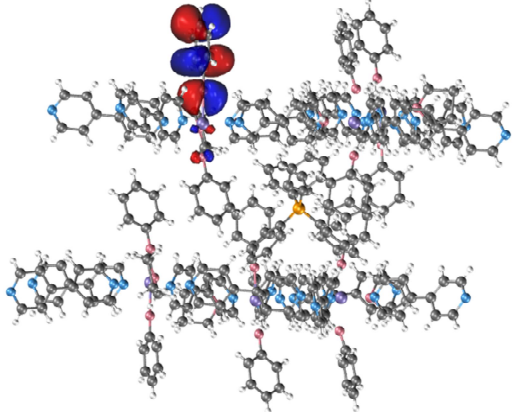
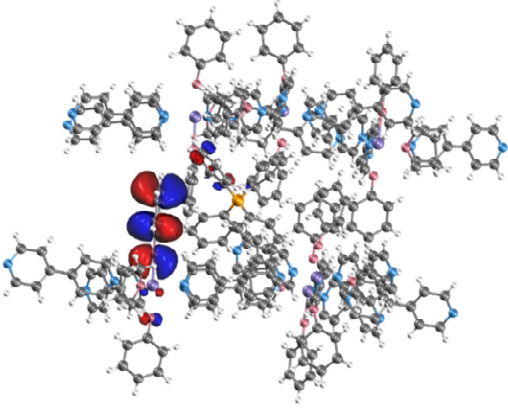
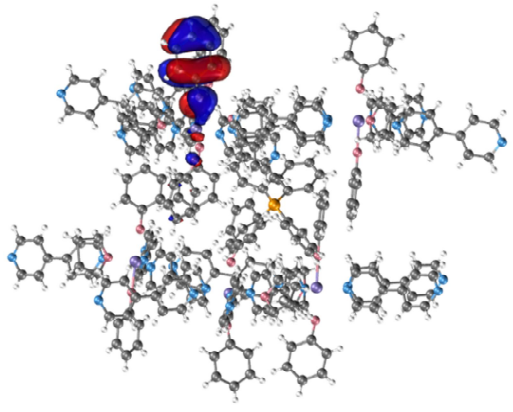
	
1218 (HOMO-49) -12.7008 eV	1219 (HOMO-48) -12.6962 eV
	
1221 (HOMO-46) -12.6526 eV	1222 (HOMO-45) -12.6451 eV
	
1223 (HOMO-44) -12.6181 eV	1224 (HOMO-43) -12.614 eV

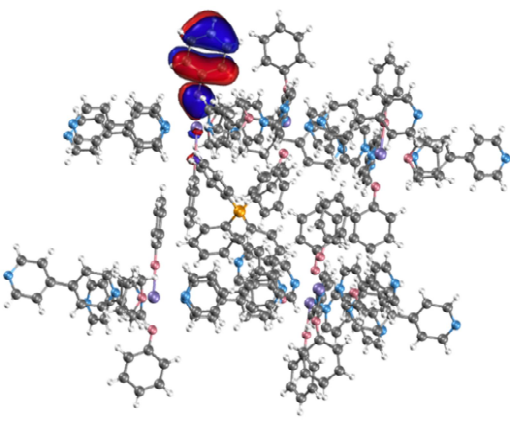
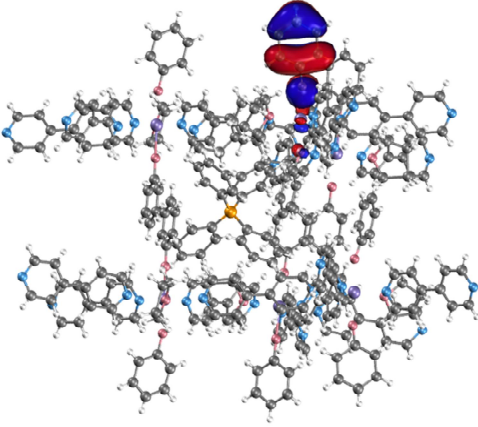
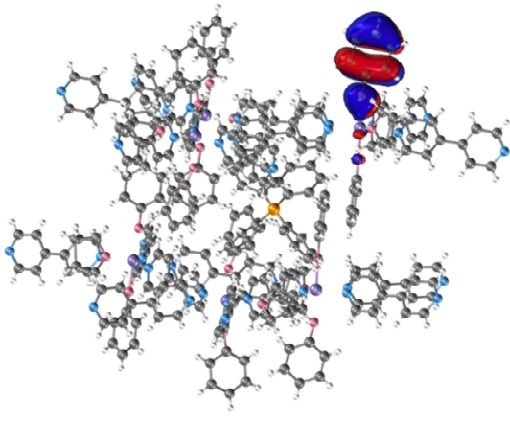
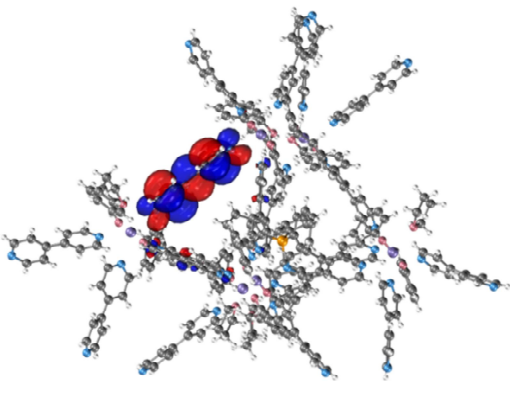
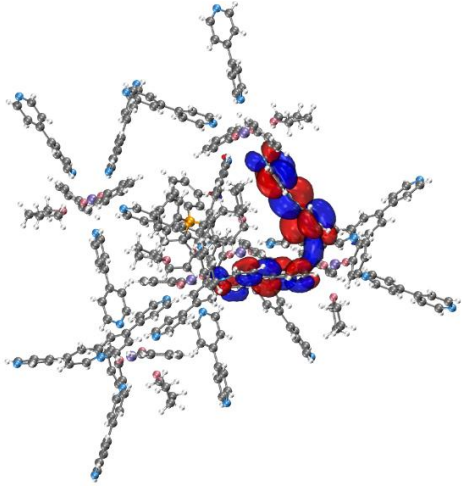
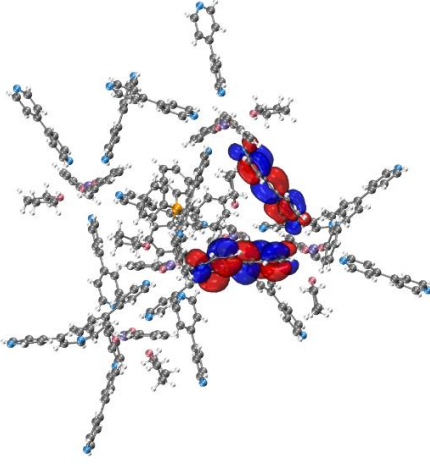
	
1225 (HOMO-42) -12.6012 eV	1226 (HOMO-41) -12.562 eV
	
1227 (HOMO-40) -12.516 eV	1228 (HOMO-39) -12.4812 eV
	
1229 (HOMO-38) -12.47 eV	1230 (HOMO-37) -12.4564 eV

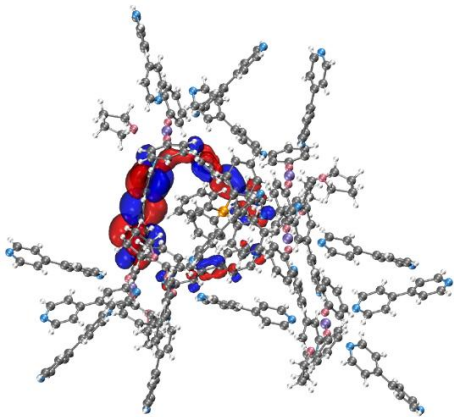
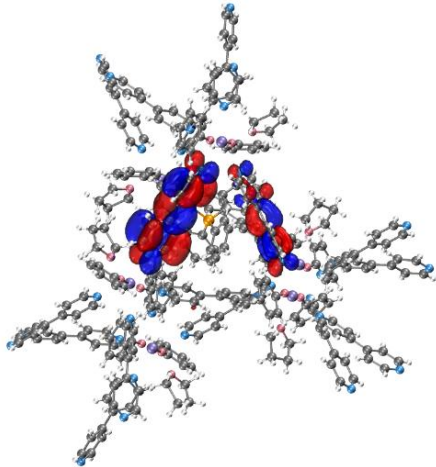
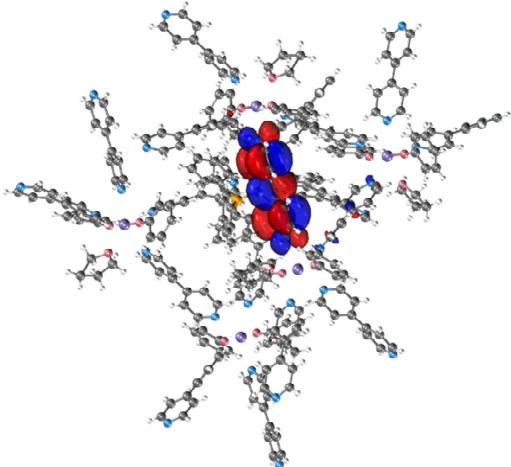
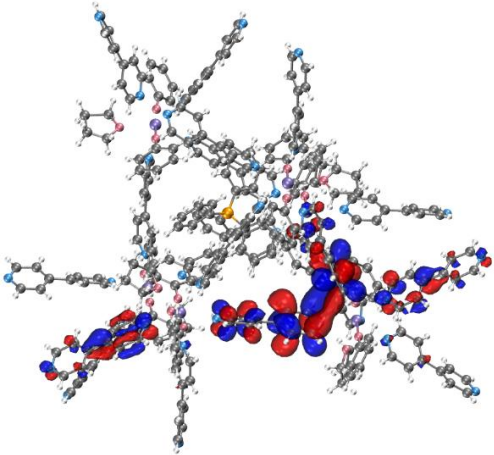
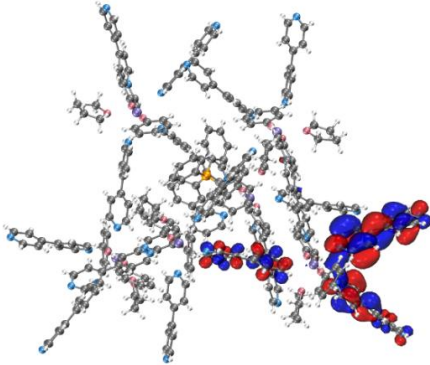
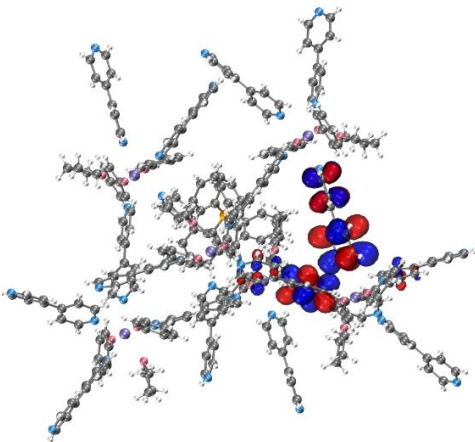
	
1231 (HOMO-36) -12.4525 eV	1232 (HOMO-35) -12.4255 eV
	
1233 (HOMO-34) -12.3894 eV	1234 (HOMO-33) -12.2498 eV
	
1235 (HOMO-32) -12.2498 eV	1236 (HOMO-31) -12.2476 eV

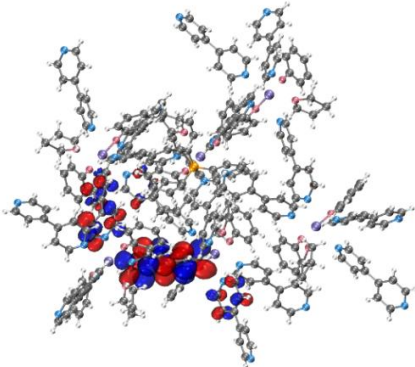
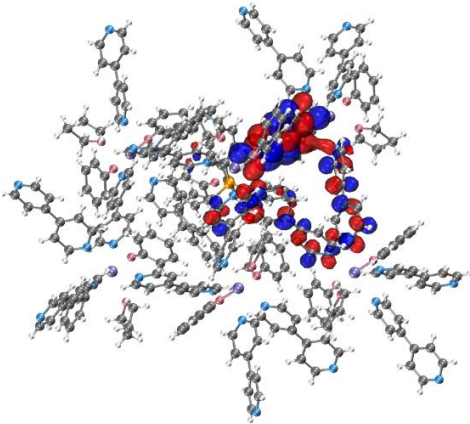
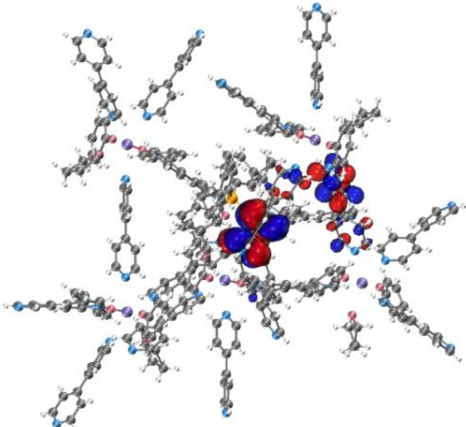
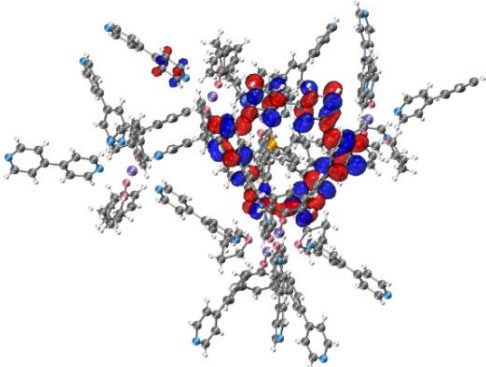
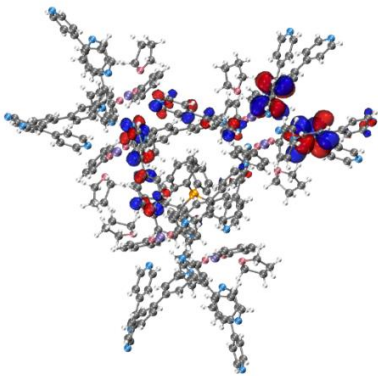
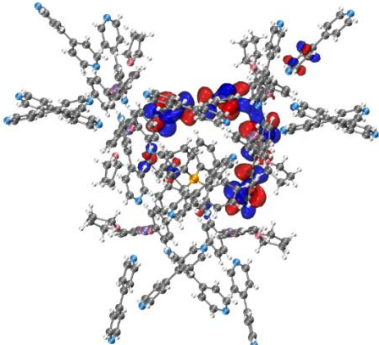
	
1239 (HOMO-28) 12.1387 eV	1241 (HOMO-26) -12.0589 eV
	
1242 (HOMO-25) -12.0427 eV	1246 (HOMO-21) -11.9562 eV
	
1247 (HOMO-20) -11.9469 eV	1248 (HOMO-19) -11.9376 eV

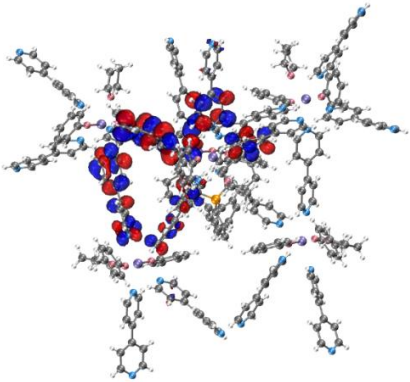
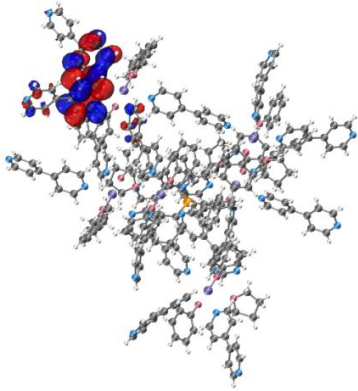
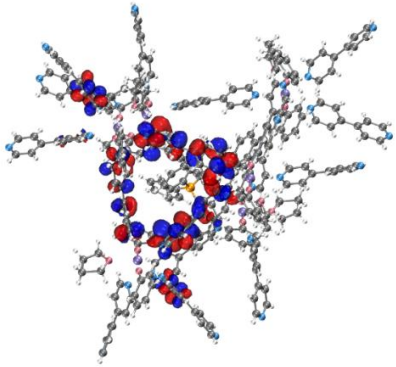
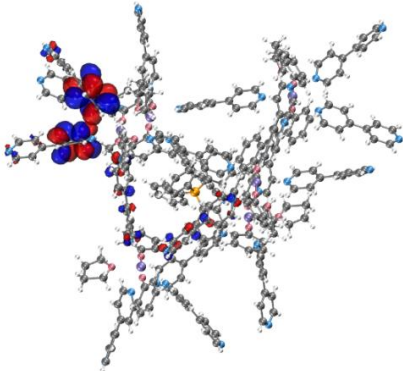
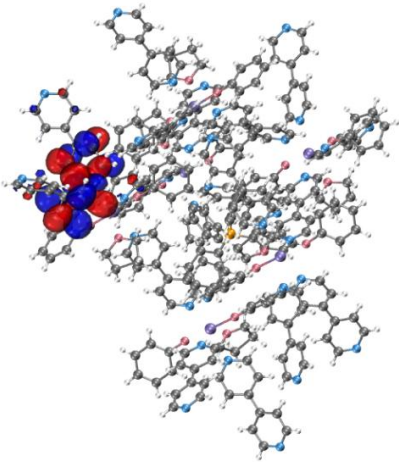
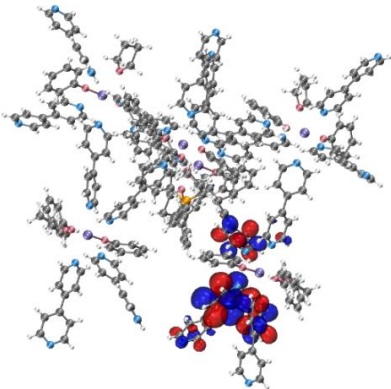
	
1249 (HOMO-18) -11.9194 eV	1252 (HOMO-15) -11.733 eV
	
1253 (HOMO-14) -11.6996 eV	1255 (HOMO-12) -11.6452 eV
	
1256 (HOMO-11) -11.6025 eV	1258 (HOMO-9) -11.5478 eV

	
1259 (HOMO-8) -11.4859 eV	1260 (HOMO-7) -11.4787 eV
	
1261 (HOMO-6) -11.4693 eV	1262 (HOMO-5) -11.4562 eV
	
1263 (HOMO-4) -11.4117 eV	1264 (HOMO-3) -11.1377 eV

	
1265 (HOMO-2) -10.9694 eV	1266 (HOMO-1) -10.9602 eV
	
1267 (HOMO) -10.902 eV	1268 (LUMO) -8.8948 eV
	
1269 (LUMO+1) -8.846 eV	1270 (LUMO+2) -8.6101 eV

	
1271 (LUMO+3) -8.5488 eV	1272 (LUMO+4) -8.3447 eV
	
1273 (LUMO+5) -8.2744 eV	1277 (LUMO+9) -7.8196 eV
	
1278 (LUMO+10) -7.7587 eV	1291 (LUMO+23) -7.281 eV

	
1292 (LUMO+24) -7.2742 eV	1293 (LUMO+25) -7.2075 eV
	
1294 (LUMO+26) -7.207 eV	1296 (LUMO+28) -7.0746 eV
	
1300 (LUMO+32) -6.9461 eV	1301 (LUMO+33) -6.9263 eV

	
1303 (LUMO+35) -6.8911 eV	1304 (LUMO+36) -6.8632 eV
	
1305 (LUMO+37) -6.8426 eV	1306 (LUMO+38) -6.7942 eV
	
1309 (LUMO+41) -6.6996 eV	1310 (LUMO+42) -6.6534 eV

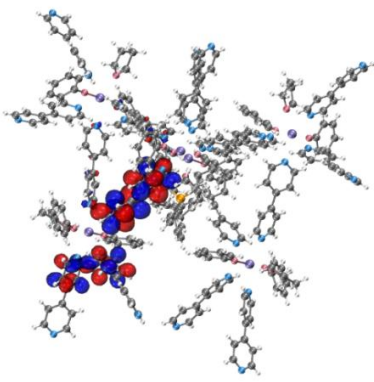
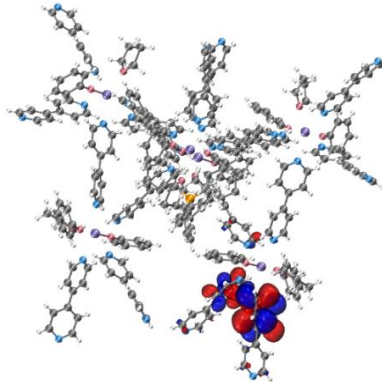
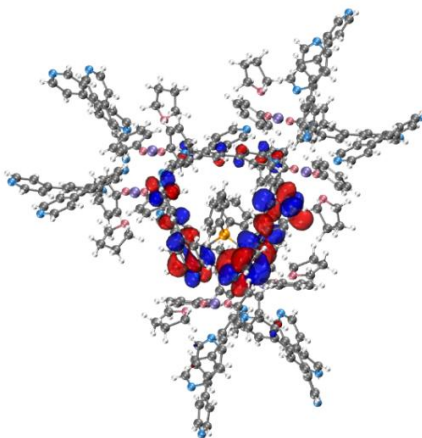
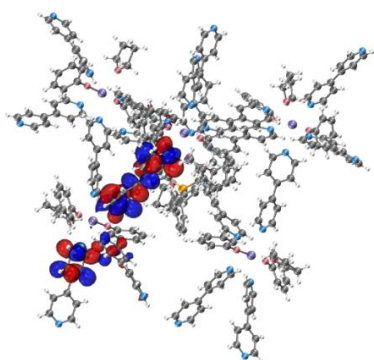
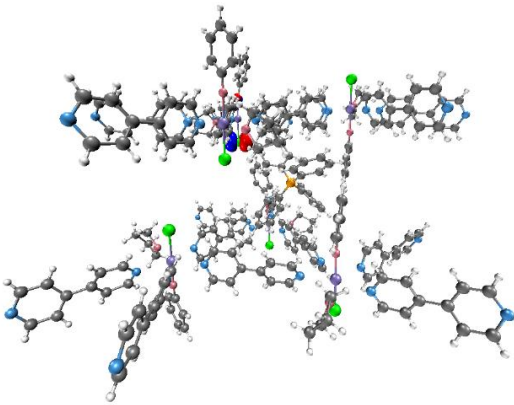
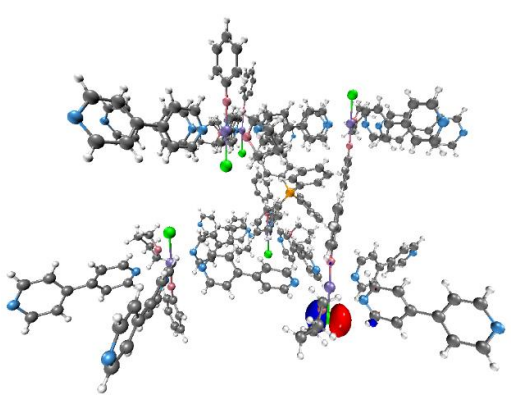
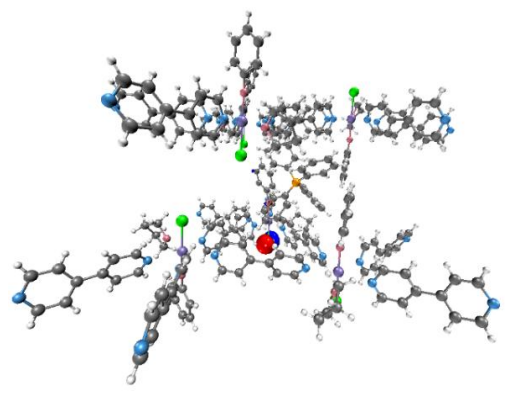
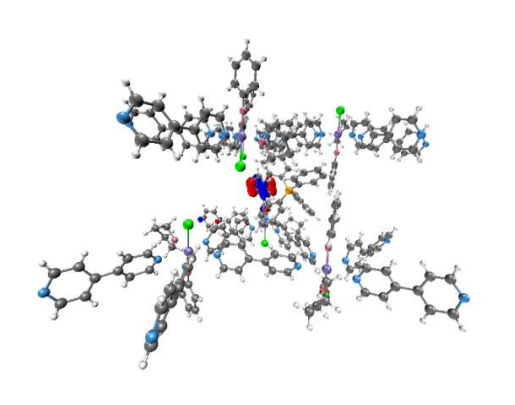
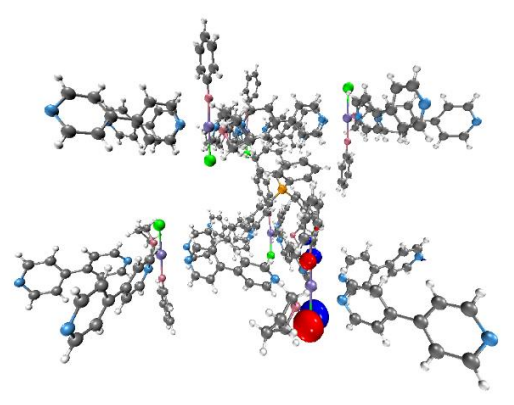
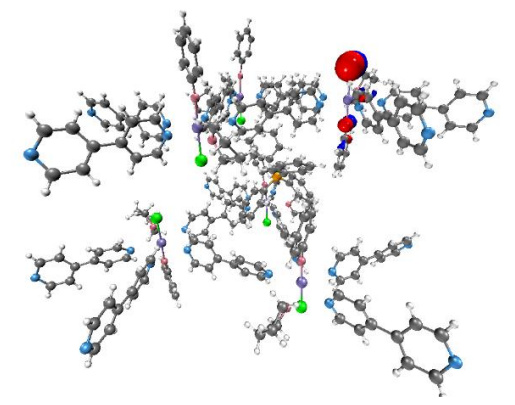
	
1311 (LUMO+43) -6.6349 eV	1312 (LUMO+44) -6.5735 eV
	
1313 (LUMO+45) -6.5567 eV	1314 (LUMO+46) -6.3404 eV

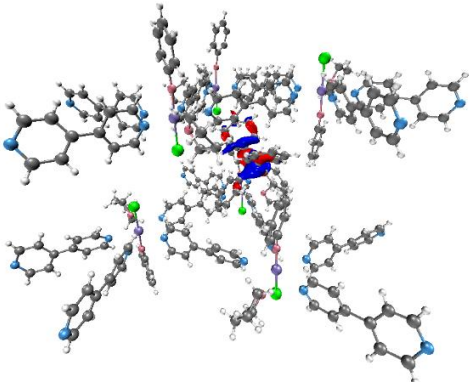
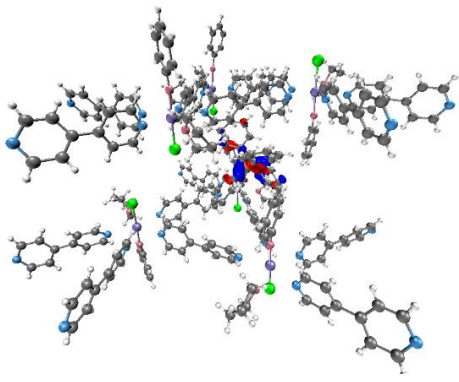
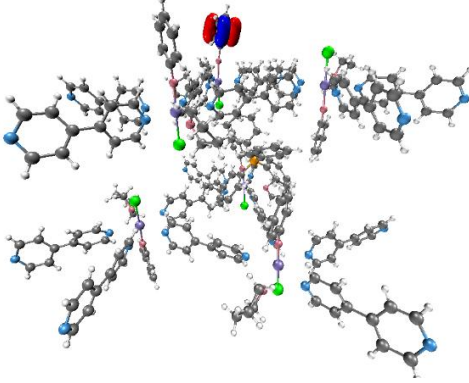
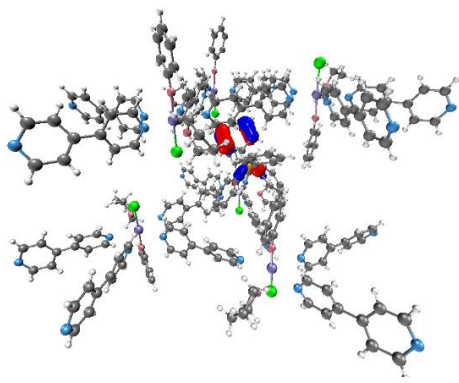
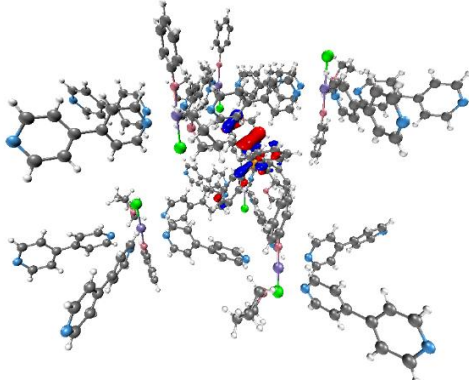
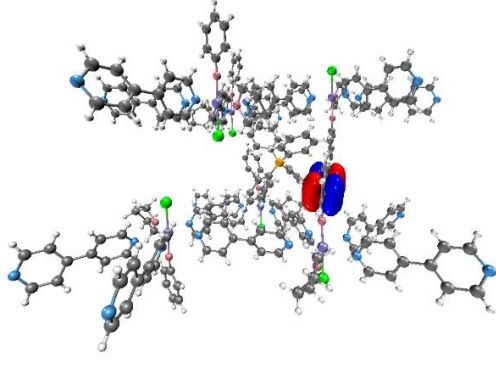
Table S14. Frontier molecular orbitals and the energy level for **1B**.

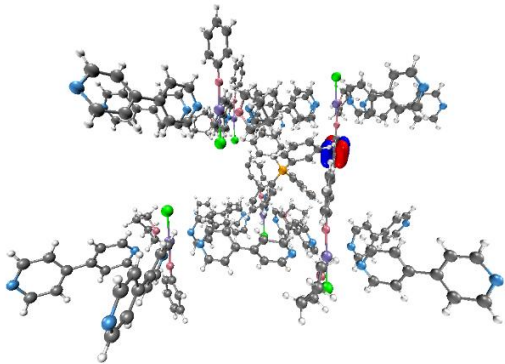
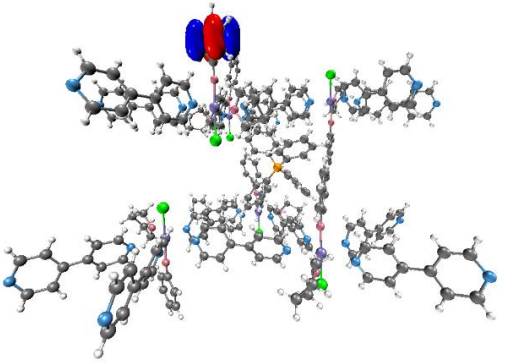
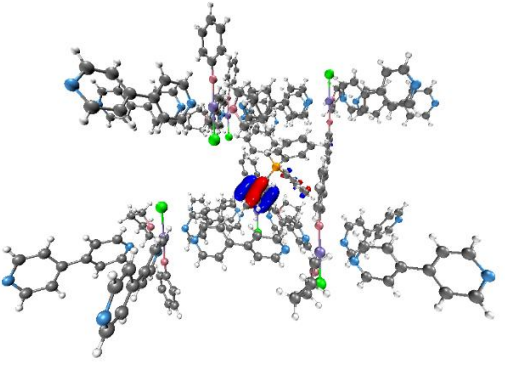
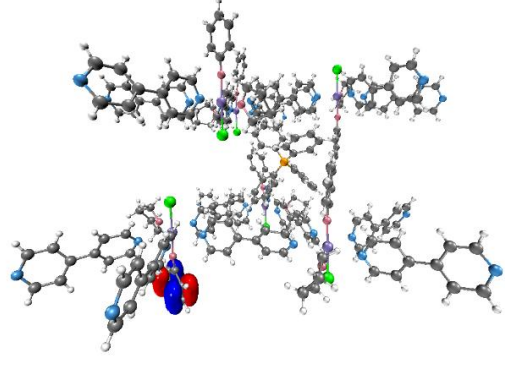
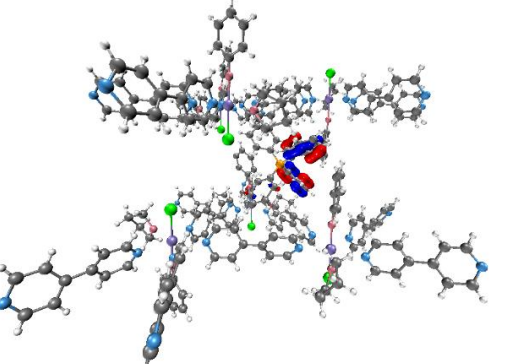
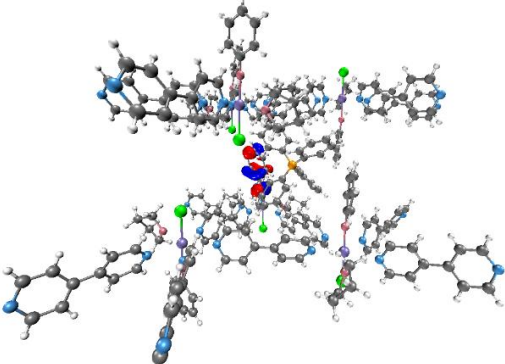
Orbital Number	Orbital Name	energy (eV)	Orbital Number	Orbital Name	energy (eV)
1076	HOMO-93	-13.9728	1170	LUMO	-9.5532
1077	HOMO-92	-13.9702	1171	LUMO+1	-9.2330
1078	HOMO-91	-13.9695	1172	LUMO+2	-9.1373
1079	HOMO-90	-13.9203	1173	LUMO+3	-9.1236
1080	HOMO-89	-13.9182	1174	LUMO+4	-8.9629
1081	HOMO-88	-13.9090	1175	LUMO+5	-8.9382
1082	HOMO-87	-13.9027	1176	LUMO+6	-8.6703
1083	HOMO-86	-13.8737	1177	LUMO+7	-8.4308
1084	HOMO-85	-13.8318	1178	LUMO+8	-8.3587
1085	HOMO-84	-13.8265	1179	LUMO+9	-8.1656
1086	HOMO-83	-13.8235	1180	LUMO+10	-7.9530
1087	HOMO-82	-13.8034	1181	LUMO+11	-7.9273
1088	HOMO-81	-13.7780	1182	LUMO+12	-7.8531
1089	HOMO-80	-13.7663	1183	LUMO+13	-7.8479
1090	HOMO-79	-13.7524	1184	LUMO+14	-7.8116
1091	HOMO-78	-13.7290	1185	LUMO+15	-7.8074
1092	HOMO-77	-13.6990	1186	LUMO+16	-7.7526
1093	HOMO-76	-13.6689	1187	LUMO+17	-7.7477
1094	HOMO-75	-13.6637	1188	LUMO+18	-7.6755
1095	HOMO-74	-13.6592	1189	LUMO+19	-7.6597
1096	HOMO-73	-13.6472	1190	LUMO+20	-7.6582
1097	HOMO-72	-13.6374	1191	LUMO+21	-7.6004
1098	HOMO-71	-13.6321	1192	LUMO+22	-7.5424
1099	HOMO-70	-13.5991	1193	LUMO+23	-7.5030
1100	HOMO-69	-13.5951	1194	LUMO+24	-7.4626
1101	HOMO-68	-13.5785	1195	LUMO+25	-7.4179
1102	HOMO-67	-13.5510	1196	LUMO+26	-7.3780
1103	HOMO-66	-13.5258	1197	LUMO+27	-7.3728
1104	HOMO-65	-13.5219	1198	LUMO+28	-7.3676
1105	HOMO-64	-13.5022	1199	LUMO+29	-7.3535
1106	HOMO-63	-13.4576	1200	LUMO+30	-7.2806
1107	HOMO-62	-13.4535	1201	LUMO+31	-7.2441
1108	HOMO-61	-13.4212	1202	LUMO+32	-7.2122
1109	HOMO-60	-13.3870	1203	LUMO+33	-7.1981
1110	HOMO-59	-13.3839	1204	LUMO+34	-7.1577
1111	HOMO-58	-13.3600	1205	LUMO+35	-7.1311
1112	HOMO-57	-13.3487	1206	LUMO+36	-7.1155
1113	HOMO-56	-13.3216	1207	LUMO+37	-7.0064
1114	HOMO-55	-13.2778	1208	LUMO+38	-6.9968
1115	HOMO-54	-13.2764	1209	LUMO+39	-6.9732
1116	HOMO-53	-13.2007	1210	LUMO+40	-6.9097
1117	HOMO-52	-13.1734	1211	LUMO+41	-6.8438
1118	HOMO-51	-13.1551	1212	LUMO+42	-6.8325
1119	HOMO-50	-13.1525	1213	LUMO+43	-6.8024
1120	HOMO-49	-13.1312	1214	LUMO+44	-6.7914
1121	HOMO-48	-13.0970	1215	LUMO+45	-6.7570
1122	HOMO-47	-13.0903	1216	LUMO+46	-6.7200
1123	HOMO-46	-13.0888	1217	LUMO+47	-6.5261

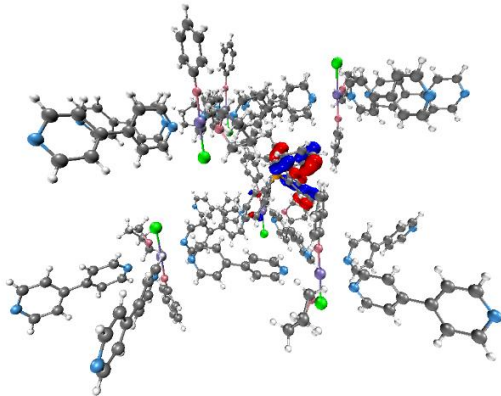
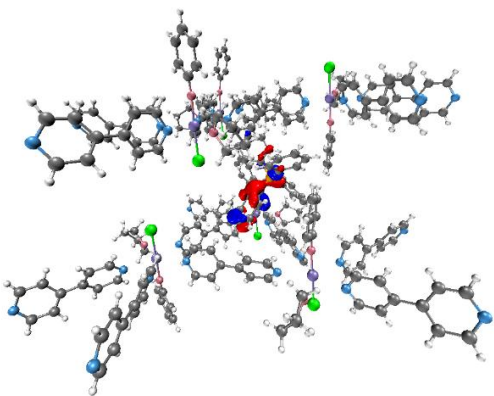
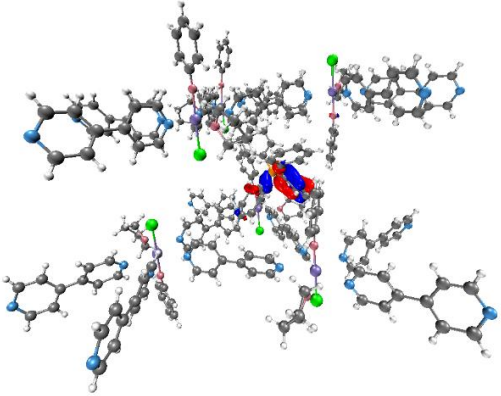
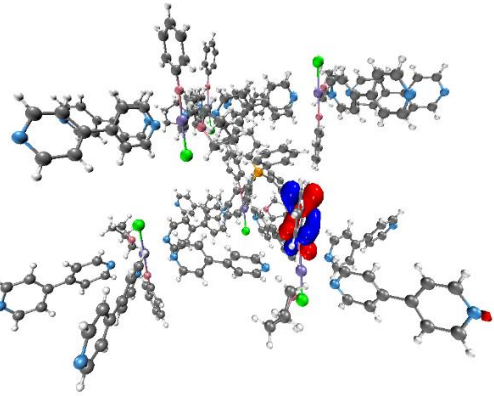
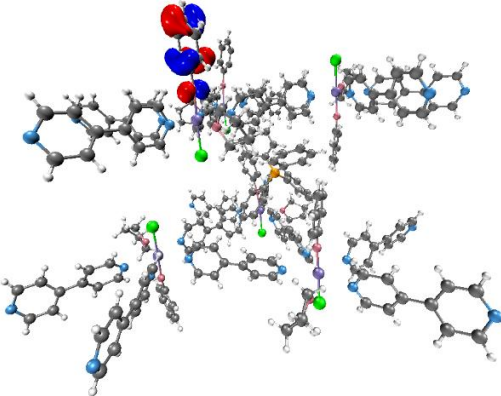
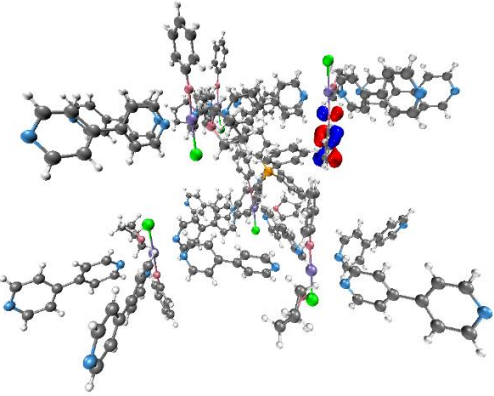
1124	HOMO-45	-13.0555	1218	LUMO+48	-6.5052
1125	HOMO-44	-13.0029	1219	LUMO+49	-6.4583
1126	HOMO-43	-12.9932	1220	LUMO+50	-6.4481
1127	HOMO-42	-12.9697	1221	LUMO+51	-6.4377
1128	HOMO-41	-12.9558	1222	LUMO+52	-6.4375
1129	HOMO-40	-12.9168	1223	LUMO+53	-6.2320
1130	HOMO-39	-12.9059	1224	LUMO+54	-6.2018
1131	HOMO-38	-12.8980	1225	LUMO+55	-6.1760
1132	HOMO-37	-12.8694	1226	LUMO+56	-6.1643
1133	HOMO-36	-12.8643	1227	LUMO+57	-6.1362
1134	HOMO-35	-12.8630	1228	LUMO+58	-6.0999
1135	HOMO-34	-12.8520	1229	LUMO+59	-6.0954
1136	HOMO-33	-12.8225	1230	LUMO+60	-6.0801
1137	HOMO-32	-12.7606	1231	LUMO+61	-6.0734
1138	HOMO-31	-12.7344	1232	LUMO+62	-6.0522
1139	HOMO-30	-12.7248	1233	LUMO+63	-5.9582
1140	HOMO-29	-12.7078	1234	LUMO+64	-5.9384
1141	HOMO-28	-12.6866	1235	LUMO+65	-5.9289
1142	HOMO-27	-12.6768	1236	LUMO+66	-5.8901
1143	HOMO-26	-12.6499	1237	LUMO+67	-5.8694
1144	HOMO-25	-12.6238	1238	LUMO+68	-5.8536
1145	HOMO-24	-12.5881	1239	LUMO+69	-5.8452
1146	HOMO-23	-12.5097	1240	LUMO+70	-5.8208
1147	HOMO-22	-12.4524	1241	LUMO+71	-5.8073
1148	HOMO-21	-12.4418	1242	LUMO+72	-5.7959
1149	HOMO-20	-12.4231	1243	LUMO+73	-5.7922
1150	HOMO-19	-12.2541	1244	LUMO+74	-5.7761
1151	HOMO-18	-12.2303	1245	LUMO+75	-5.7637
1152	HOMO-17	-12.2097	1246	LUMO+76	-5.7292
1153	HOMO-16	-12.1916	1247	LUMO+77	-5.7180
1154	HOMO-15	-12.1726	1248	LUMO+78	-5.6845
1155	HOMO-14	-12.1510	1249	LUMO+79	-5.6773
1156	HOMO-13	-12.1305	1250	LUMO+80	-5.6616
1157	HOMO-12	-12.1174	1251	LUMO+81	-5.6480
1158	HOMO-11	-11.9666	1252	LUMO+82	-5.6315
1159	HOMO-10	-11.8294	1253	LUMO+83	-5.6276
1160	HOMO-9	-11.8057	1254	LUMO+84	-5.6231
1161	HOMO-8	-11.7324	1255	LUMO+85	-5.5901
1162	HOMO-7	-11.7129	1256	LUMO+86	-5.5823
1163	HOMO-6	-11.6956	1257	LUMO+87	-5.5468
1164	HOMO-5	-11.6777	1258	LUMO+88	-5.5138
1165	HOMO-4	-11.6383	1259	LUMO+89	-5.5059
1166	HOMO-3	-11.6270			
1167	HOMO-2	-11.6153			
1168	HOMO-1	-11.5509			
1169	HOMO	-11.5468			

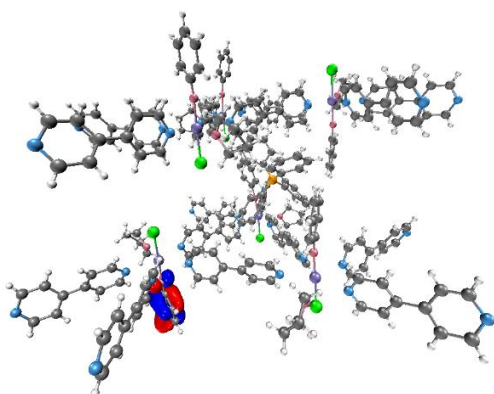
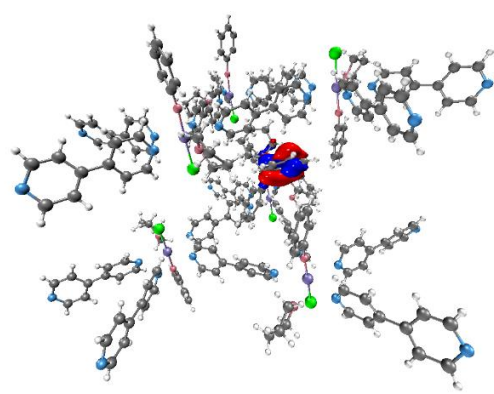
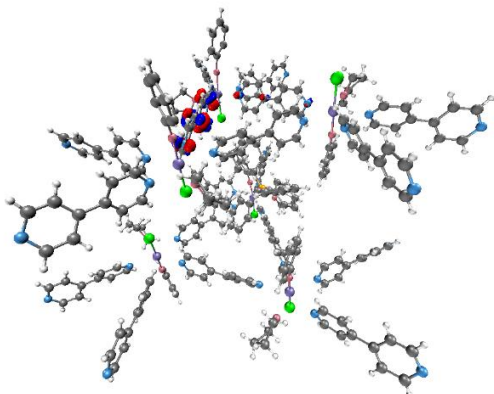
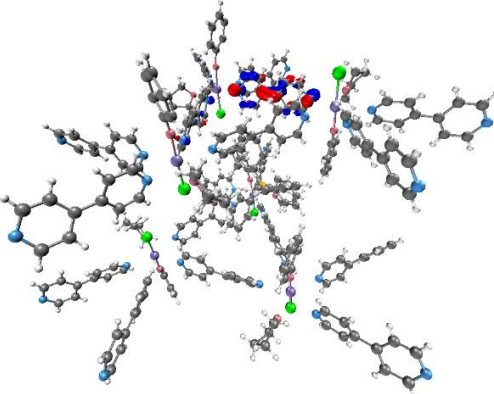
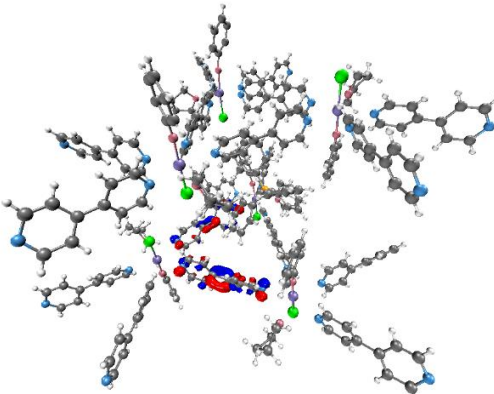
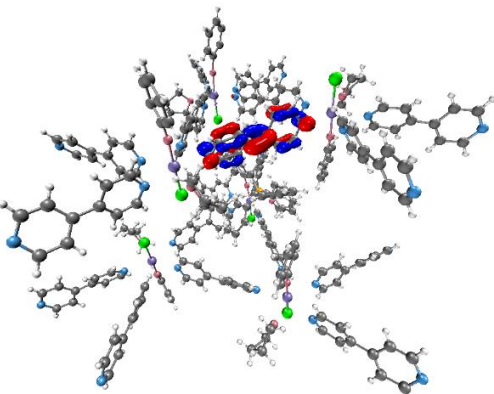
Table S15. Partial HOMOs and LUMOs orbital plots with **1B**. The densities for HOMOs are predominantly localized on BPh_4^- , OPh^- and Cl^- , while the densities for LUMOs are mainly distributed over BPy.

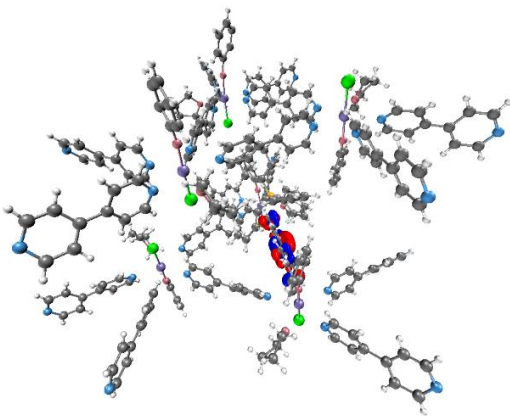
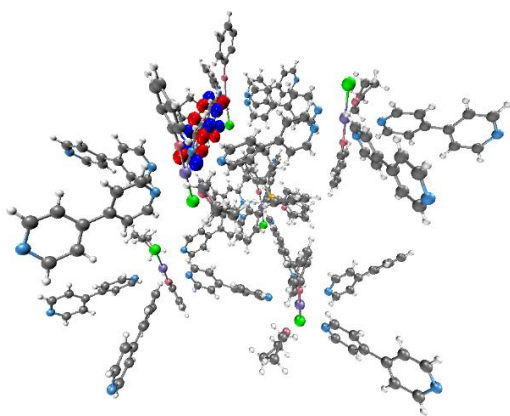
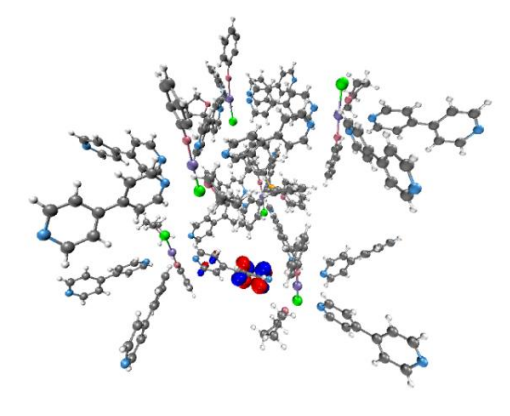
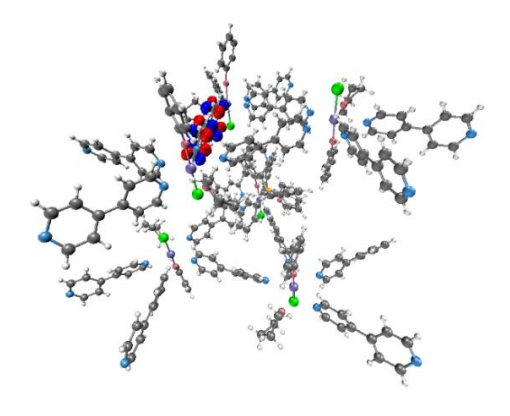
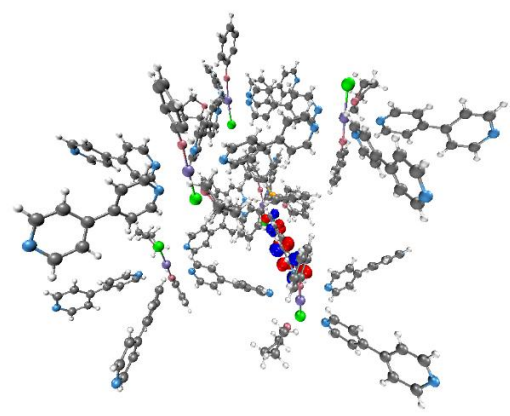
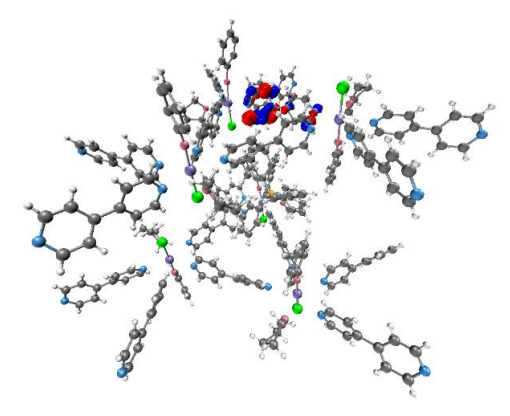
	
1090 (HOMO-79) -13.7524 eV	1104 (HOMO-65) -13.5219 eV
	
1107 (HOMO-62) -13.4535 eV	1109 (HOMO-60) -13.3870 eV
	
1110 (HOMO-59) -13.3839 eV	1112 (HOMO-57) -13.3487 eV

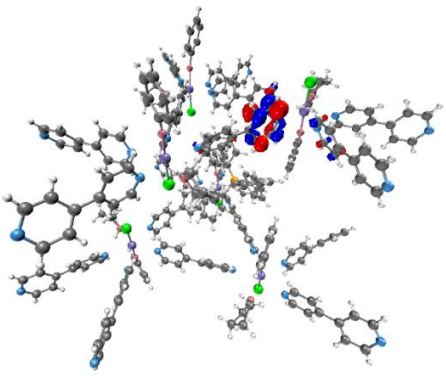
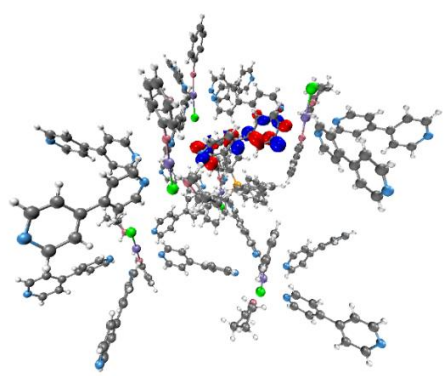
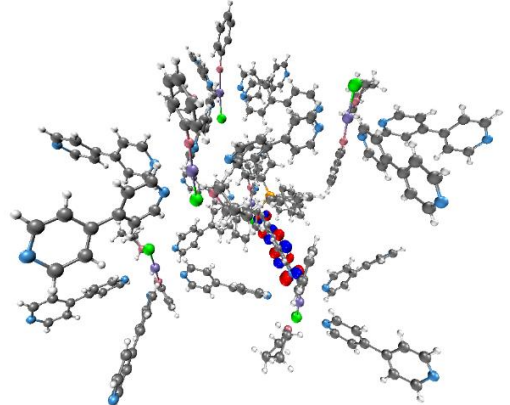
	
1116(HOMO-53) -13.2007 eV	1123 (HOMO-46) -13.0888 eV
	
1124 (HOMO-45) -13.0555 eV	1125(HOMO-44) -13.0029 eV
	
1130 (HOMO-39) -12.9059 eV	1137 (HOMO-32) -12.7606 eV

	
1140 (HOMO-29) -12.7078 eV	1141 (HOMO-28) -12.6866 eV
	
1142 (HOMO-27) -12.6768 eV	1144 (HOMO-25) -12.6238 eV
	
1145 (HOMO-24) -12.5881 eV	1146 (HOMO-23) -12.5097 eV

	
1147 (HOMO-22) -12.4524 eV	1152 (HOMO-17) -12.2097 eV
	
1159 (HOMO-10) -11.8294 eV	1161 (HOMO-8) -11.7324 eV
	
1164 (HOMO-5) -11.6777 eV	1165 (HOMO-4) -11.6383 eV

	
1167 (HOMO-2) -11.6153 eV	1169 (HOMO) -11.5468 eV
	
1172 (LUMO+2) -9.1373 eV	1174 (LUMO+4) -8.9629 eV
	
1175 (LUMO+5) -8.9382 eV	1176 (LUMO+6) -8.6703 eV

	
1177 (LUMO+7) -8.4308 eV	1178 (LUMO+8) -8.3587 eV
	
1180 (LUMO+10) -7.9530 eV	1184 (LUMO+14) -7.8116 eV
	
1189 (LUMO+19) -7.6597 eV	1206 (LUMO+36) -7.1155 eV

	
1209 (LUMO+39) -6.9732 eV	1215 (LUMO+45) -6.7570 eV
	
1216 (LUMO+46) -6.7200 eV	

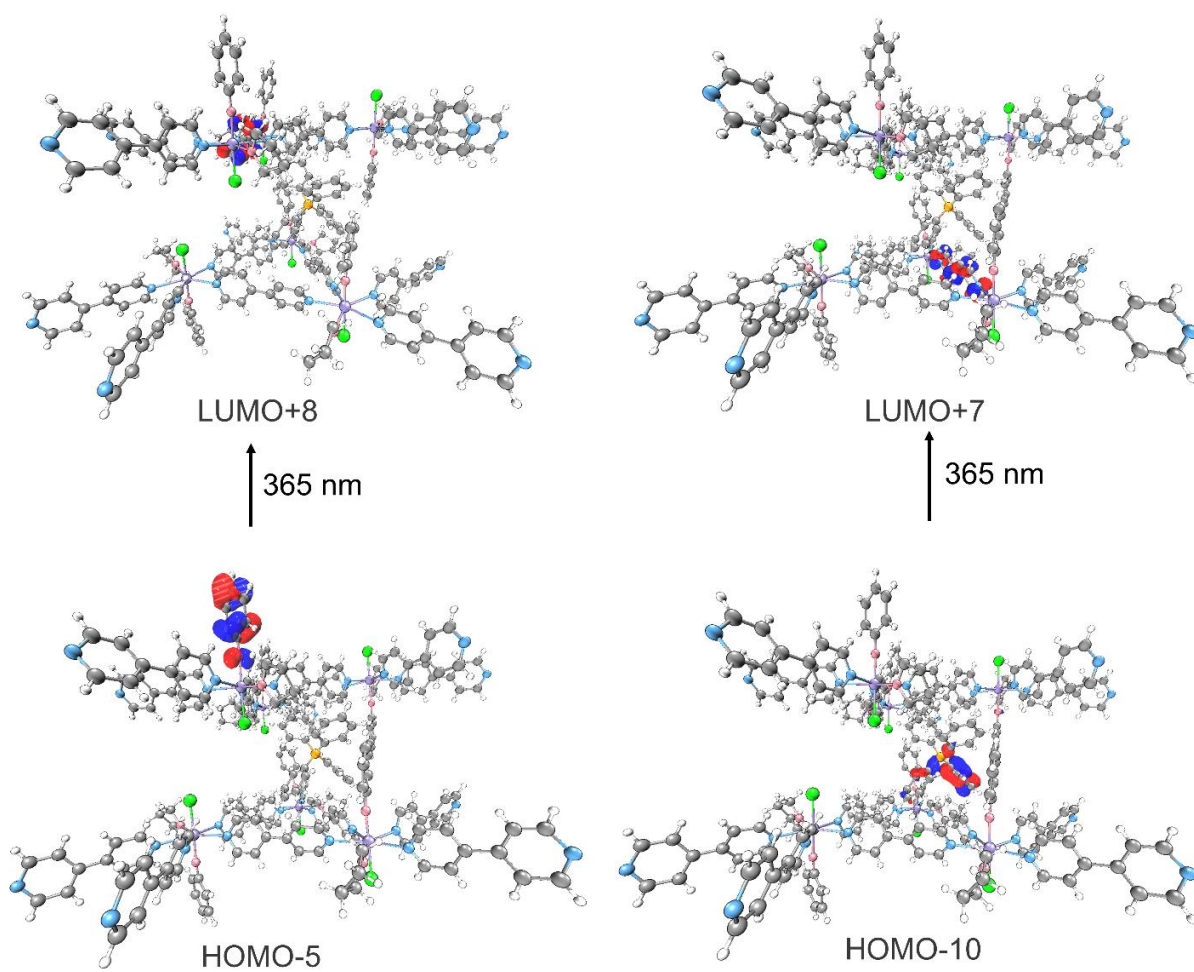


Figure S26. The partial orbitals involved in the 365 nm electronic transition within **1B**.

6. Magnetic Data

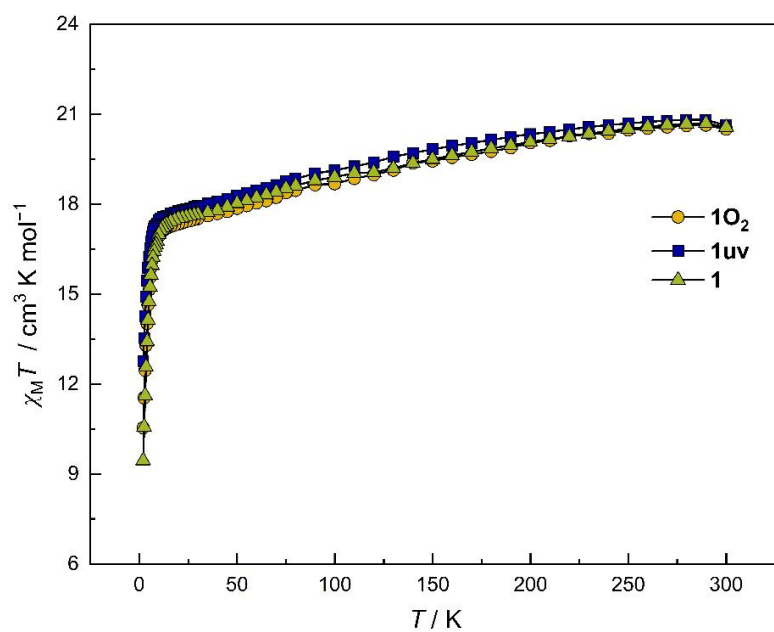


Figure S27. Temperature-dependent susceptibilities for **1**, **1uv** and **1O₂** under a dc magnetic field of 1000 Oe.

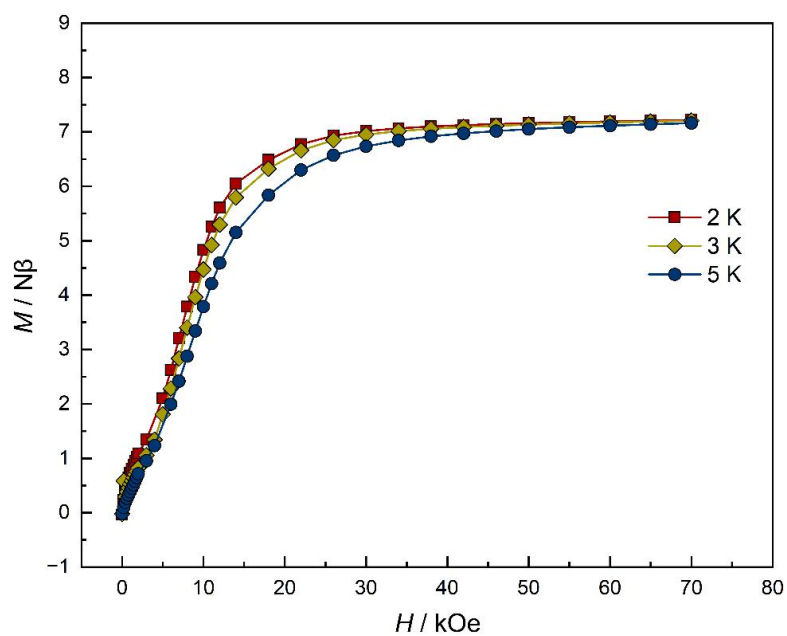


Figure S28. The plots of M vs. H at various temperatures for **1**.

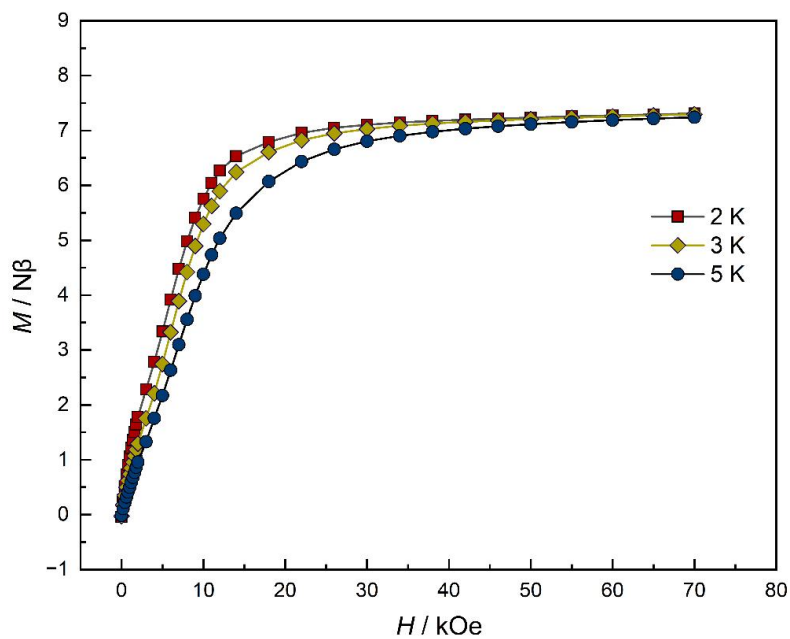


Figure S29. The plots of M vs. H at various temperatures for **1uv**.

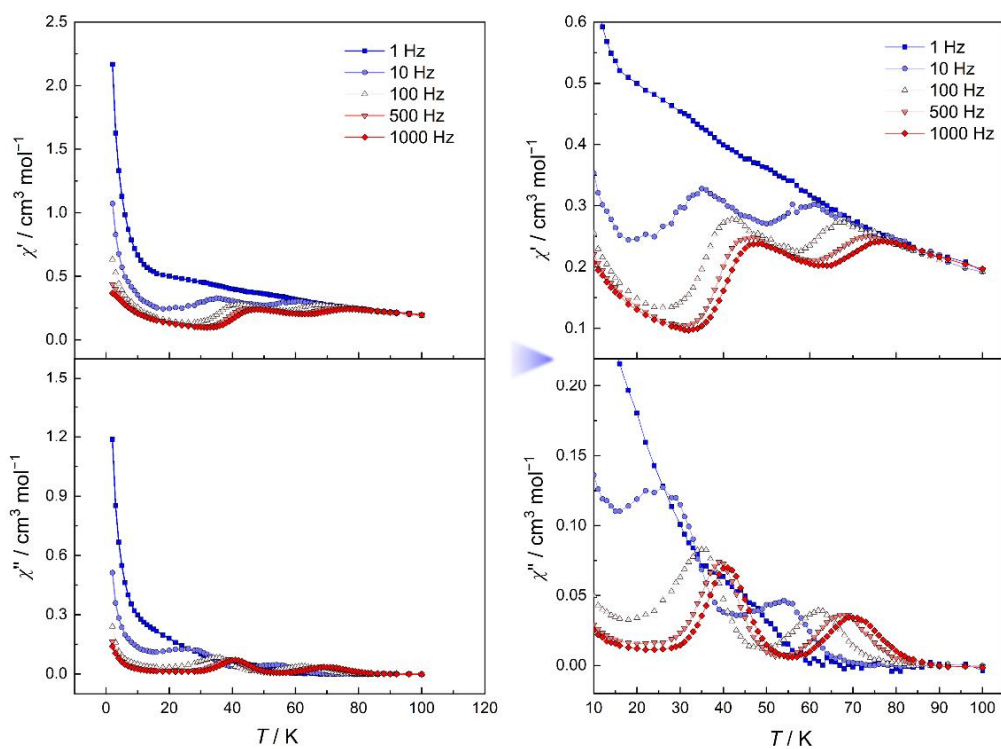


Figure S30. Temperature-dependence of the in-phase (χ') and out-of-phase (χ'') ac susceptibility for **1** under zero dc field in the full temperature range (left) and in the temperature range of 10–100 K (right).

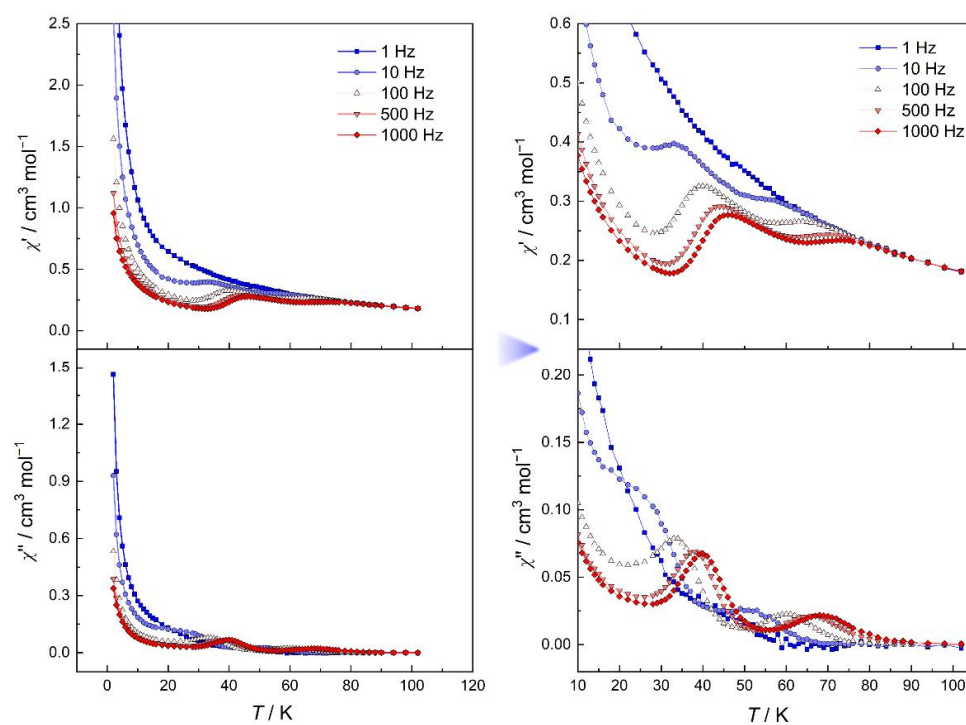


Figure S31. Temperature-dependence of the in-phase (χ') and out-of-phase (χ'') ac susceptibility for **1uv** under zero dc field in the full temperature range (left) and in the temperature range of 10–100 K (right).

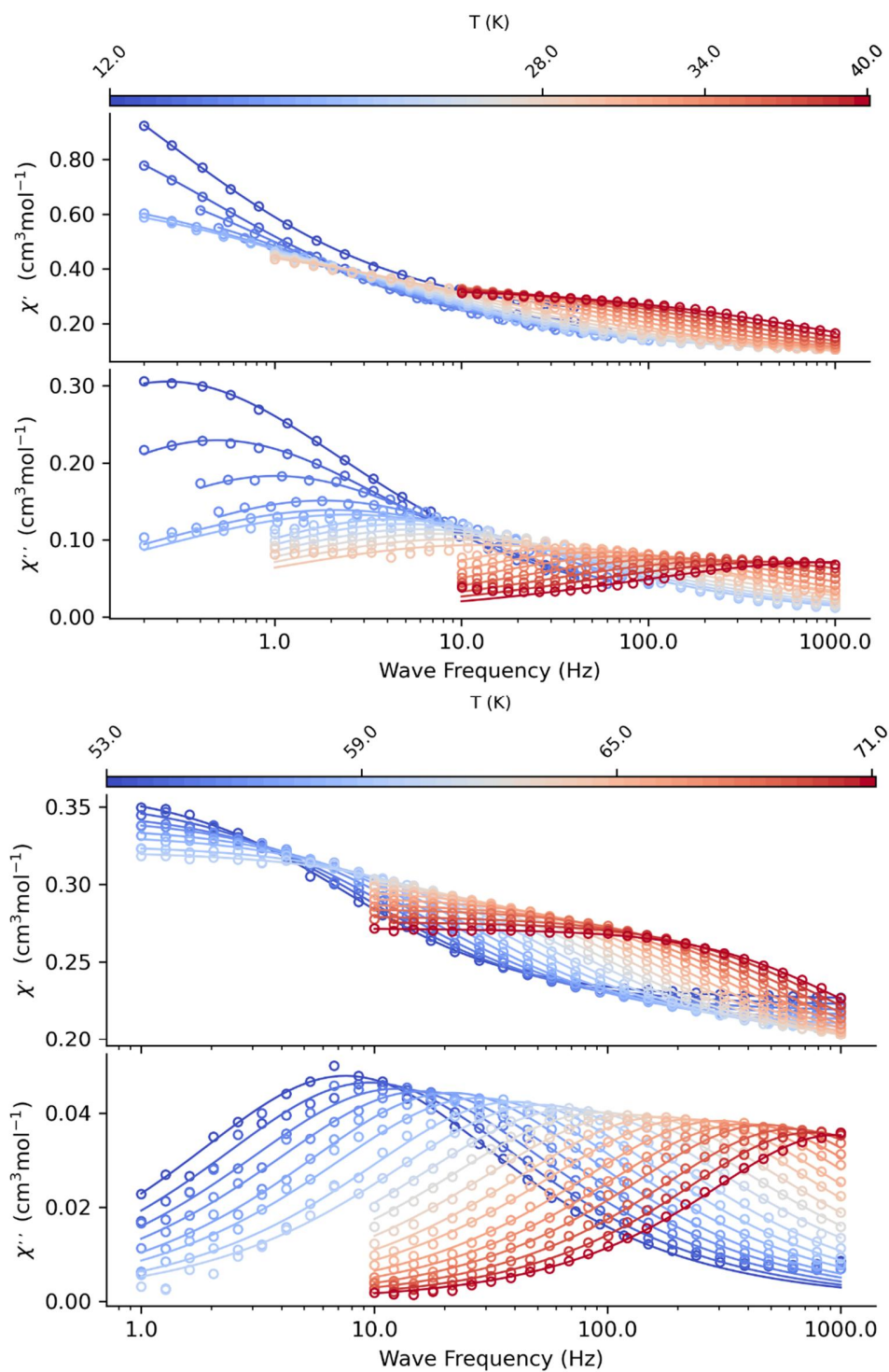


Figure S32. Frequency-dependence of the in-phase (χ') and out-of-phase (χ'') ac susceptibility for **1** from 12.0 to 40.0 K (top) and from 53.0 to 71.0 K (bottom).

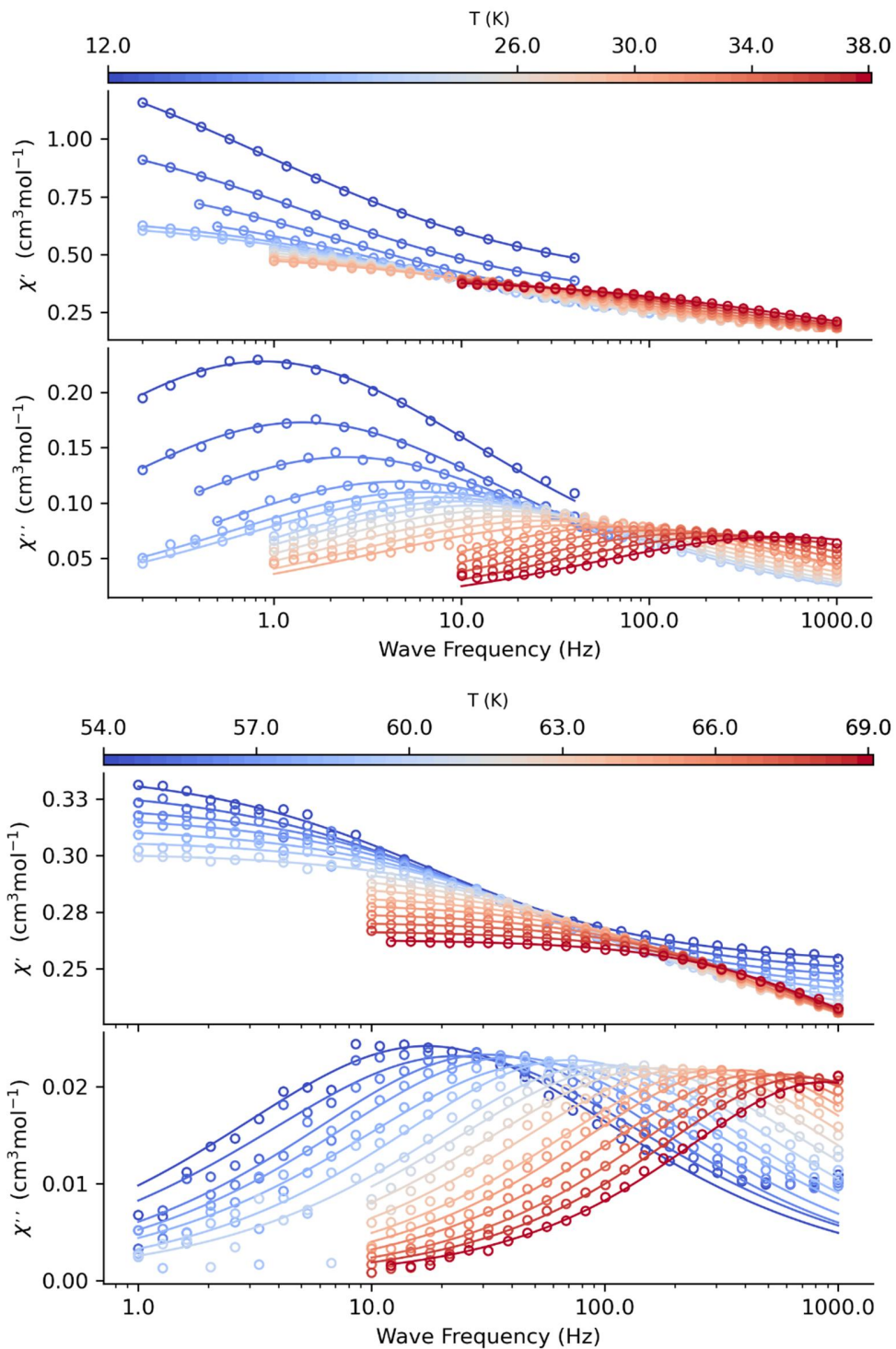


Figure S33. Frequency-dependence of the in-phase (χ') and out-of-phase (χ'') ac susceptibility for **1uv** from 12.0 to 38.0 K (top) and from 54.0 to 69.0 K (bottom).

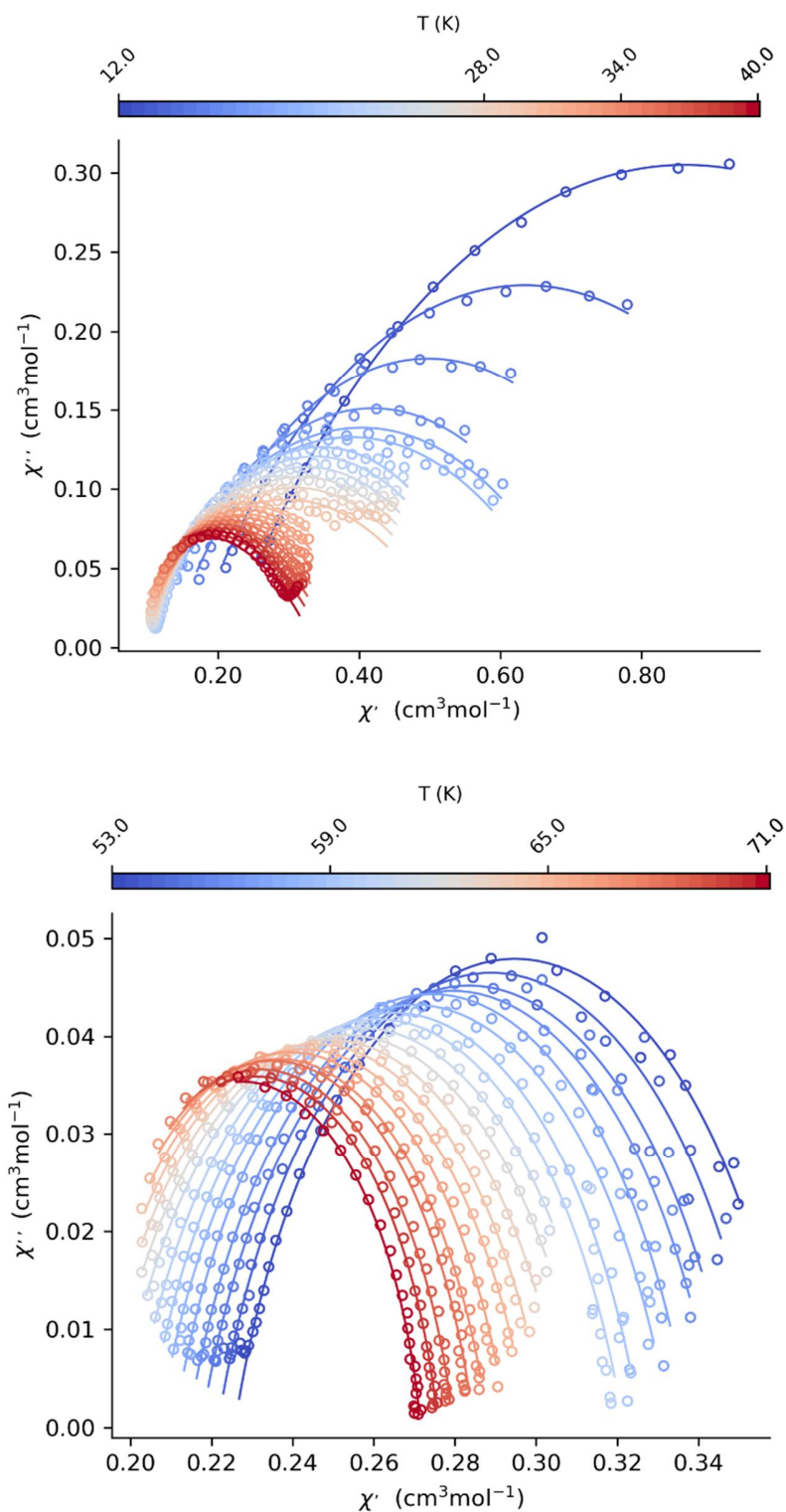


Figure S34. Cole–Cole plots of **1** at various temperatures from 12.0 to 40.0 K (top) and from 53.0 to 71.0 K (bottom). The solid lines represent simultaneous fit of χ' and χ'' data to the generalized Debye model.

Table S16. The best results fitted for **1** at zero dc field by the generalized Debye model.

<i>T</i> / K	τ / s	α	<i>T</i> / K	τ / s	α
12.0	5.94E-01	0.44	53.0	2.11E-02	0.23
16.0	3.23E-01	0.43	54.0	1.66E-02	0.23
20.0	1.58E-01	0.42	55.0	1.28E-02	0.24
24.0	8.73E-02	0.45	56.0	9.94E-03	0.24
26.0	7.60E-02	0.50	57.0	7.46E-03	0.23
27.0	6.45E-02	0.50	58.0	5.69E-03	0.23
28.0	3.79E-02	0.40	59.0	4.07E-03	0.22
29.0	3.23E-02	0.42	60.0	3.08E-03	0.23
30.0	2.68E-02	0.44	61.0	2.26E-03	0.24
31.0	2.18E-02	0.46	62.0	1.70E-03	0.25
32.0	1.73E-02	0.50	63.0	1.25E-03	0.24
33.0	1.28E-02	0.53	64.0	9.28E-04	0.23
34.0	4.39E-03	0.39	65.0	7.07E-04	0.23
35.0	2.90E-03	0.40	66.0	5.27E-04	0.21
36.0	1.82E-03	0.42	67.0	4.13E-04	0.20
37.0	1.08E-03	0.44	68.0	3.33E-04	0.18
38.0	5.88E-04	0.46	69.0	2.59E-04	0.17
39.0	3.03E-04	0.49	70.0	2.13E-04	0.14
40.0	1.61E-04	0.49	71.0	1.67E-04	0.14

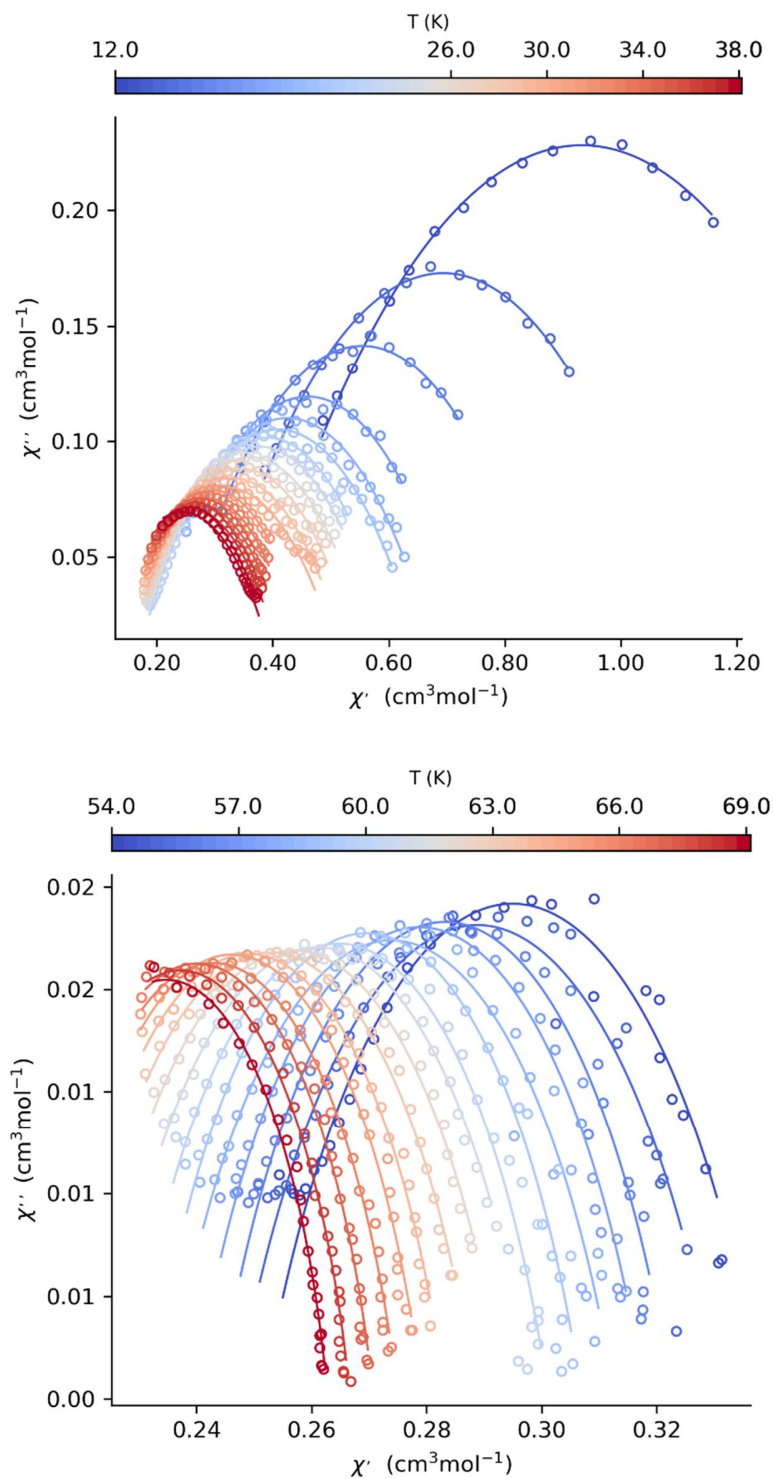


Figure S35. Cole–Cole plots of **1uv** at various temperatures from 12.0 to 38.0 K (top) and from 54.0 to 69.0 K (bottom). The solid lines represent simultaneous fit of χ' and χ'' data to the generalized Debye model.

Table S17. The best results fitted for **1uv** at zero dc field by the generalized Debye model.

<i>T</i> / K	τ / s	α	<i>T</i> / K	τ / s	α
12.0	1.80E-01	0.53	54.0	9.24E-03	0.35
16.0	1.12E-01	0.49	55.0	7.02E-03	0.36
20.0	6.51E-02	0.48	56.0	5.09E-03	0.32
24.0	3.49E-02	0.48	57.0	4.01E-03	0.33
26.0	2.56E-02	0.49	58.0	2.94E-03	0.34
27.0	2.13E-02	0.50	59.0	2.24E-03	0.31
28.0	1.66E-02	0.47	60.0	1.63E-03	0.30
29.0	1.37E-02	0.47	61.0	1.26E-03	0.32
30.0	1.05E-02	0.48	62.0	9.44E-04	0.32
31.0	7.80E-03	0.49	63.0	7.06E-04	0.30
32.0	5.56E-03	0.49	64.0	5.27E-04	0.28
33.0	3.90E-03	0.51	65.0	4.34E-04	0.26
34.0	2.23E-03	0.44	66.0	3.40E-04	0.26
35.0	1.44E-03	0.45	67.0	2.70E-04	0.23
36.0	9.10E-04	0.45	68.0	2.22E-04	0.21
37.0	5.50E-04	0.46	69.0	1.82E-04	0.20
38.0	3.12E-04	0.48			

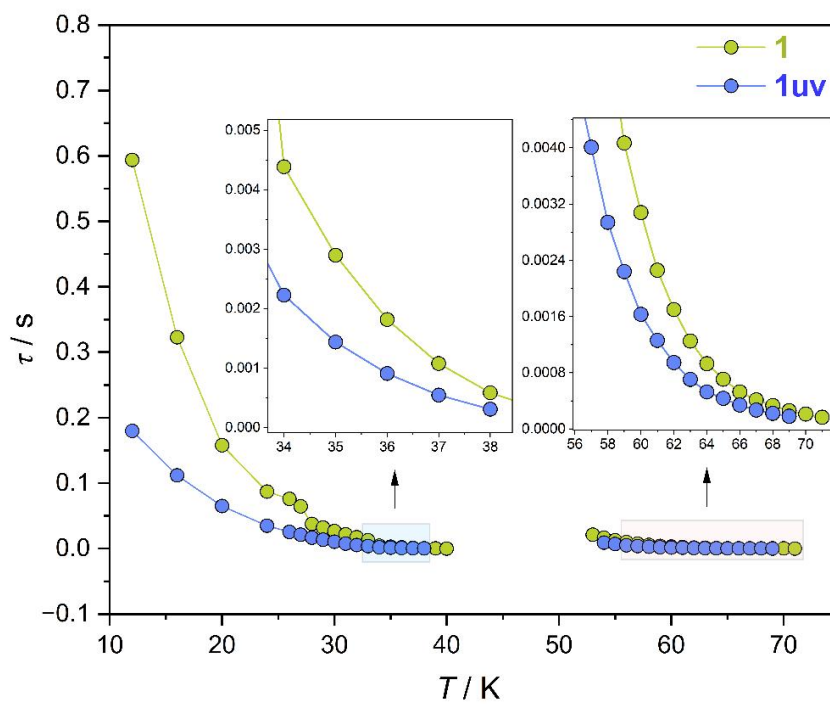


Figure S36. Comparison of relaxation times between **1** and **1uv**.

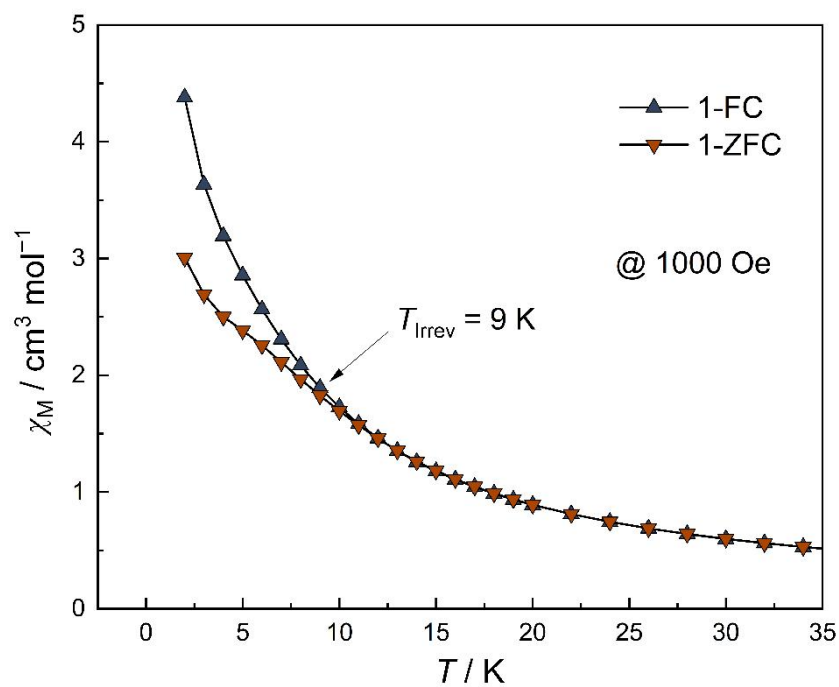


Figure S37. FC and ZFC of **1** at an applied field of 1000 Oe.

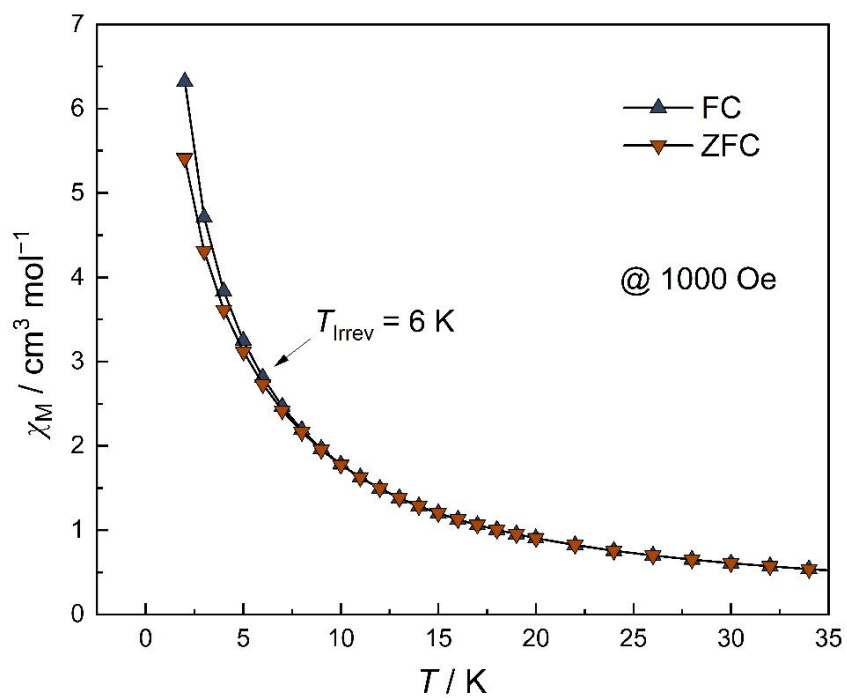


Figure S38. FC and ZFC of **1uv** at an applied field of 1000 Oe.

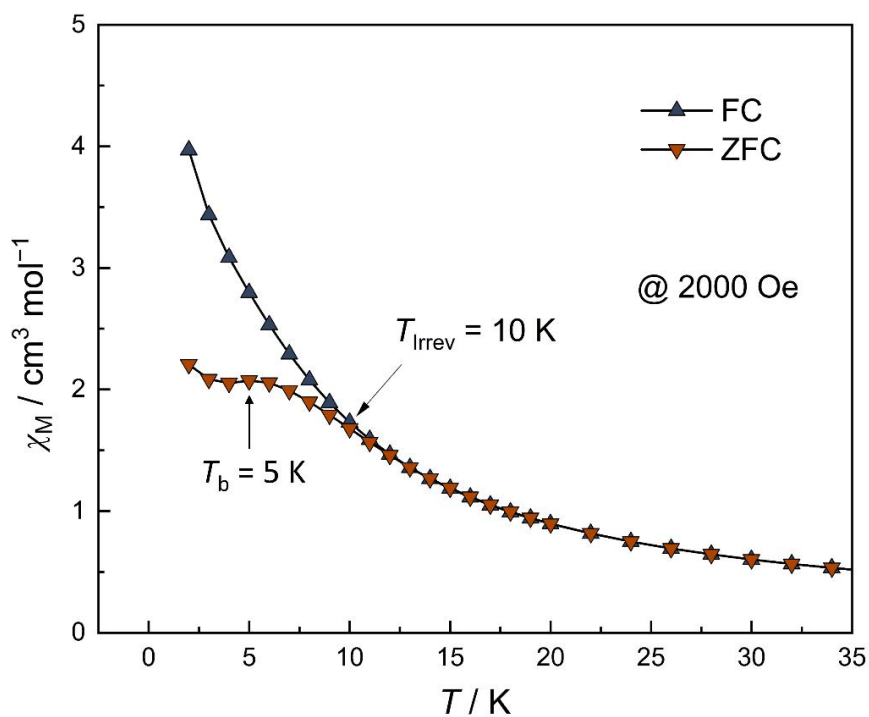


Figure S39. FC and ZFC of **1** at an applied field of 2000 Oe.

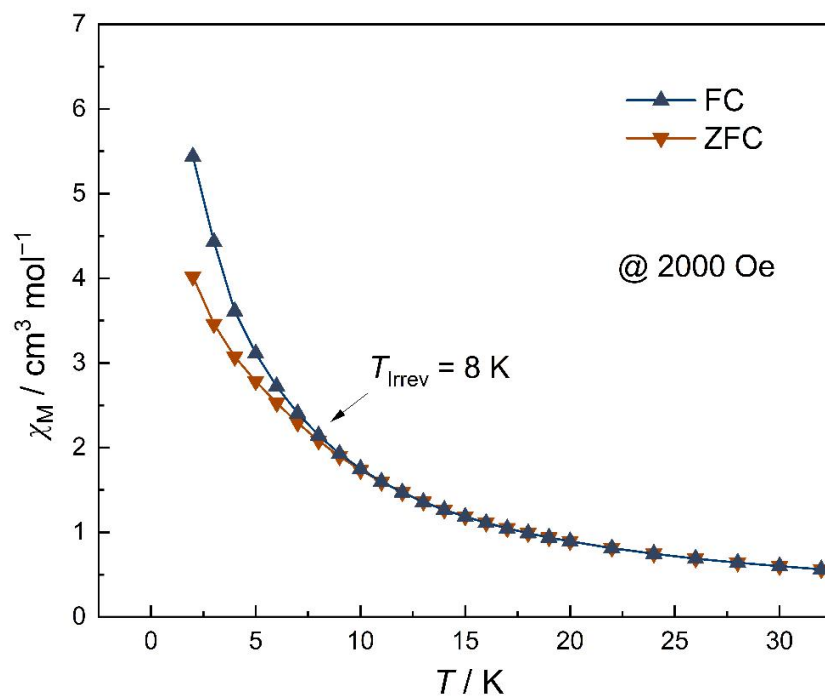


Figure S40. FC and ZFC of **1uv** at an applied field of 2000 Oe.

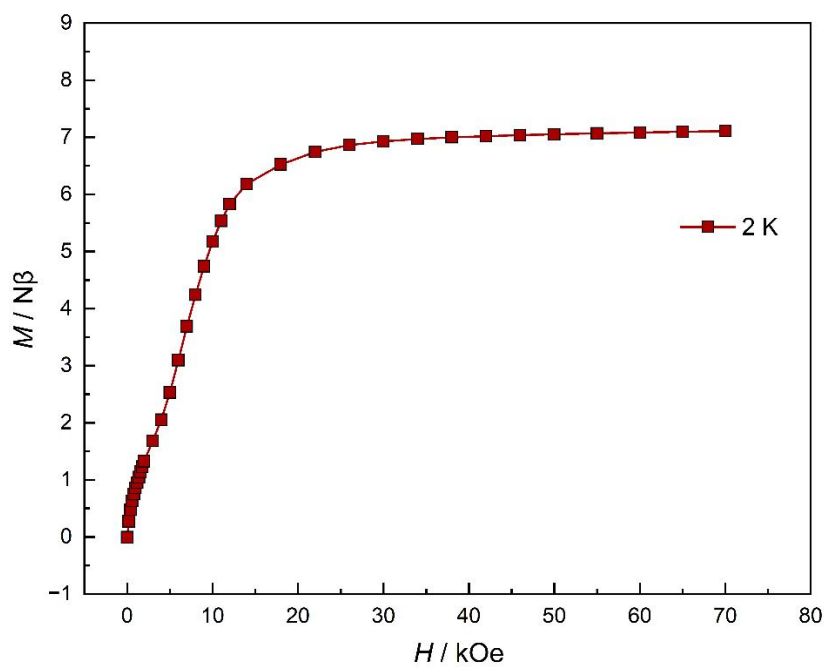


Figure S41. The plot of M vs. H at 2 K for **1Oz**.

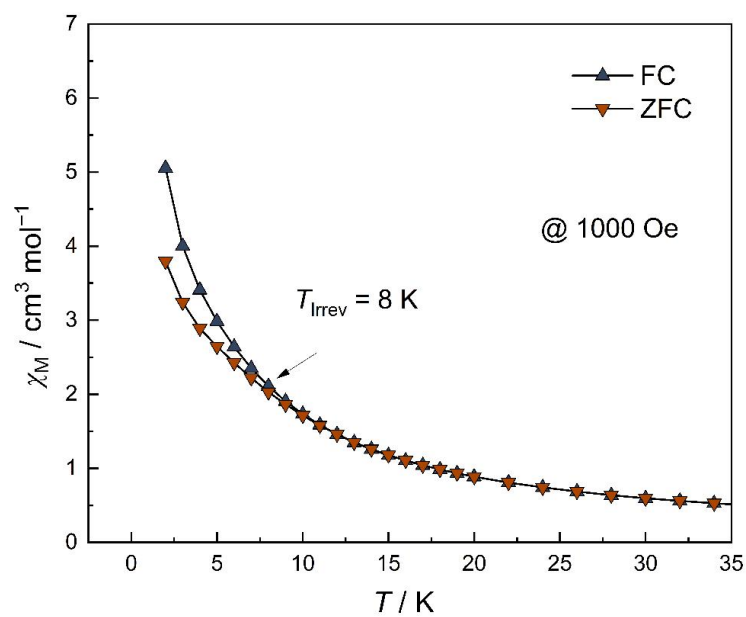


Figure S42. FC and ZFC of **1O₂** at an applied field of 1000 Oe.

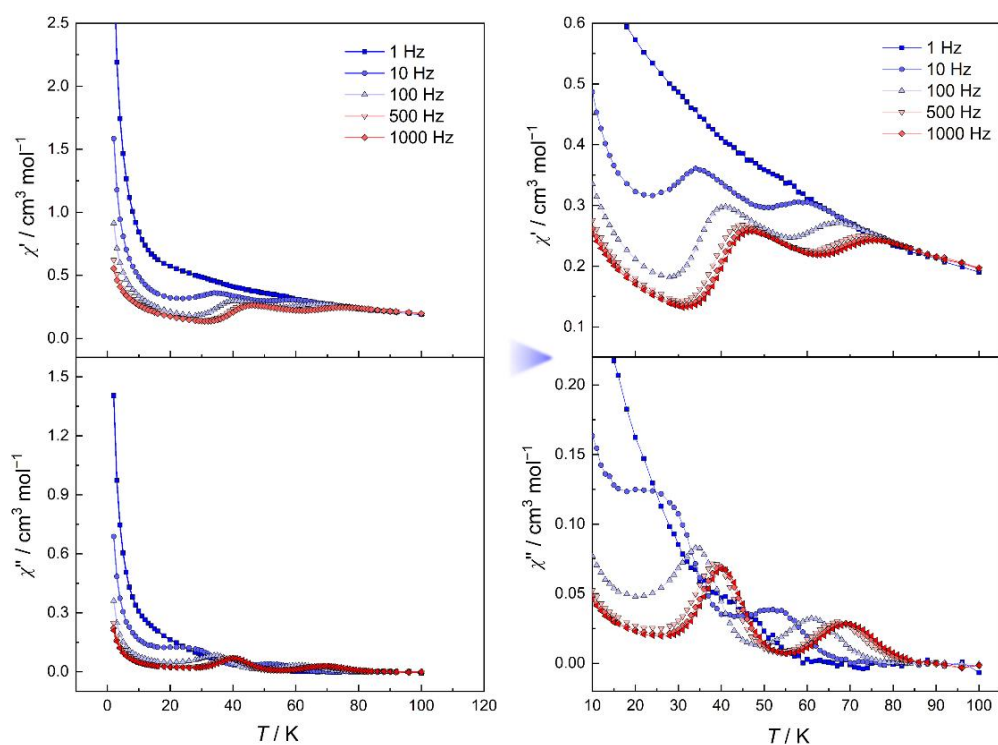


Figure S43. Temperature-dependence of the in-phase (χ') and out-of-phase (χ'') ac susceptibility for **1O₂** under zero dc field in the full temperature range (left) and in the temperature range of 10–100 K (right).

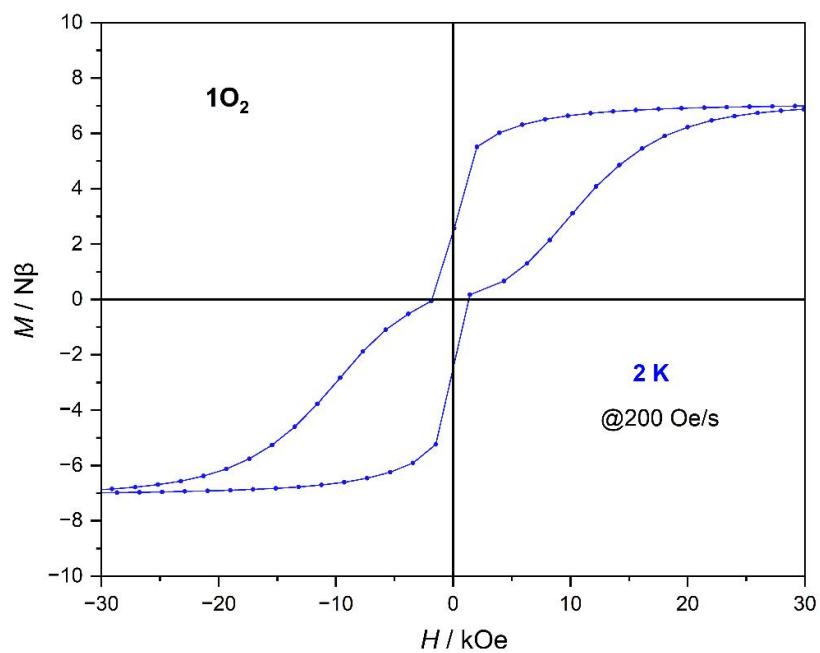


Figure S44. Hysteresis loop for **1O₂** measured at 2 K with a sweep rate of 200 Oe/s.

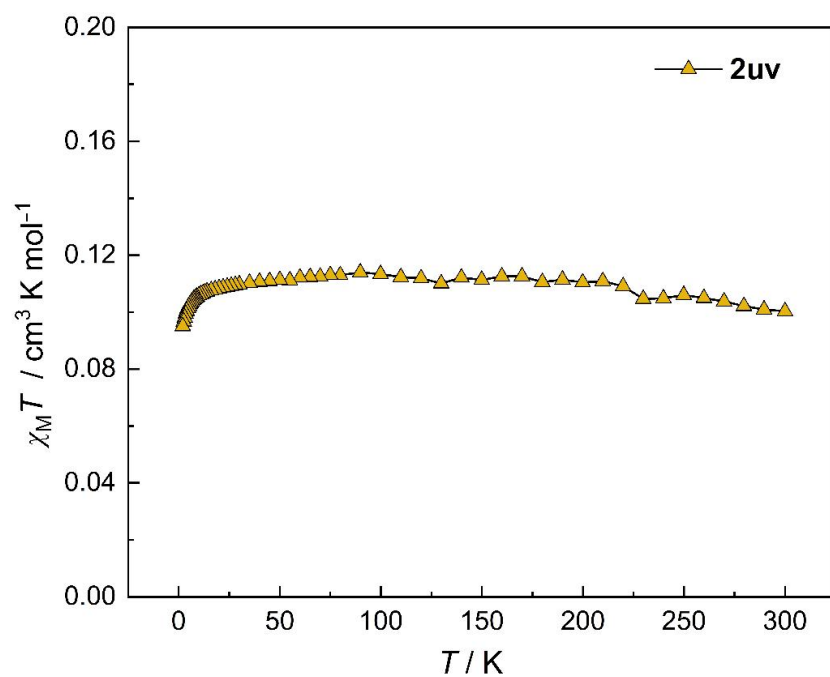


Figure S45. Temperature-dependent susceptibilities for **2uv** under a dc magnetic field of 1000 Oe.

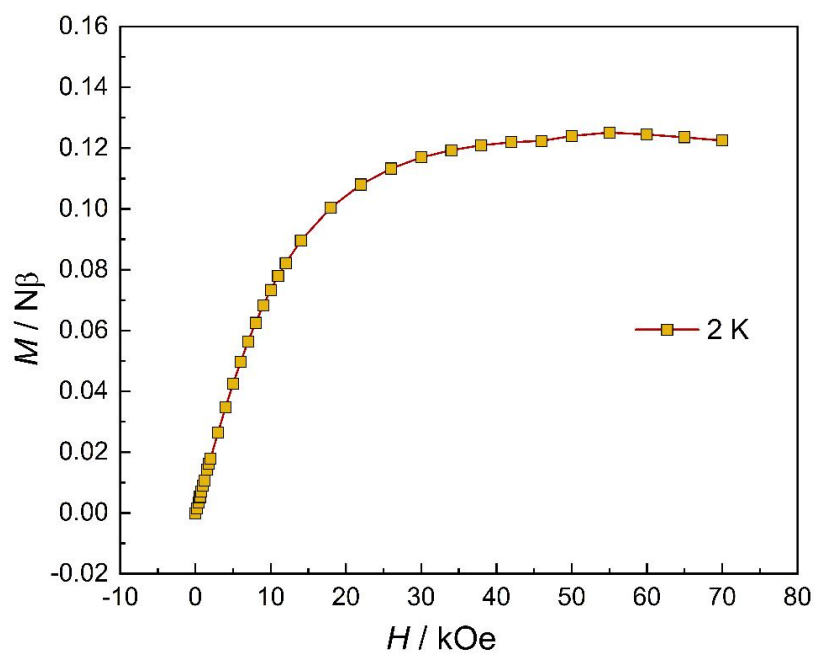


Figure S46. The plot of M vs. H at 2 K for **2uv**.

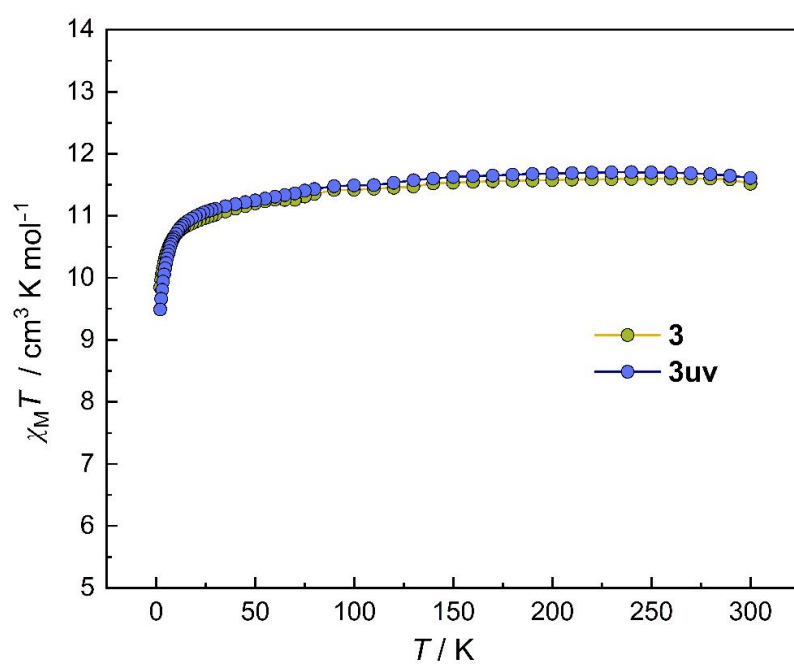


Figure S47. Temperature-dependent susceptibilities for **3** and **3uv** under a dc magnetic field of 1000 Oe.

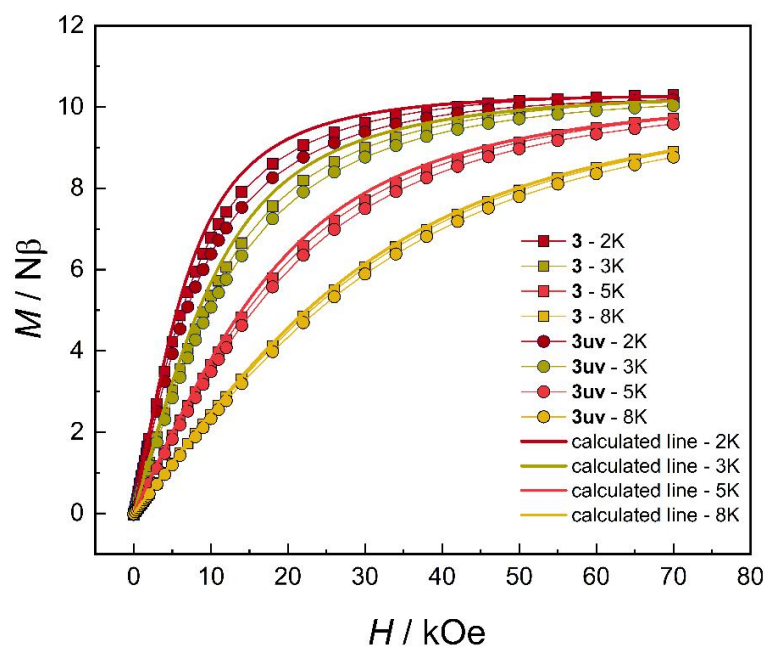


Figure S48. The plots of M vs. H at various temperatures for **3** and **3uv**, with Brillouin calculated curves for 1.5 Gd^{3+} (non-interacting) at corresponding temperatures.

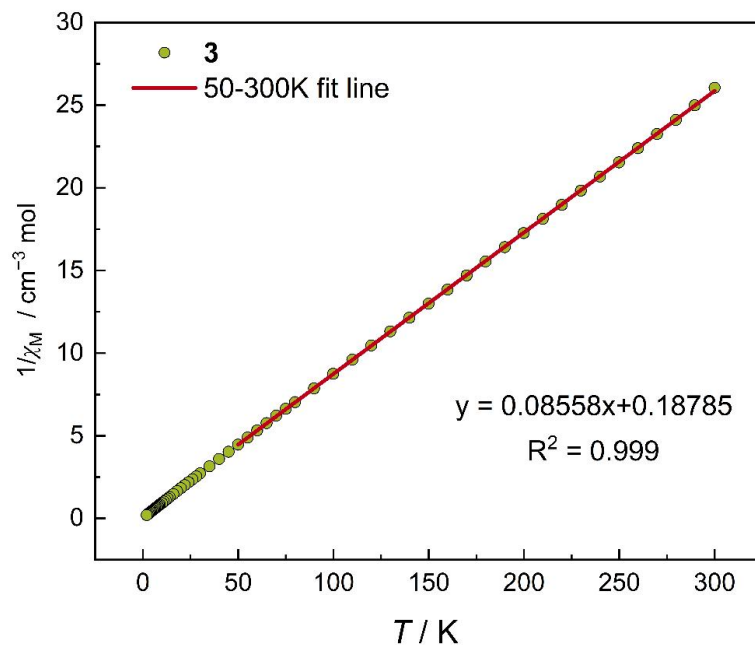


Figure S49. $1/\chi_M$ vs. T of **3**, with a linear fit in the range of 50–300 K.

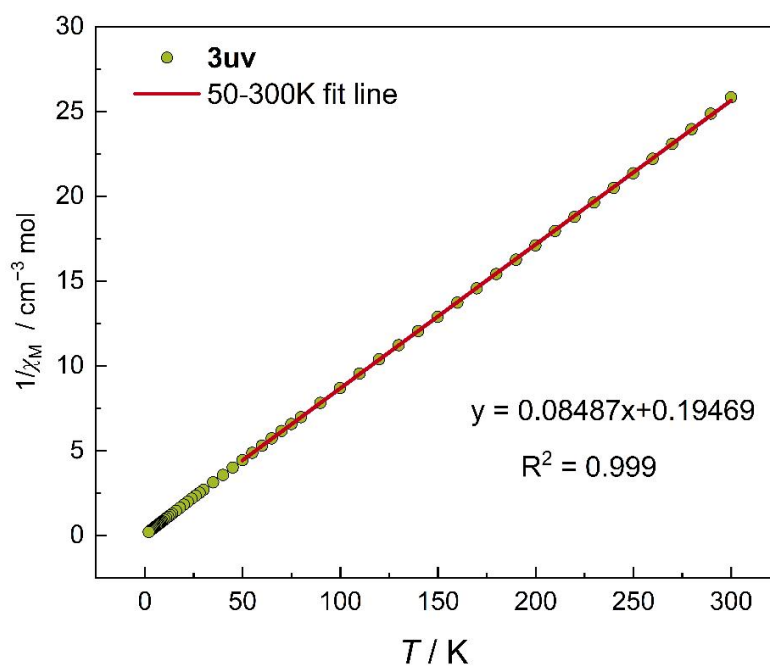


Figure S50. $1/\chi_M$ vs. T of **3uv**, with a linear fit in the range of 50–300 K.

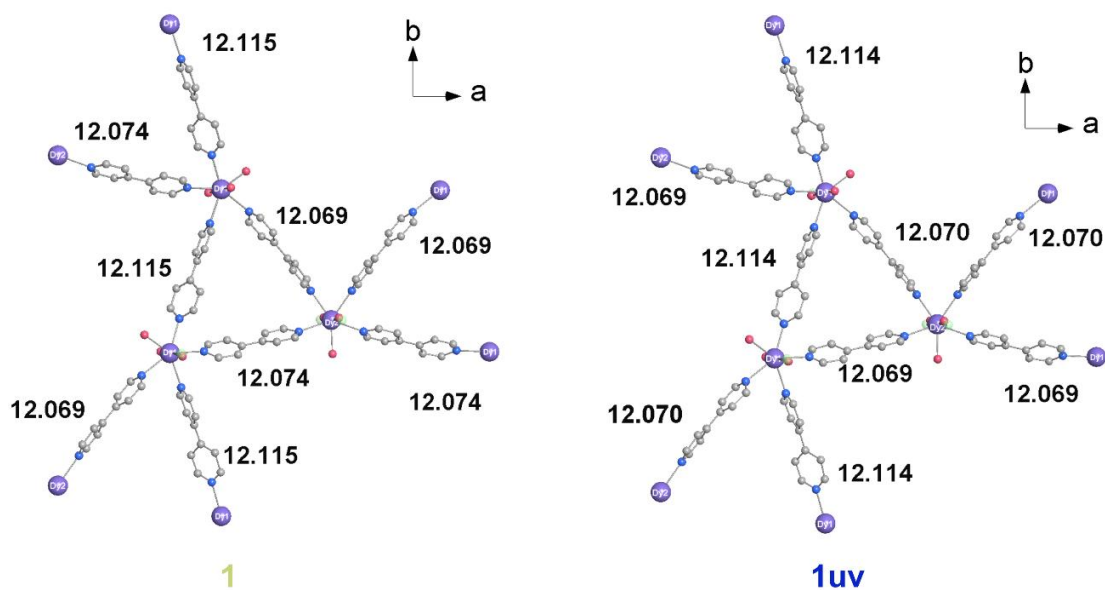


Figure S51. The intralayer Dy...Dy distances (Å) in **1** and **1uv**. H atoms and some C atoms are omitted for clarity.

7. *Ab initio* Calculations

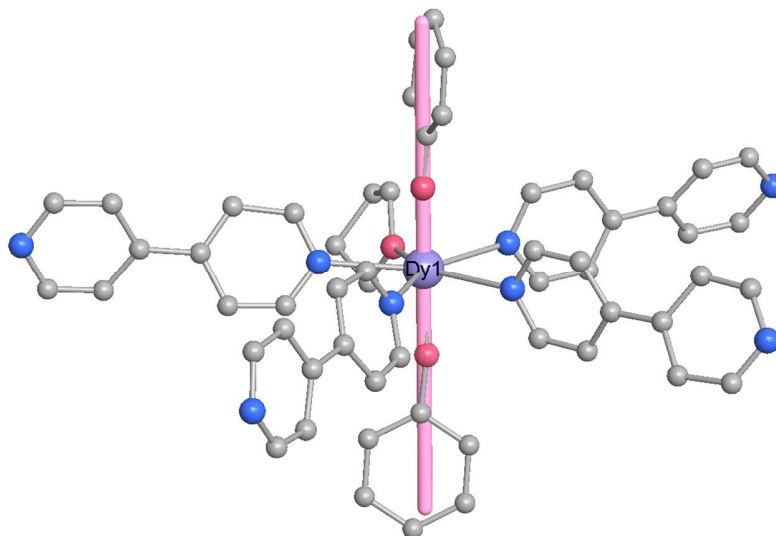


Figure S52. Structure of the $[\text{Dy}(\text{OPh})_2(\text{BPy})_4(\text{THF})]^+$ cation in Dy1 (**1a**-Dy1) for **1**. The pink solid line represents *ab initio* calculated orientation of the main magnetic axis on Dy1 in the ground Kramer's doublets.

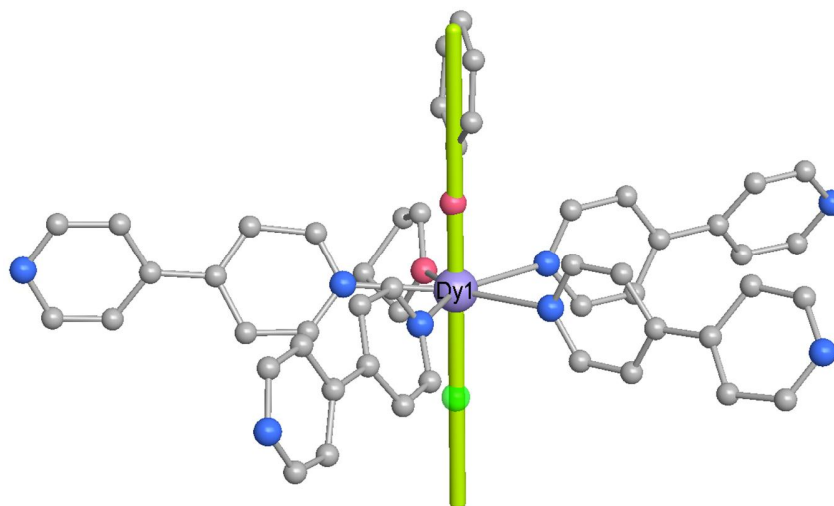


Figure S53. Structure of the $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4(\text{THF})]^+$ cation in Dy1 (**1b**-Dy1) for **1**. The green solid line represents *ab initio* calculated orientation of the main magnetic axis on Dy1 in the ground Kramer's doublets.

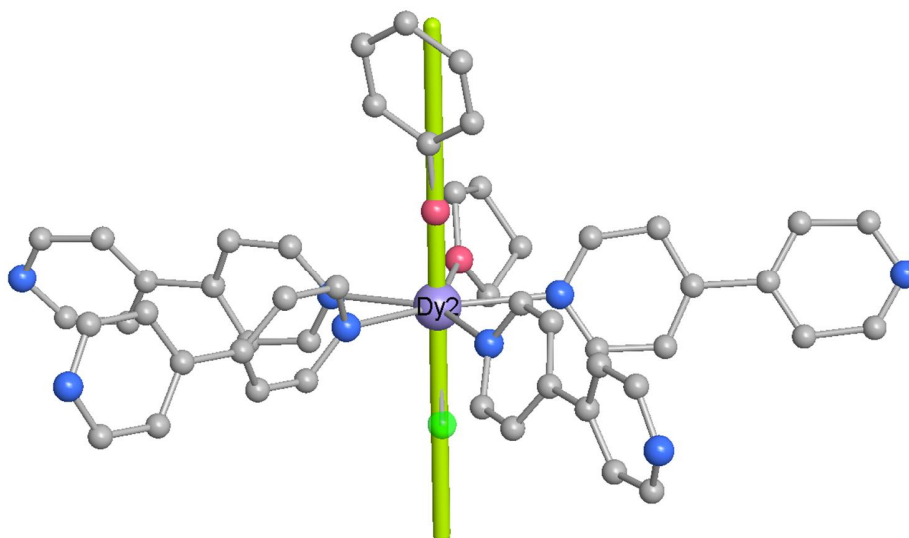


Figure S54. Structure of the $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4(\text{THF})]^+$ cation in Dy2 (**1b**-Dy2) for **1**. The green solid line represents *ab initio* calculated orientation of the main magnetic axis on Dy2 in the ground Kramer's doublets.

Table S18. SA-CASSCF/RASSI calculated electronic states for Dy1 in $[\text{Dy}(\text{OPh})_2(\text{BPy})_4\text{THF}]^+$ in **1**.

Energy (cm^{-1})	g_x	g_y	g_z	g_z Angle ($^\circ$)	Wavefunction
0.0	0.00	0.00	19.88	--	99.9% $ \pm 15/2\rangle$
528.8	0.00	0.00	16.97	1.2	99.8% $ \pm 13/2\rangle$
861.6	0.27	0.56	13.91	2.6	95.6% $ \pm 11/2\rangle$
912.0	0.44	2.16	17.60	89.1	12.0% $ \pm 3/2\rangle$ + 16.5% $ \pm 1/2\rangle$ + 54.8% $ \mp 1/2\rangle$
970.2	3.11	5.88	9.98	86.9	70.2% $ \pm 3/2\rangle$ + 15.3% $ \mp 1/2\rangle$
1003.0	1.56	2.83	9.33	25.6	44.5% $ \pm 9/2\rangle$ + 37.9% $ \pm 5/2\rangle$
1046.9	0.17	2.32	13.23	49.8	23.3% $ \pm 9/2\rangle$ + 18.6% $ \pm 5/2\rangle$ + 18.6% $ \mp 5/2\rangle$ + 10.2% $ \mp 7/2\rangle$ + 21.9% $ \mp 9/2\rangle$
1068.5	0.43	2.54	13.76	54.1	74.0% $ \pm 7/2\rangle$

^a Only components with > 10% contribution are given, rounded to the nearest percent.

Table S19. *Ab initio* calculated crystal field parameters for Dy1 in [Dy(OPh)₂(BPy)₄THF]⁺ in **1**.

Crystal Field Parameter	Value / cm ⁻¹
B_2^{-2}	1.4849409922191E-02
B_2^{-1}	4.9192720353192E-02
B_2^0	-5.2256029390494E+00
B_2^1	-3.9509589272969E-02
B_2^2	8.2401793674304E-01
B_4^{-4}	-1.6092252354631E-04
B_4^{-3}	1.3985597135604E-02
B_4^{-2}	-3.6266866220268E-04
B_4^{-1}	-4.9546542610419E-04
B_4^0	-1.6149893516589E-02
B_4^1	2.4761085043656E-03
B_4^2	-1.0394730213525E-03
B_4^3	-1.1014142234818E-02
B_4^4	-1.7136090114534E-03
B_6^{-6}	1.5000867186545E-05
B_6^{-5}	1.6645436478492E-04
B_6^{-4}	-2.0007963121469E-06
B_6^{-3}	8.6364882927016E-05
B_6^{-2}	-1.4184969297858E-05
B_6^{-1}	-4.1262610466264E-05
B_6^0	1.5390828551581E-05
B_6^1	-4.8301013267473E-05
B_6^2	-3.3575764056966E-05
B_6^3	-1.0452947683045E-04
B_6^4	-1.2332569118228E-05
B_6^5	1.2565318898288E-04
B_6^6	3.1378368208893E-05

Table S20. SA-CASSCF/RASSI calculated electronic states for Dy1 in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1**.

Energy (cm ⁻¹)	<i>g_x</i>	<i>g_y</i>	<i>g_z</i>	<i>g_z</i> Angle (°)	Wavefunction
0.0	0.00	0.00	19.88	--	89.9% ±15/2>
397.2	0.01	0.02	17.03	6.3	81.5% ±13/2>
528.5	0.46	1.61	18.09	83.0	25.1% ±5/2> + 35.3% ±3/2> + 20.3% ±1/2>
579.2	3.40	6.24	10.28	63.8	21.6% ±11/2> + 13.5% ±7/2> + 42.4% ±1/2>
614.5	1.40	2.44	9.46	9.5	49.2% ±11/2> + 11.6% ±3/2> + 24.4% ±1/2>
659.2	0.18	1.75	8.94	12.5	10.1% ±11/2> + 19.6% ±9/2> + 13.1% ±7/2> + 25.6% ±3/2>
711.2	0.75	0.84	10.62	22.5	40.1% ±9/2> + 18.3% ±5/2> + 13.2% ± 9/2>
768.9	0.14	0.34	16.47	49.0	39.8% ±7/2> + 21.8% ±5/2> + 15.8% ± 7/2>

^a Only components with > 10% contribution are given, rounded to the nearest percent.

Table S21. *Ab initio* calculated crystal field parameters for Dy1 in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1**.

Crystal Field Parameter	Value / cm ⁻¹
B_2^{-2}	4.9431846575993E-02
B_2^{-1}	5.7261465526174E-01
B_2^0	-3.0421087780959E+00
B_2^1	-8.5669133866917E-02
B_2^2	6.1156196469244E-01
B_4^{-4}	1.2854344981160E-04
B_4^{-3}	9.2481701062001E-03
B_4^{-2}	3.4619739382249E-03
B_4^{-1}	5.1681736134136E-02
B_4^0	-1.1282367726393E-02
B_4^1	-6.5597922472814E-03
B_4^2	9.9567284645921E-03
B_4^3	-1.1713695971288E-02
B_4^4	-7.4834170348923E-04
B_6^{-6}	4.4485532559403E-05
B_6^{-5}	1.3165693422128E-04
B_6^{-4}	3.2278293360530E-05
B_6^{-3}	5.0942379229237E-05
B_6^{-2}	3.7517205436488E-05
B_6^{-1}	-5.9146943305867E-05
B_6^0	9.9096973870462E-06
B_6^1	-8.5702926284373E-07
B_6^2	1.6942329849400E-05
B_6^3	-9.7335623119898E-05
B_6^4	2.5959638130627E-05
B_6^5	4.3952145242781E-05
B_6^6	3.1679139754616E-05

Table S22. SA-CASSCF/RASSI calculated electronic states for Dy2 in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1**.

Energy (cm ⁻¹)	<i>g_x</i>	<i>g_y</i>	<i>g_z</i>	<i>g_z</i> Angle (°)	Wavefunction
0.0	0.00	0.00	19.88	--	98.1% ±15/2>
365.3	0.02	0.02	16.98	1.9	94.8% ±13/2>
447.3	0.64	2.23	17.61	89.8	19.8% ±3/2> + 43.1% ±1/2> + 26.5% ∓1/2>
493.2	2.88	4.86	10.02	87.7	23.8% ±5/2> + 42.9% ±3/2> + 10.3% ∓1/2> + 11.3% ∓3/2>
534.2	2.07	3.33	8.50	16.0	21.9% ±11/2> + 18.5% ±7/2> + 34.3% ±5/2>
556.8	0.04	1.12	12.53	15.9	68.5% ±11/2> + 12.9% ±5/2>
586.1	0.89	1.67	11.22	27.6	50.3% ±9/2> + 23.4% ±7/2>
631.9	0.08	0.13	17.39	51.3	36.3% ±9/2> + 33.6% ±7/2> + 11.6% ±5/2>

^a Only components with > 10% contribution are given, rounded to the nearest percent.

Table S23. *Ab initio* calculated crystal field parameters for Dy2 in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1**.

Crystal Field Parameter	Value / cm ⁻¹
B_2^{-2}	-3.2985233396591E-03
B_2^{-1}	-2.6478061817730E-02
B_2^0	-2.3825432517132E+00
B_2^1	-2.2501537070577E-01
B_2^2	5.6077915292519E-01
B_4^{-4}	-2.0919309229508E-03
B_4^{-3}	-1.4249831123503E-02
B_4^{-2}	3.7416397100964E-04
B_4^{-1}	-4.8763238087275E-03
B_4^0	-1.1915030357729E-02
B_4^1	-2.1706998169216E-02
B_4^2	-3.3517682297944E-03
B_4^3	2.8457727493733E-03
B_4^4	1.8912564061457E-03
B_6^{-6}	-6.4945963732210E-06
B_6^{-5}	2.0489388075696E-05
B_6^{-4}	-3.2661055239410E-05
B_6^{-3}	-1.0727182550415E-04
B_6^{-2}	1.1279426839443E-05
B_6^{-1}	-1.4576271940804E-05
B_6^0	-7.5884758969187E-08
B_6^1	1.3624056079517E-05
B_6^2	-1.3444083831168E-05
B_6^3	3.7280249319522E-05
B_6^4	2.6093195626383E-05
B_6^5	-1.8223372229418E-04
B_6^6	-2.0572015302074E-06

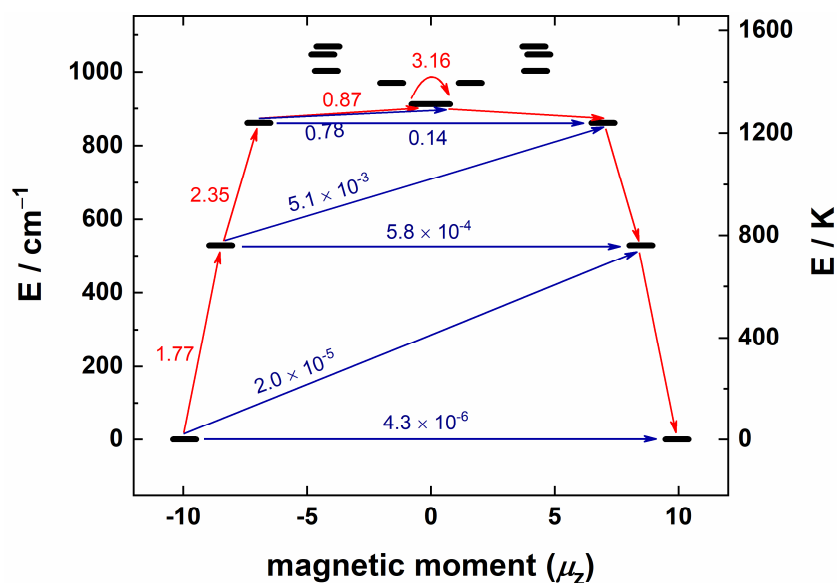


Figure S55. *Ab initio* calculated electronic states of Dy1 in $[\text{Dy}(\text{OPh})_2(\text{BPy})_4(\text{THF})]^+$ in **1** with the probability of transition between different sub-states. The thick black lines represent the Kramer's doublets as a function of their magnetic moment along the main anisotropy axis. The path shown by the red arrows represents the most probable path for magnetic relaxation.

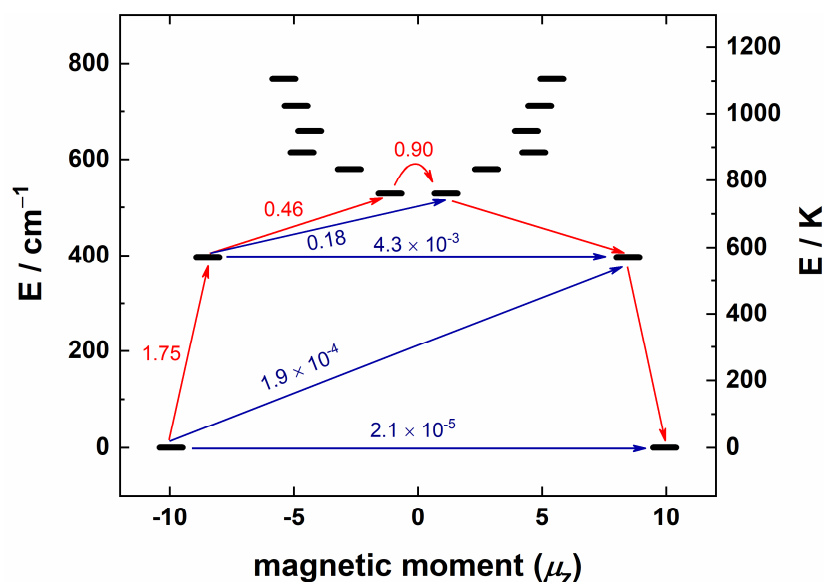


Figure S56. *Ab initio* calculated electronic states of Dy1 in $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4(\text{THF})]^+$ in **1** with the probability of transition between different sub-states. The thick black lines represent the Kramer's doublets as a function of their magnetic moment along the main anisotropy axis. The path shown by the red arrows represents the most probable path for magnetic relaxation.

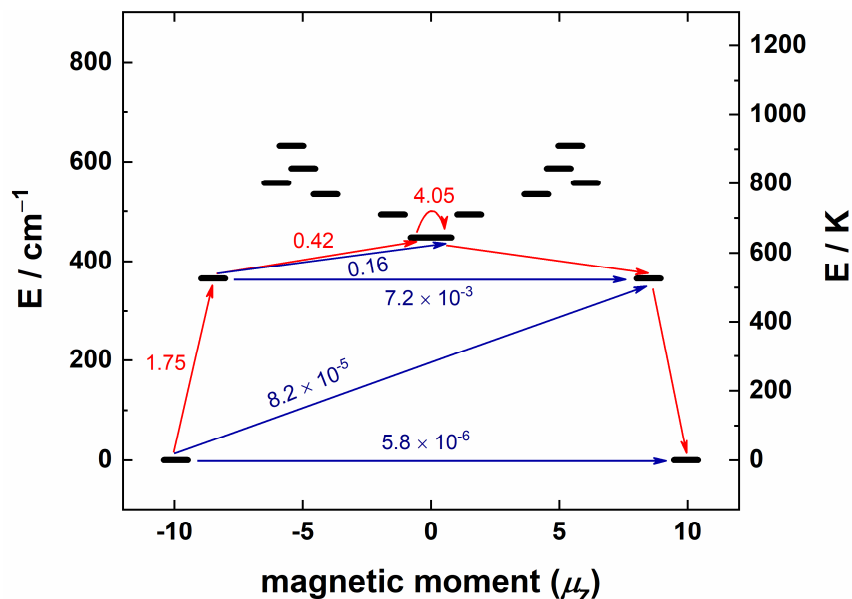


Figure S57. *Ab initio* calculated electronic states of Dy2 in $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4\text{THF}]^+$ in **1** with the probability of transition between different sub-states. The thick black lines represent the Kramer's doublets as a function of their magnetic moment along the main anisotropy axis. The path shown by the red arrows represents the most probable path for magnetic relaxation.

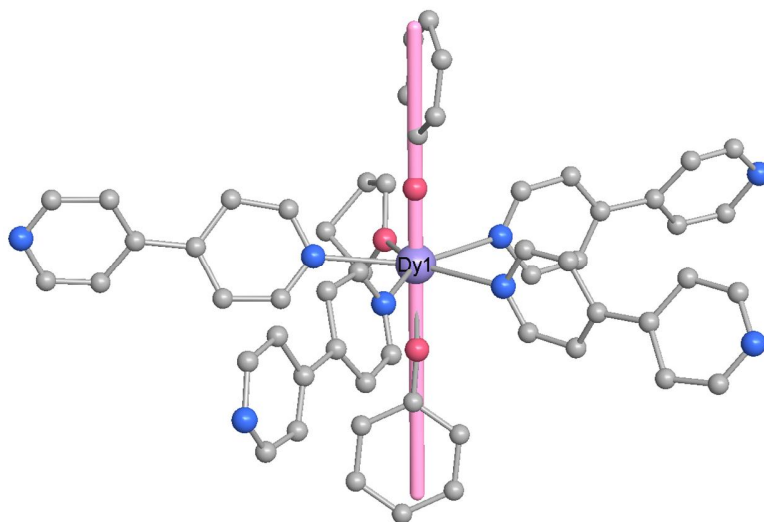


Figure S58. Structure of the $[\text{Dy}(\text{OPh})_2(\text{BPy})_4(\text{THF})]^+$ cation in Dy1 (**1c-Dy1**) for **1uv**. The pink solid line represents *ab initio* calculated orientation of the main magnetic axis on Dy1 in the ground Kramer's doublets.

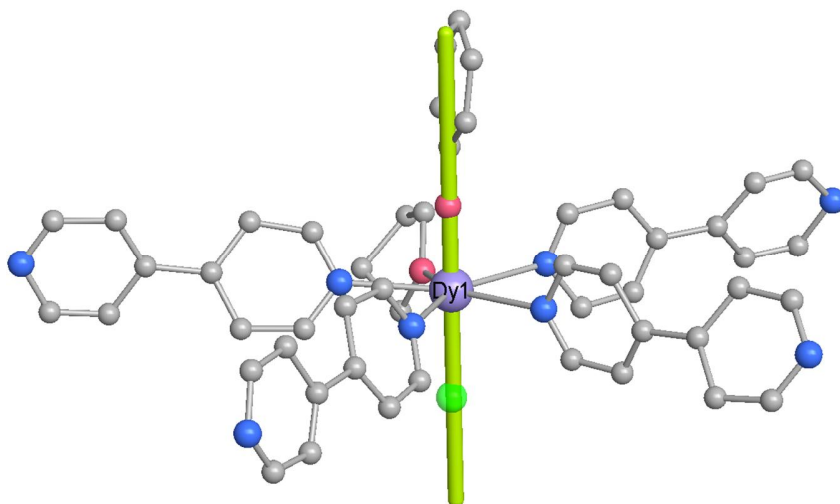


Figure S59. Structure of the $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4(\text{THF})]^+$ cation in Dy1 (**1d-Dy1**) for **1uv**. The green solid line represents *ab initio* calculated orientation of the main magnetic axis on Dy1 in the ground Kramer's doublets.

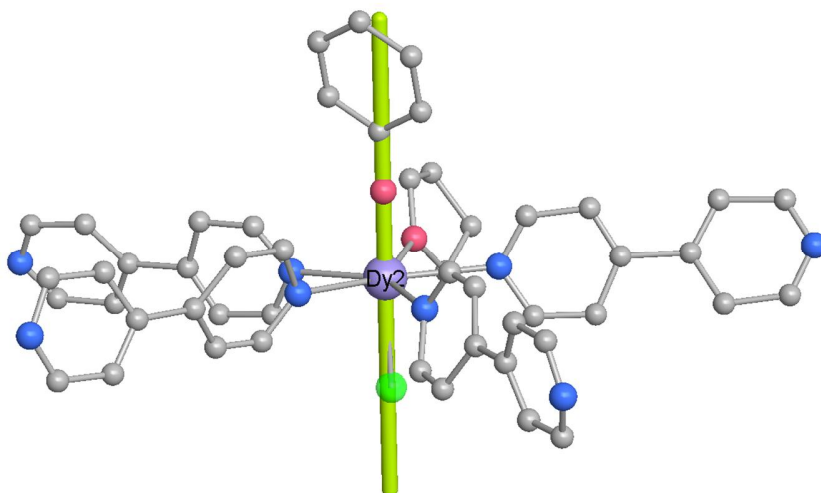


Figure S60. Structure of the $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4(\text{THF})]^+$ cation in Dy2 (**1d-Dy2**) for **1uv**. The green solid line represents *ab initio* calculated orientation of the main magnetic axis on Dy2 in the ground Kramer's doublets.

Table S24. SA-CASSCF/RASSI calculated electronic states for Dy1 in [Dy(OPh)₂(BPy)₄(THF)]⁺ in **1uv**.

Energy (cm ⁻¹)	<i>g_x</i>	<i>g_y</i>	<i>g_z</i>	<i>g_z</i> Angle (°)	Wavefunction
0.0	0.00	0.00	19.88	--	99.8% ±15/2>
527.3	0.00	0.00	16.97	0.5	99.3% ±13/2>
863.4	0.24	0.45	13.98	3.4	95.9% ±11/2>
921.0	0.45	2.35	17.42	86.4	58.9% ±1/2> + 14.7% ±3/2>
977.4	3.44	6.24	9.50	86.4	52.4% ±3/2> + 12.7% ±1/2> + 12.6% ±5/2> + 12.6% ±7/2>
1017.8	0.66	3.12	8.78	19.3	45.8% ±9/2> + 19.3% ±5/2> + 13.1% ±3/2>
1039.9	0.55	0.88	14.57	54.0	33.3% ±9/2> + 21.7% ±7/2> + 26.0% ±5/2>
1085.0	0.33	0.72	16.42	57.8	53.4% ±7/2> + 19.8% ±5/2> + 10.1% ±3/2> + 10.1% ±1/2>

^a Only components with > 10% contribution are given, rounded to the nearest percent.

Table S25. *Ab initio* calculated crystal field parameters for Dy1 in [Dy(OPh)₂(BPy)₄(THF)]⁺ in **1uv**.

Crystal Field Parameter	Value / cm ⁻¹
B_2^{-2}	1.7643977367588E-02
B_2^{-1}	6.6241207387463E-02
B_2^0	-5.2986588020591E+00
B_2^1	1.1488961930310E-02
B_2^2	7.0624261687562E-01
B_4^{-4}	2.2741446597198E-03
B_4^{-3}	2.1540478591032E-02
B_4^{-2}	-5.9611388972086E-04
B_4^{-1}	8.6819889094288E-03
B_4^0	-1.5961452936612E-02
B_4^1	1.0201304205511E-02
B_4^2	-1.2545259500810E-03
B_4^3	-1.5243927602262E-03
B_4^4	-5.1999529183696E-04
B_6^{-6}	-3.5472220546927E-05
B_6^{-5}	-4.4485140129135E-05
B_6^{-4}	8.2459419250243E-06
B_6^{-3}	1.5404377592158E-04
B_6^{-2}	1.3730657250008E-06
B_6^{-1}	-5.6866039883715E-05
B_6^0	1.5889367014911E-05
B_6^1	-3.8642627428096E-05
B_6^2	-1.3277530717105E-05
B_6^3	-3.6994403731617E-05
B_6^4	9.0680121881789E-06
B_6^5	1.7637204291185E-04
B_6^6	2.2225230164437E-05

Table S26. SA-CASSCF/RASSI calculated electronic states for Dy1 in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1uv**.

Energy (cm ⁻¹)	<i>g_x</i>	<i>g_y</i>	<i>g_z</i>	<i>g_z</i> Angle (°)	Wavefunction
0.0	0.00	0.00	19.88	--	91.1% ±15/2>
416.2	0.01	0.01	17.02	5.6	81.1% ±13/2>
595.8	0.14	0.92	18.19	79.3	14.2% ±7/2> + 28.2% ±5/2> + 30.3% ±3/2> + 10.6% ±1/2>
637.8	3.28	6.04	10.80	43.3	48.5% ±11/2> + 11.6% ±3/2> + 20.1% ±1/2>
672.4	1.58	4.25	7.43	16.1	21.2% ±11/2> + 18.0% ±1/2> + 24.4% ± 1/2>
718.8	0.64	1.90	9.28	31.0	36.9% ±9/2> + 11.4% ±7/2> + 24.0% ±3/2> + 12.0% ±1/2>
775.6	0.84	1.18	11.00	30.6	31.7% ±9/2> + 12.4% ±7/2> + 11.4% ±5/2> + 17.0% ±3/2> + 10.4% ± 9/2>
847.8	0.11	0.26	17.12	53.6	35.7% ±7/2> + 33.4% ±5/2>

^a Only components with > 10% contribution are given, rounded to the nearest percent.

Table S27. *Ab initio* calculated crystal field parameters for Dy1 in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1uv**.

Crystal Field Parameter	Value / cm ⁻¹
B_2^{-2}	-3.5331158372259E-02
B_2^{-1}	6.2103435828693E-01
B_2^0	-3.5254692363270E+00
B_2^1	2.7128263586069E-01
B_2^2	5.8283250286453E-01
B_4^{-4}	1.4326993886780E-03
B_4^{-3}	1.8655430110973E-02
B_4^{-2}	-8.2712071508784E-03
B_4^{-1}	5.0371195272062E-02
B_4^0	-1.1138014321438E-02
B_4^1	2.4314473019276E-02
B_4^2	8.3735605119028E-03
B_4^3	2.3386984902228E-03
B_4^4	1.8105371291978E-03
B_6^{-6}	-5.9603692467310E-05
B_6^{-5}	-1.4397020630402E-04
B_6^{-4}	-3.2099737456020E-05
B_6^{-3}	1.0790900301297E-04
B_6^{-2}	1.2999971657995E-05
B_6^{-1}	-8.9452170145603E-05
B_6^0	9.0162377663969E-06
B_6^1	-3.5464943389871E-05
B_6^2	3.4232161994715E-05
B_6^3	1.1583889601479E-05
B_6^4	4.5507731427602E-05
B_6^5	4.4394249305382E-05
B_6^6	-1.2303160990837E-05

Table S28. SA-CASSCF/RASSI calculated electronic states for Dy2 in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1uv**.

Energy (cm ⁻¹)	<i>g</i> _x	<i>g</i> _y	<i>g</i> _z	<i>g</i> _z Angle (°)	Wavefunction
0.0	0.00	0.00	19.88	--	96.6% ±15/2>
363.1	0.01	0.02	17.01	2.8	91.5% ±13/2>
452.9	0.41	1.23	18.28	87.9	28.6% ±3/2> + 15.0% ±1/2> + 43.7% ∓1/2>
517.4	2.38	4.33	11.58	87.9	32.9% ±5/2> + 17.2% ±3/2> + 13.9% ∓1/2> + 20.5% ∓3/2>
558.2	0.99	1.79	12.31	9.8	68.0% ±11/2>
574.5	0.81	4.94	7.69	27.7	15.7% ±11/2> + 32.4% ±7/2> + 18.3% ±5/2> + 13.6% ±1/2>
610.5	2.08	3.21	10.46	21.8	66.7% ±9/2>
686.9	0.04	0.08	18.56	56.9	15.4% ±9/2> + 26.5% ±7/2> + 16.8% ±5/2> + 14.8% ∓7/2>

^a Only components with > 10% contribution are given, rounded to the nearest percent.

Table S29. *Ab initio* calculated crystal field parameters for Dy2 in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1uv**.

Crystal Field Parameter	Value / cm ⁻¹
B_2^{-2}	9.3424944176858E-03
B_2^{-1}	1.3631442896774E-02
B_2^0	-2.5886083713958E+00
B_2^1	-3.1251675169597E-01
B_2^2	8.9846520530440E-01
B_4^{-4}	-4.0516613018575E-04
B_4^{-3}	-9.8174845995242E-03
B_4^{-2}	1.6400706542017E-03
B_4^{-1}	3.8447976667621E-03
B_4^0	-1.1661543663191E-02
B_4^1	-3.1501535989181E-02
B_4^2	-7.0456184443471E-03
B_4^3	5.3700369630042E-03
B_4^4	4.0156082696482E-03
B_6^{-6}	5.9392300053168E-06
B_6^{-5}	-1.3564340318788E-04
B_6^{-4}	-1.2147174907379E-05
B_6^{-3}	-7.4369972863440E-05
B_6^{-2}	2.8984308557400E-06
B_6^{-1}	-4.9178999333240E-05
B_6^0	5.1857145716882E-06
B_6^1	4.9760109765404E-05
B_6^2	-2.3311923786070E-05
B_6^3	6.1845664633409E-05
B_6^4	5.3717085264399E-05
B_6^5	-9.3536440325424E-05
B_6^6	2.3725215562222E-06

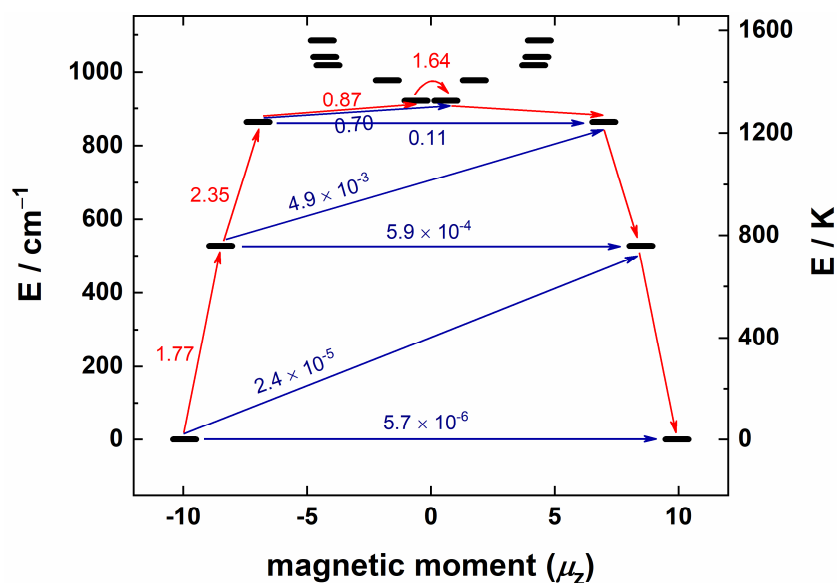


Figure S61. *Ab initio* calculated electronic states of Dy1 in $[\text{Dy}(\text{OPh})_2(\text{BPy})_4(\text{THF})]^+$ in **1uv** with the probability of transition between different sub-states. The thick black lines represent the Kramer's doublets as a function of their magnetic moment along the main anisotropy axis. The path shown by the red arrows represents the most probable path for magnetic relaxation.

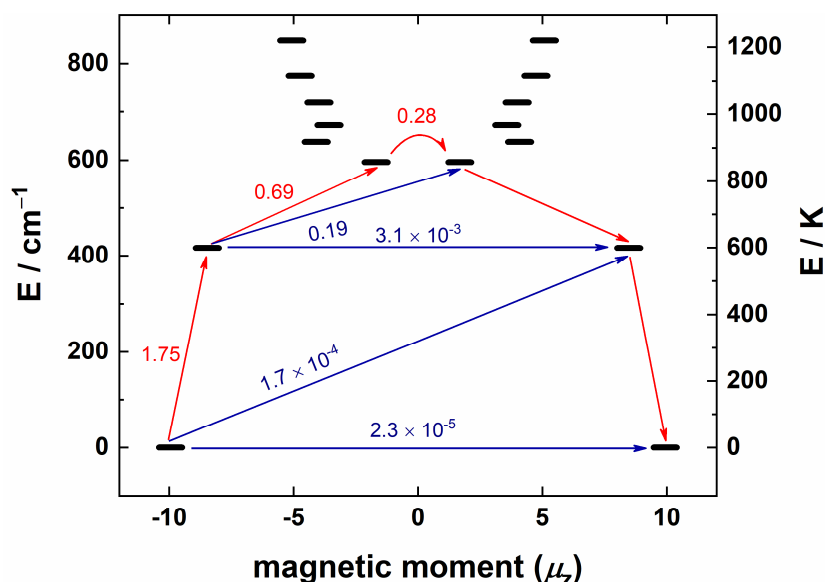


Figure S62. *Ab initio* calculated electronic states of Dy1 in $[\text{Dy}(\text{OPh})\text{Cl}(\text{BPy})_4\text{THF}]^+$ in **1uv** with the probability of transition between different sub-states. The thick black lines represent the Kramer's doublets as a function of their magnetic moment along the main anisotropy axis. The path shown by the red arrows represents the most probable path for magnetic relaxation.

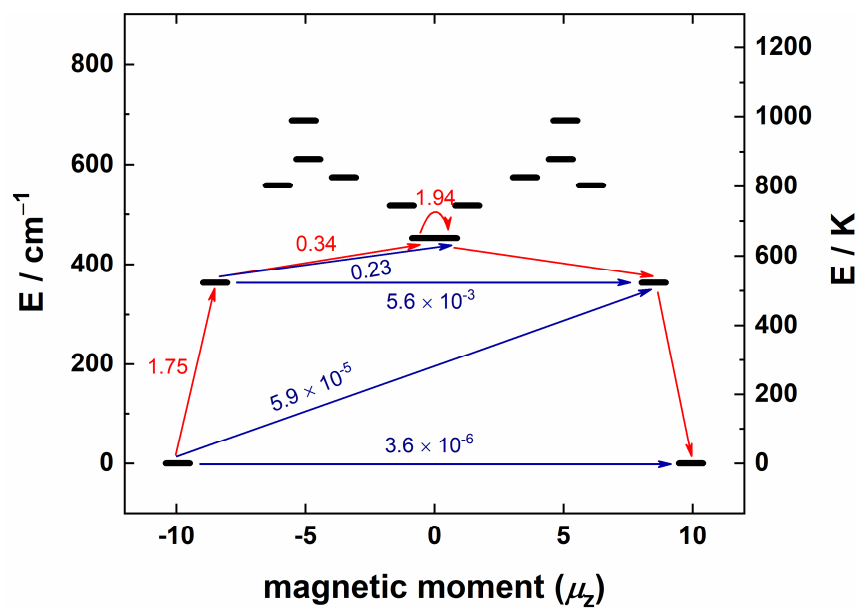


Figure S63. *Ab initio* calculated electronic states of Dy² in [Dy(OPh)Cl(BPy)₄THF]⁺ in **1uv** with the probability of transition between different sub-states. The thick black lines represent the Kramer's doublets as a function of their magnetic moment along the main anisotropy axis. The path shown by the red arrows represents the most probable path for magnetic relaxation.

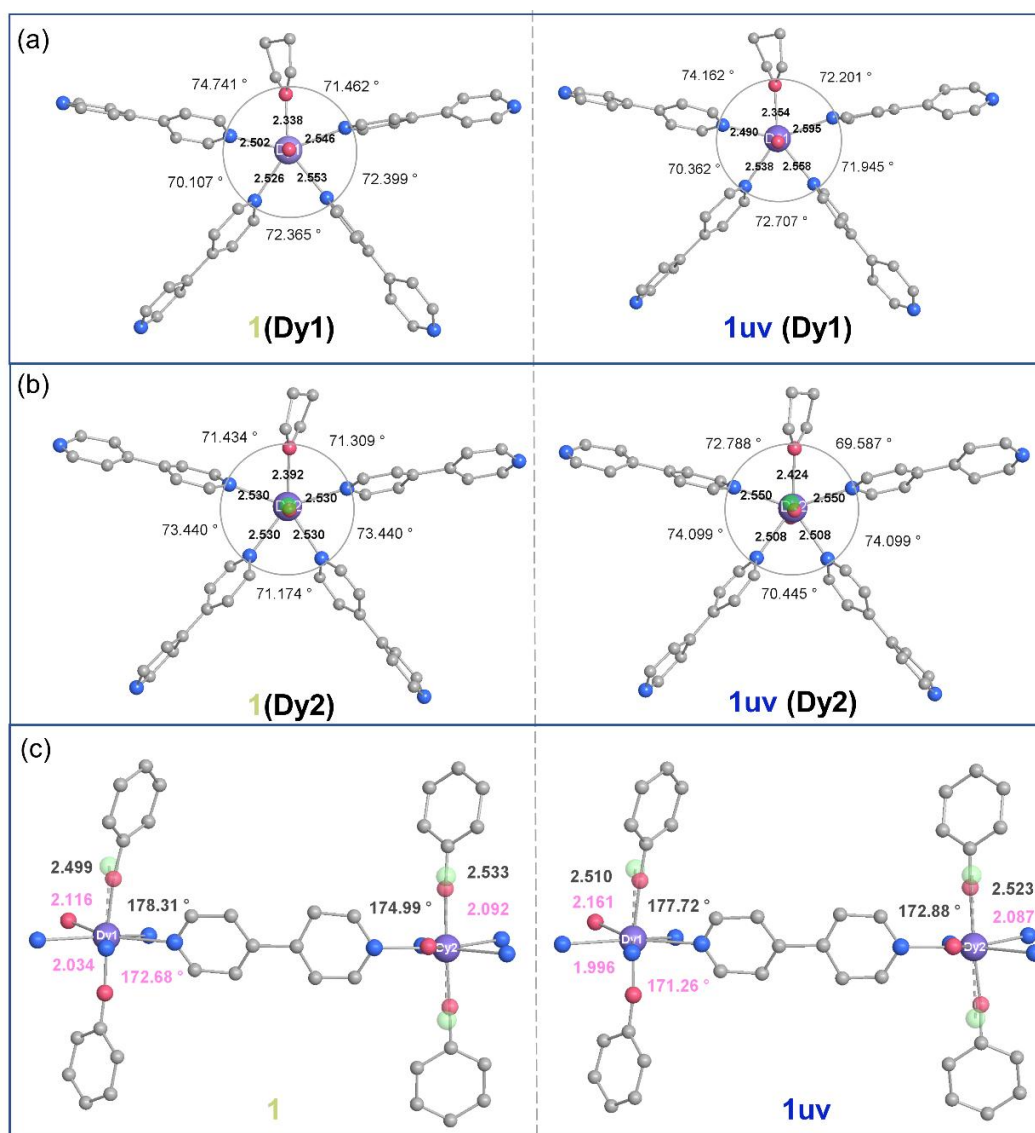


Figure S64. Selected bond lengths (Å) and angles (°) around Dy1 and Dy2 centers in **1** and **1uv**. The Dy–O distances and O–Dy–O angles are highlighted in pink, while the Dy–Cl distances and O–Dy–Cl angles are shown in black. H atoms and some C atoms are omitted for clarity.

1. Reed, J. B.; Hopkins, B.; Audrieth, L.; Selwood, P.; Ward, R.; Dejong, J. *Inorg. Synth.* Anhydrous rare earth chlorides. **1939**, 1, 28-33.
2. Goodwin, C. A. P.; Ortu, F.; Reta, D.; Chilton, N. F.; Mills, D. P. Molecular magnetic hysteresis at 60 kelvin in dysprosocenium. *Nature* **2017**, 548, 439-442.
3. Becke, A. D. Density-functional thermochemistry. III. The role of exact exchange. *J. Chem. Phys.* **1993**, 98, 5648-5652.
4. Lee, C.; Yang, W.; Parr, R. G. Development of the Colle-Salvetti correlation-energy formula into a functional of the electron density. *Phys. Rev. B* **1988**, 37, 785-789.
5. Grimme, S.; Antony, J.; Ehrlich, S.; Krieg, H. A consistent and accurate ab initio parametrization of density functional dispersion correction (DFT-D) for the 94 elements H-Pu. *J. Chem. Phys.* **2010**, 132.
6. Weigend, F.; Ahlrichs, R. Balanced basis sets of split valence, triple zeta valence and quadruple zeta valence quality for H to Rn: Design and assessment of accuracy. *Phys. Chem. Chem. Phys.* **2005**, 7, 3297-3305.
7. Krishnan, R.; Binkley, J. S.; Seeger, R.; Pople, J. A. Self - consistent molecular orbital methods. XX. A basis set for correlated wave functions. *J. Chem. Phys.* **1980**, 72, 650-654.
8. Gaussian 16, Revision C.01, Frisch, M. J.; Trucks, G. W.; Schlegel, H. B.; Scuseria, G. E.; Robb, M. A.; Cheeseman, J. R.; Scalmani, G.; Barone, V.; Petersson, G. A.; Nakatsuji, H.; Li, X.; Caricato, M.; Marenich, A. V.; Bloino, J.; Janesko, B. G.; Gomperts, R.; Mennucci, B.; Hratchian, H. P.; Ortiz, J. V.; Izmaylov, A. F.; Sonnenberg, J. L.; Williams-Young, D.; Ding, F.; Lipparini, F.; Egidi, F.; Goings, J.; Peng, B.; Petrone, A.; Henderson, T.; Ranasinghe, D.; Zakrzewski, V. G.; Gao, J.; Rega, N.; Zheng, G.; Liang, W.; Hada, M.; Ehara, M.; Toyota, K.; Fukuda, R.; Hasegawa, J.; Ishida, M.; Nakajima, T.; Honda, Y.; Kitao, O.; Nakai, H.; Vreven, T.; Throssell, K.; Montgomery, J. A., Jr.; Peralta, J. E.; Ogliaro, F.; Bearpark, M. J.; Heyd, J. J.; Brothers, E. N.; Kudin, K. N.; Staroverov, V. N.; Keith, T. A.; Kobayashi, R.; Normand, J.; Raghavachari, K.; Rendell, A. P.; Burant, J. C.; Iyengar, S. S.; Tomasi, J.; Cossi, M.; Millam, J. M.; Klene, M.; Adamo, C.; Cammi, R.; Ochterski, J. W.; Martin, R. L.; Morokuma, K.; Farkas, O.; Foresman, J. B.; Fox, D. J. Gaussian, Inc., Wallingford CT, 2016.
9. a) Roos, B. O.; Lindh, R.; Malmqvist, P.-Å.; Veryazov, V.; Widmark, P.-O. *J. Phys. Chem.* A main group atoms and dimers studied with a new relativistic ANO basis set. **2004**, 108, 2851-2858; b) Roos, B. O.; Lindh, R.; Malmqvist, P.-Å.; Veryazov, V.; Widmark, P.-O. New relativistic ANO basis sets for transition metal atoms. *J. Phys. Chem. A* **2005**, 109, 6575-6579; c) Roos, B. O.; Lindh, R.; Malmqvist, P.-Å.; Veryazov, V.; Widmark, P.-O. New relativistic ANO basis sets for actinide atoms. *Chem. Phys. Lett.* **2005**, 409, 295-299.
10. Dolomanov, O. V.; Bourhis, L. J.; Gildea, R. J.; Howard, J. A. K.; Puschmann, H. Olex2: a complete structure solution, refinement and analysis program. *J. Appl. Cryst.* **2009**, 42, 339-341.